

Project Data Page:

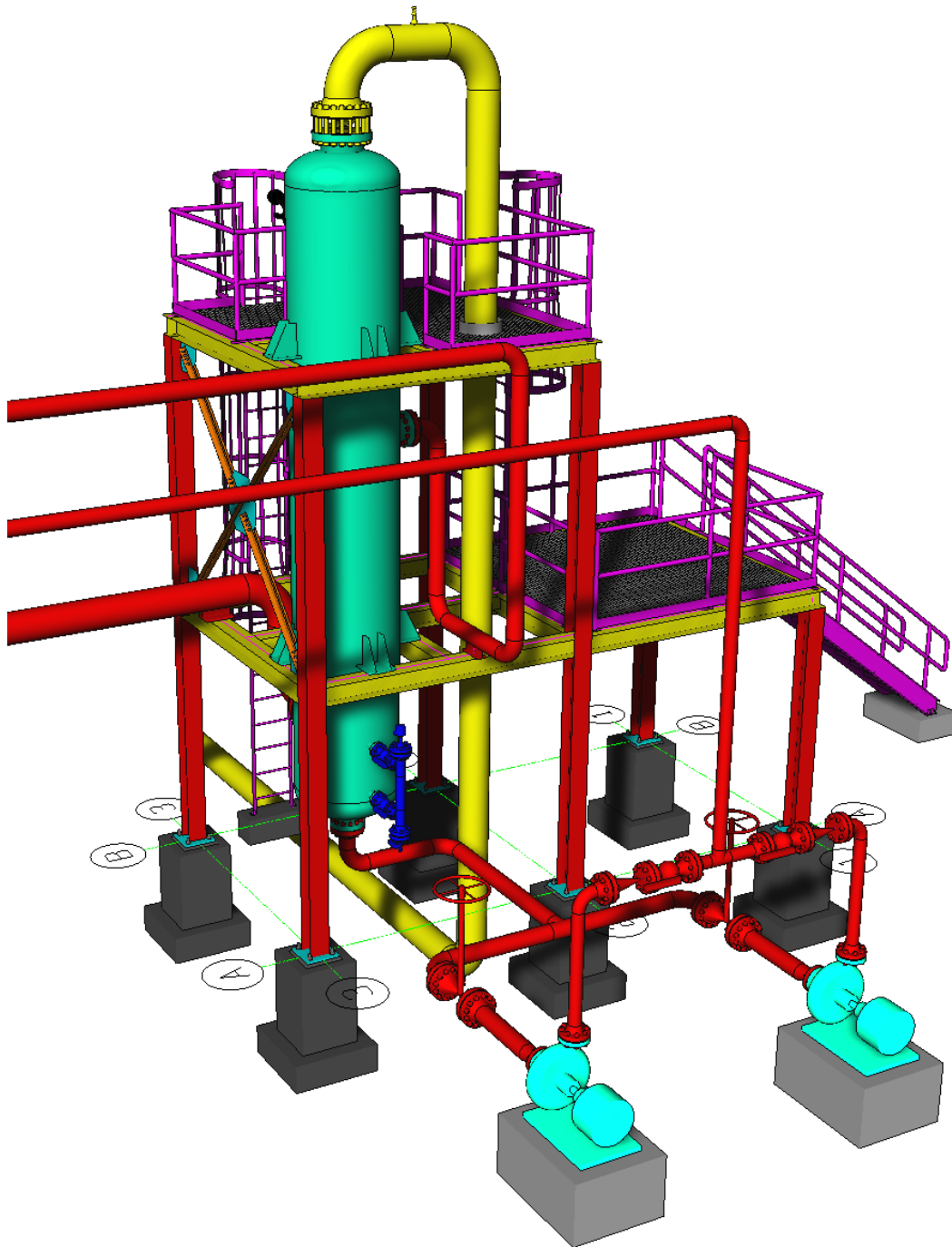




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\123C1 NVS Rev 02.pvdb

Date of Analysis : Apr 10,2026 4:12pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 123_C1_NVS Rev 02 Page 2 of 241
 Vessel Design Summary: Step: 38 4:12pm Apr 10,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1756.000 x 914.400 mm ID
 Vessel Design Length, Tangent to Tangent 13729.18 mm

 Specified Datum Line Distance 112.00 mm

 Shell Side Design Temperature 370 C
 Channel Side Design Temperature 343 C

 Shell Side Design Pressure 18.000 MPa
 Channel Side Design Pressure 14.280 MPa

 Shop Shell Side Test Pressure 24.301 MPa
 Field Shell Side Test Pressure 25.059 MPa
 Shop Channel Side Test Pressure 21.240 MPa
 Field Channel Side Test Pressure 19.726 MPa

 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Shell	SA-387 22	2	...	K21590	
Shell	SA-336 F22	3	...	K21590	
Head	SA-516 70	...	...	K02700	
Cone	SA-336 F22	3	...	K21590	
Flange	SA-336 F22	3	...	K21590	
Nozzle	SA-336 F22	3	...	K21590	+IT
Nozzle	SA-266 2	...	...	K03506	
Nozzle	SA-105	...	...	K03504	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 TP321	...	...	S32100	
Tubesheet	SA-336 F22	3	...	K21590	
Flg Bolting	SA-540 B21	1	<= 100	K14073	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-387 22	2	...	K21590	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel head	14.280	0.10	14.700	3.0000	N/A	No
Channel shell	14.297	0.10	14.900	3.0000	N/A	No
Shell	18.001	0.00	18.600	3.0000	N/A	No
Shell	18.001	0.00	18.600	3.0000	N/A	No
Shell	18.001	0.00	18.400	2.0000	N/A	No
Cone-1	18.001	0.10	18.200	2.0000	N/A	No
Cone-2	18.000	0.10	18.500	2.0000	N/A	No
Forged pipe	18.000	0.10	18.600	2.0000	N/A	No



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Vessel Design Summary:

Step: 38 4:12pm Apr 10,2026

Liquid Level: 1766.00 mm Dens.: 0.000 kg/cm3 Sp. Gr.: 0.029

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Sphere	0.0	112.0	64.0	54.0	52.5	5.5	1.00	1.00
Cylinder	2000.0	2000.0	112.0	112.0	107.5	9.5	1.00	1.00
Body Flg	3117.2	664.0	363.0	355.0	347.9	237.1	1.00	1.00
Cylinder	4717.2	1600.0	132.0	132.0	127.5	...	1.00	1.00
Cylinder	11917.2	7200.0	164.0	164.0	160.5	...	1.00	1.00
Conical	12717.2	800.0	170.0	170.0	167.6	6.1	1.00	1.00
Conical	13217.2	500.0	106.0	106.0	103.1	4.5	1.00	1.00
Cylinder	13617.2	400.0	88.0	88.0	84.7	4.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Tube inlet	WRC-297	0.976	Shell	Passed
N2- Tube outle	WRC-297	0.976	Shell	Passed
Shell Bypass N	WRC-297	0.970	Shell	Passed
Shell inlet-3	WRC-297	0.970	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width 288.000 mm
 Saddle Bearing Angle 120.000 deg.
 Centerline Dimension 2133.000 mm
 Wear Pad Width 300.000 mm
 Wear Pad Thickness 30.000 mm
 Wear Pad Bearing Angle 132.000 deg.
 Distance from Saddle to Tangent 3400.000 mm

Baseplate Length 2616.000 mm
 Baseplate Thickness 55.000 mm
 Baseplate Width 420.000 mm
 Number of Ribs (including outside ribs) 5
 Rib Thickness 32.000 mm
 Web Thickness 32.000 mm
 Height of Center Web 1085.000 mm
 Number of Bolts in Baseplate 8

Baseplate Sketch

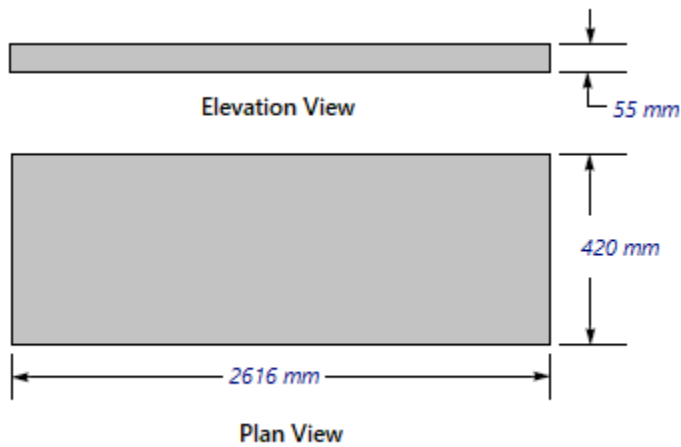
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Vessel Design Summary:

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Maximum Tensile Bolt Load	86439. N
---------------------------	----------

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	2045735.12	N
Maximum Transverse Saddle Shear Load	89572.17	N
Maximum Longitudinal Saddle Shear Load	103238.98	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	16636.71	N
Maximum Transverse Saddle Shear Load	3512.42	N
Maximum Longitudinal Saddle Shear Load	2069.50	N

Weights:

Fabricated	- Bare w/o removable internals	173221.8	kg
Shop Test	- Fabricated + water (full)	203265.3	kg
Shipping	- Fab. + rem. intls.+ shipping app.	173221.8	kg
Erected	- Fab. + rem. intls.+ insul. (etc)	174671.8	kg
Empty	- Fab. + intls. + details + wghts.	174671.8	kg
Operating	- Empty + operating liquid (no ca)	182900.2	kg
Field Test	- Empty weight + test medium (full)	203265.3	kg

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Nozzle Summary:

Step: 36 4:12pm Apr 10,2026

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Manhole	14.7	OK	...	OK	37.69	OK	Passed
N1- Tube inlet	14.9	OK	...	OK	29.59	OK	Passed
N2- Tube outlet	14.9	OK	...	OK	23.04	OK	Passed
Shell Bypass No	18.5	...	...	OK	40.57	OK	Passed
Shell inlet-3	18.4	...	...	OK	61.72	OK	Passed
BD3	18.4	...	...	OK	16.25	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 18.4 Nozzle : BD3 [Shellside]
 Minimum MAWP Nozzles : 14.7 Nozzle : Manhole [Tubeside]

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	Manhole	0.000	0.000	1231.200
20	N1- Tube inlet	1012.000	0.000	824.800
20	N2- Tube outlet	1012.000	180.000	621.600
40	Shell Bypass No	4070.564	0.000	824.800
40	Shell inlet-3	4109.175	180.000	1236.000
50	BD3	7160.156	180.000	428.648

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 35 4:12pm Apr 10,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
BD3	2.000 in	Actual	LWN	4.120	26.924	...	...	356.06	...
N2- Tube outlet	12.000 in	Actual	WNF	13.890	24.000	...	...	618.85	...
N1- Tube inlet	16.000 in	Actual	WNF	18.362	30.000	...	...	964.88	1500
Shell Bypass No	16.000 in	Actual	None	19.250	41.280	...	...	730.06	...
Manhole	24.000 in	Actual	WNF	27.984	50.600	...	...	562.63	1500
Shell inlet-3	30.000 in	Actual	None	30.000	75.000	...	...	1303.97	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
BD3	SA-105	155.000	10.000	...	...	...
N2- Tube ou	SA-266 2	64.000	15.000	...	...	...
N1- Tube in	SA-266 2	64.000	15.000	...	...	...
Shell Bypas	SA-336 F22	155.000	10.000	...	...	...
Manhole	SA-336 F22	56.000	10.000	...	...	...
Shell inlet	SA-336 F22	132.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
BD3	7048.156	180.0	190.50	0.00	Shell
N2- Tube outlet	900.000	180.0	450.00	0.00	Channel shell
N1- Tube inlet	900.000	0.0	750.00	0.00	Channel shell
Shell Bypass No	3958.564	0.0	500.00	0.00	Shell
Manhole	...	0.0	350.00	0.00	Channel head
Shell inlet-3	3997.175	180.0	1085.00	0.00	Shell

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Bill of Material:

Step: 34 4:12pm Apr 10,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 64.0mm THK X 1765.0mm ID X 112.0mm	SA-516 70
1	WELD NECK FLANGE: 363.0mm THK X 2610.0mm OD	SA-336 F22
1	CONCENTRIC CONE: 106.0mm THK X 1100.0mm X 914.4mm ID X 500.0 mm	SA-336 F22
1	CYLINDER: 88.0mm THK X 914.4mm ID X 400.0mm	SA-336 F22
12	BAFFLES: 1747.6mm DIA X 19.0mm THK	SA-387 22
16	TIE RODS: 12mm DIA X 8500mm	SA-193 B7M
16	SPACERS: 24.0mm DIA X 3.0mm THK X 8500.0mm	SA-179
1	TUBESHEET: 2610.0mm X 450.0mm THK	SA-336 F22
2156	TUBES: 8500.0mm X 19.0mm DIA X 3.0mm THK	SA-213 TP321
1	CLADDING: 0.000mm X 6.000mm THK	...
1	LINING: 1600.000mm X 3.000mm THK	...
2	SADDLE: 288.0mm X 120 DEG	SA-387 22
1	CLADDING: 7200.000mm X 5.000mm THK	...
1	CLADDING: 800.000mm X 5.000mm THK	...
1	CLADDING: 500.000mm X 5.000mm THK	...
1	CLADDING: 400.000mm X 5.000mm THK	...
1	CLASS 1500 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	GASKET: 1826.0mm OD X 1780.0mm ID	...
36	BODY FLANGE BOLTS: 95.2mm DIA	SA-540 B21
36	NUTS FOR BODY FLANGE BOLTS: 95.2mm DIA	...
2	CYLINDRICAL SEGMENT 112.0mm THK X 2191.3mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 112.0mm THK X 1134.0mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 132.0mm THK X 2485.9mm X 1600.0mm	SA-387 22
1	CYLINDRICAL SEGMENT 132.0mm THK X 3030.7mm X 1600.0mm	SA-387 22
2	CYLINDRICAL SEGMENT 164.0mm THK X 2123.9mm X 7200.0mm	SA-336 F22
1	CYLINDRICAL SEGMENT 164.0mm THK X 1268.8mm X 7200.0mm	SA-336 F22
1	CON. CONICAL SEGMENT 170.0mm THK X 2338.2mm X 800.0mm X 1557.2mm	SA-336 F22
1	CON. CONICAL SEGMENT 170.0mm THK X 2850.7mm X 800.0mm X 1898.5mm	SA-336 F22
1	NAMEPLATE	...



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MDMT Summary: Step: 37 4:12pm Apr 10,2026

Minimum Design Metal Temperature Results Summary (C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Bolting [17]					Impact Testing Required			
Forged pipe [8]	B	29	-104		0.321	88.000	1.00	Yes
Left Saddle [24]	B	4						Yes
Sdl 2 Fr40 [24]	B	4						Yes
BD3 [1]	A	21	-104		0.168	26.924	1.00	No
Tubesheet: SS[13]	B	34	32		0.962	112.500	1.00	No
Warmest MDMT:		34	32					
Channel head [9]	B	18	15		0.935	54.000	1.00	Yes
Manhole [1]	B	37	19		0.683	50.600	1.00	Yes
Nozzle Flg [5]	A	-18	-42					
N1- Tube inlet[1]	B	24	16		0.870	30.000	1.00	Yes
Nozzle Flg [5]	A	-18	-42					
N2- Tube outle[1]	B	18	9		0.843	24.000	1.00	Yes
Tubesheet: CS[14]	B	34	32		0.962	112.500	1.00	No
Warmest MDMT:		37	32					

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
- The Basic MDMT includes the (30F) PWHT credit if applicable.



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Warnings and Errors:

Step: 0 4:12pm Apr 10,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 4:12pm Apr 10,2026

Units used in this Analysis (SI):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kg
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1602E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1602E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in. ⁴	416231.0000	mm ⁴
G Load	G's	1.0000	Gs
Wind Load	lbs./sq.ft.	47.8790	N/m ²
Elevation	Feet	304.8000	mm
Volume	in. ³	0.1639E-04	m ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	18	MPa
Channel Side Design Pressure	14.28	MPa
Shell Side Design Temperature	370.0	C
Channel Side Design Temperature	343.0	C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-12.0	C
MDMT (CET), Tube Side	-12.0	C
User defined MAWP, Shell Side	0	MPa
User defined MAWP, Channel Side	0	MPa
User defined MAPnc, Shell Side	0	MPa
User defined MAPnc, Channel Side	0	MPa
User defined Test Pres., Shell Side	0	MPa
User defined Test Pres., Channel Side	0	MPa

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 4:12pm Apr 10,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P. + Static Head to Element Bottom	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 Yes

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/8
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Channel head
Distance "FROM" to "TO"	112 mm
Inside Diameter	1765 mm
Element Thickness	54 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	64 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	14.28 MPa
Design Temperature Internal Pressure	343 C
Design External Pressure	0.103 MPa
Design Temperature External Pressure	200 C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	129.12 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm3
P Number Thickness	32.004 mm
Yield Stress, Operating	194.68 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Nozzle
Detail ID	Manhole
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	39508 N



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Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-336 F22 +IT

Element From Node	10
Detail Type	Weight
Detail ID	DAVIT WEIGHT
Dist. from "FROM" Node / Offset dist	-650 mm
Miscellaneous Weight	1961.2 N
Offset from Element Centerline	300 mm

Element#	2/8
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Channel shell
Distance "FROM" to "TO"	2000 mm
Inside Diameter	1756 mm
Element Thickness	112 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	112 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	14.28 MPa
Design Temperature Internal Pressure	343 C
Design External Pressure	0.103 MPa
Design Temperature External Pressure	200 C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1756 mm
Liquid Density	0.0009996 kg/cm3

Element From Node	20
Detail Type	Nozzle
Detail ID	N1- Tube inlet
Dist. from "FROM" Node / Offset dist	900 mm
Nozzle Diameter	16 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	14294 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	20
Detail Type	Nozzle
Detail ID	N2- Tube outlet
Dist. from "FROM" Node / Offset dist	900 mm
Nozzle Diameter	12 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	8770.5 N
Grade of Attached Flange	None
Nozzle Matl	SA-266 2



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Element From Node	20
Detail Type	Weight
Detail ID	Pass Partition
Dist. from "FROM" Node / Offset dist	800 mm
Miscellaneous Weight	4412.7 N
Offset from Element Centerline	0 mm

Element#	3/8
Element From Node	30
Element To Node	40
Element Type	Flange
Description	Shell
Distance "FROM" to "TO"	664 mm
Flange Inside Diameter	1760 mm
Element Thickness	355 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	363 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18 MPa
Design Temperature Internal Pressure	370 C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F22
Allowable Stress, Ambient	148 MPa
Allowable Stress, Operating	138.2 MPa
Allowable Stress, Hydrotest	278.99 MPa
Material Density	0.00775 kg/cm3
P Number Thickness	0 mm
Yield Stress, Operating	245.6 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K21590
Class / Thickness / Grade	3 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1766 mm
Liquid Density	0.29387E-04 kg/cm3

Element From Node	30
Detail Type	Lining
Detail ID	caldding
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	0 mm
Thickness of Lining	6 mm
Density	0.00801 kg/cm3

Element#	4/8
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	1600 mm
Inside Diameter	1756 mm



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Element Thickness	132	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	132	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	18	MPa
Design Temperature Internal Pressure	370	C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 22	
Allowable Stress, Ambient	148	MPa
Allowable Stress, Operating	138.2	MPa
Allowable Stress, Hydrotest	278.99	MPa
Material Density	0.00775	kg/cm3
P Number Thickness	0	mm
Yield Stress, Operating	245.6	MPa
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K21590	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1756	mm
Liquid Density	0.29387E-04	kg/cm3

Element From Node	40	
Detail Type	Lining	
Detail ID	Cladding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1600	mm
Thickness of Lining	3	mm
Density	0.00801	kg/cm3

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell Bypass No	
Dist. from "FROM" Node / Offset dist	841.39	mm
Nozzle Diameter	16	in.
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12919	N
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell inlet-3	
Dist. from "FROM" Node / Offset dist	880	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	25949	N
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22	



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Input Echo:

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Element#	5/8
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	7200 mm
Inside Diameter	1756 mm
Element Thickness	164 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	164 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18 MPa
Design Temperature Internal Pressure	480 C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F22
Allowable Stress, Ambient	148 MPa
Allowable Stress, Operating	110.72 MPa
Allowable Stress, Hydrotest	278.99 MPa
Material Density	0.00775 kg/cm3
P Number Thickness	0 mm
Yield Stress, Operating	223.61 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K21590
Class / Thickness / Grade	3 ::
Product Form	Forgings
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1100 mm
Width of Saddle	288 mm
Height of Saddle at Bottom	2133 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	300 mm
Thickness of Wear Plate	30 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.40000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	50
Detail Type	Saddle
Detail ID	Sd1 2 Fr40
Dist. from "FROM" Node / Offset dist	6500 mm
Width of Saddle	288 mm
Height of Saddle at Bottom	2133 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	300 mm
Thickness of Wear Plate	30 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0



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Input Echo: Step: 1 4:12pm Apr 10,2026

Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1756	mm
Liquid Density	0.29390E-04	kg/cm3
Element From Node	50	
Detail Type	Lining	
Detail ID	Caldding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	7200	mm
Thickness of Lining	5	mm
Density	0.00801	kg/cm3
Element From Node	50	
Detail Type	Nozzle	
Detail ID	BD3	
Dist. from "FROM" Node / Offset dist	2331	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	297.53	N
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	50	
Detail Type	Weight	
Detail ID	Fire prooving w	
Dist. from "FROM" Node / Offset dist	1100	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	2133	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Fire prroving w	
Dist. from "FROM" Node / Offset dist	6500	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	2133	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Bundle skid bar	
Dist. from "FROM" Node / Offset dist	2800	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element#	6/8	
Element From Node	60	
Element To Node	70	
Element Type	Conical	
Description	Cone-1	
Distance "FROM" to "TO"	800	mm
Inside Diameter	1756	mm
Element Thickness	170	mm



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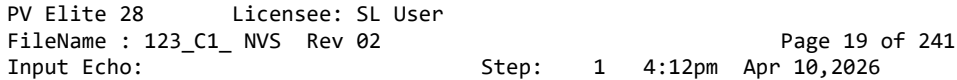
Input Echo:

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Internal Corrosion Allowance	2	mm
Nominal Thickness	170	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	18	MPa
Design Temperature Internal Pressure	480	C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F22	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	1100	mm
Design Length of Cone	800	mm
Large knuckle radius	525	mm
Large knuckle thickness	175	mm
Small knuckle radius	0	mm
Small knuckle thickness	0	mm
Half Apex Angle of Cone	25.750565	degrees
Toriconical (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1756	mm
Liquid Density	0.29387E-04	kg/cm3
Element From Node	60	
Detail Type	Lining	
Detail ID	Cladding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	800	mm
Thickness of Lining	5	mm
Density	0.00801	kg/cm3

Element#	7/8	
Element From Node	70	
Element To Node	80	
Element Type	Conical	
Description	Cone-2	
Distance "FROM" to "TO"	500	mm
Inside Diameter	1100	mm
Element Thickness	106	mm
Internal Corrosion Allowance	2	mm
Nominal Thickness	106	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	18	MPa
Design Temperature Internal Pressure	480	C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F22	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	914.4	mm
Design Length of Cone	500	mm
Half Apex Angle of Cone	10.514455	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Liquid	



Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1100 mm
Liquid Density	0.29387E-04 kg/cm3

Element From Node	70
Detail Type	Lining
Detail ID	cladding
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	500 mm
Thickness of Lining	5 mm
Density	0.00801 kg/cm3

Element From Node	70	
Detail Type	Weight	
Detail ID	weigt for incon	
Dist. from "FROM" Node / Offset dist	225.04	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element#	8/8	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Forged pipe	
Distance "FROM" to "TO"	400	mm
Inside Diameter	914.4	mm
Element Thickness	88	mm
Internal Corrosion Allowance	2	mm
Nominal Thickness	88	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	18	MPa
Design Temperature Internal Pressure	480	C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F22	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	914.4 mm
Liquid Density	0.29390E-04 kg/cm3

Element From Node	80	
Detail Type	Lining	
Detail ID	cladding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	400	mm
Thickness of Lining	5	mm
Density	0.00801	kg/cm3



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XY Coordinate Calculations:

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Channel head		112	...	112	...
Channel shell		2112	...	2000	...
Shell		3229.18	...	664	...
Shell		4829.18	...	1600	...
Shell		12029.2	...	7200	...
Cone-1		12829.2	...	800	...
Cone-2		13329.2	...	500	...
Forged pipe		13729.2	...	400	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: FLANGE

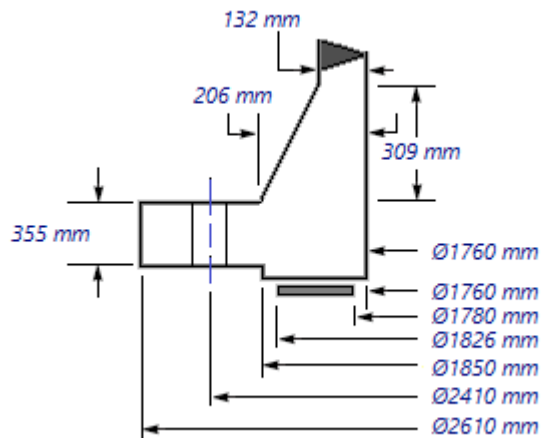
Flng: 3 4:12pm Apr 10,2026

Flange Input Data Values

Description: FLANGE :

Shell

Flange Type	Integral Weld Neck, Integral type		
Design Pressure	P	18.00	MPa
Design Temperature		370	C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1760.000	mm
Flange Outside Diameter	A	2610.000	mm
Flange Thickness	t	355.0000	mm
Thickness of Hub at Small End	go	132.0000	mm
Thickness of Hub at Large End	gl	206.0000	mm
Length of Hub	h	309.0000	mm
Flange Material		SA-336 F22	
Flange Material UNS number		K21590	
Flange Allowable Stress At Temperature	Sfo	138.20	MPa
Flange Allowable Stress At Ambient	Sfa	148.00	MPa
Bolt Material		SA-540 B21	
Bolt Allowable Stress At Temperature	Sb	228.00	MPa
Bolt Allowable Stress At Ambient	Sa	228.00	MPa
Diameter of Bolt Circle	C	2410.000	mm
Nominal Bolt Diameter	a	95.2500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1850.000	mm
Flange Face Inside Diameter	Fid	1760.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1826.000	mm
Gasket Inside Diameter	Gi	1780.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	69.00	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: FLANGE

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ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (18 \cdot (1760/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 18) + Ca$$

$$= 127.7617 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 129) / (883 + 0.6 \cdot 129)$$

$$= 18.563 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148 \cdot 1 \cdot 132) / (880 + 0.6 \cdot 132)$$

$$= 20.367 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	1766.000 mm
Corroded Large Hub,	g1Cor = g1-ci	203.000 mm
Corroded Small Hub,	g0Cor = go-ci	129.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	119.000 mm

Gasket Contact Width,	N = (Go - Gi)/2	23.000 mm
Basic Gasket Width,	bo = N/2	11.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.545 mm
Gasket Reaction Diameter,	G = Go - 2 * b	1808.909 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1809^2 \cdot 18$$

$$= 46256332.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.55 \cdot \pi \cdot 1809 \cdot 3 \cdot 18$$

$$= 5244462.500 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1766^2 \cdot 18$$

$$= 44087872.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 46256332 - 44087872$$

$$= 2168457.750 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(46256332 + 5244462 + 0, 0)$$

$$= 51500792.000 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 69 \cdot 8.55 \cdot \pi \cdot 1809 + 0 \cdot 0 \cdot 0$$

$$= 3350534.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((51500792 + 0 + 4 \cdot 0 / 1809) / 228, 3350534 / 228)$$

$$= 225903.984 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:



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Flg Calc [Int P]: FLANGE

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$$\begin{aligned}
 &= 2a + 6t/(m + 0.5) \\
 &= 2 \cdot 95.2 + 6 \cdot 355/(3 + 0.5) \\
 &= 799.071 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 2410 \cdot \sin(\pi / 36) \\
 &= 210.045 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{(Bs/(2a + t))}, 1) \\
 &= \max(\sqrt{(210/(2 \cdot 95.2 + 355))}, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.2E+06	0.2E+06	
Radial Distance between Hub and Bolts:	112.713	119.000	
Radial Distance between Bolts and Edge:	88.900	100.000	
Circ. Spacing between the Bolts:	193.675	210.045	799.071

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= Ab \cdot Sa / (y \cdot \pi (Go + Gi)) \\
 &= 234766 \cdot 228 / (69 \cdot \pi (1826 + 1780)) \\
 &= 68.476 \text{ mm [Note: Exceeds actual gasket width, 23.000]}
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= Sa (Am + Ab) / 2 \\
 &= 228 (225904 + 234766) / 2 \\
 &= 52510960.00 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= Wo - H \\
 &= 51500792 - 46256332 \\
 &= 5244463.50 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (2410 - 1809) / 2 \\
 &= 300.5455 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5 ((C - B) / 2 + hg) \\
 &= 0.5 ((2410 - 1766) / 2 + 301) \\
 &= 311.2728 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g1) / 2 \\
 &= (2410 - 1766 - 203) / 2 \\
 &= 220.5000 \text{ mm}
 \end{aligned}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	44087872.	220.5000	1.0000	9725317120.



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Flg Calc [Int P]: FLANGE

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Face Pressure, Mt	2168458.	311.2728	1.0000	675255488.
Gasket Load, Mg	5244464.	300.5455	1.0000	1576839040.
Gasket Seating, Matm	52510960.	300.5455	1.0000	15788332032.

Total Moment for Operation, Mo 11977412608. N-mm
Total Moment for Gasket seating, Mg 15788332032. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 477.299 mm
Hub Ratio, h/h0 = HL / H0 0.647
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.574

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.821
Factor V per 4.16.5 0.269
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 2610/1766
= 1.478

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169(1.48² • log10(1.48))) / (1.48² - 1)) / (1.48 - 1)
= 5.142

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.48²(1 + 8.55246 • log10(1.48)) - 1) / ((1.0472 + 1.9448 • 1.48²) • (1.48 - 1))
= 1.720

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.48²(1 + 8.55246 • log10(1.48)) - 1) / (1.36136(1.48² - 1) • (1.48 - 1))
= 5.650

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.48² + 1) / (1.48² - 1)
= 2.689

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 5.65 • 129² • 477 / 0.27
= 167036928.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.82/477
= 0.002 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (355 • 0.0017 + 1) / 1.72 + 355³/167036928
= 1.204

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1•11.98E+09/1766)/(1.2•203²)
= 136.64 MPa

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)



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Flg Calc [Int P]: FLANGE

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$$= (1 \cdot 15.79E+09 / 1766) / (1.2 \cdot 203^2)$$

$$= 180.12 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 11.98E+09 / 1766) / (1.2 \cdot 355^2)$$

$$= 81.06 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.81 \cdot 15.79E+09 / 1766) / (1.2 \cdot 355^2)$$

$$= 106.85 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (5.14 \cdot 11.98E+09 / (355^2 \cdot 1766)) - 2.69 \cdot 81.1$$

$$= 58.67 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (5.14 \cdot 15.79E+09 / (355^2 \cdot 1766)) - 2.69 \cdot 107$$

$$= 77.34 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (137 + \max(81.1, 58.7)) / 2$$

$$= 108.85 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (180 + \max(107, 77.3)) / 2$$

$$= 143.48 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 51500792 / 234766$$

$$= 219.39 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (3350534 / 234766)$$

$$= 14.27 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	136.64	207.30	180.12	222.00
Radial Flange	81.06	138.20	106.85	148.00
Tangential Flange	58.67	138.20	77.34	148.00
Maximum Average	108.85	138.20	143.48	148.00
Bolting	219.39	228.00	14.27	228.00

Minimum Required Flange Thickness 347.929 mm

Estimated M.A.W.P. (Operating) 18.6 MPa

Estimated Finished Weight of Flange at given Thk. 10479.7 kg

Estimated Unfinished Weight of Forging at given Thk 15013.7 kg

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

Operating stress	Gasket Seating stress
------------------	-----------------------



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Flg Calc [Int P]: FLANGE

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		Actual	Allowed	ratio	Actual	Allowed	ratio
Longitudinal Hub	Sh	561.37	561.37	1.000	561.37	619.99	0.905
Tangential Flange	St	241.03	421.03	0.572	241.03	464.99	0.518
Tangential Flange	Sto	178.05	280.68	0.634	178.05	309.99	0.574
Combined	Sh/f + St	802.40	842.05	0.953	802.40	929.98	0.863
Combined	Sr + St	574.04	842.05	0.682	574.04	929.98	0.617
Combined	Sh + Sto	739.42	842.05	0.878	739.42	929.98	0.795
Radial Flange	Sr	333.01	561.37	0.593	333.01	619.99	0.537
Maximum Stress Ratio				1.000			0.905

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 0.91/2.04)

= 1.0000

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/h0 < 0.35, else max (0.25; fn3)

= max(0.25; 0.77/2.3) if h/h0 < 0.35, else max (0.25; 0.25/0.8)

= 0.3172

In this case, the hub ratio h/h0, 0.6474 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



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Flg Calc [Int P]: FLANGE

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Input Data:

Maximum permissible bolt stress	Sbmax	724.0 MPa
Minimum permissible bolt stress	Sbmin	414.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	697.2 MPa
Target assembly gasket stress	SgT	207.0 MPa
Maximum permissible gasket stress	Sgmax	276.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	6521.2769 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 MPa
Flange Material Operating Yield Stress	Syo	245.6 MPa
Single flange rotation at Sfmax	Thetafmax	0.323 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1826^2 - 1780^2)$$

$$= 130278.5 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 130278 / (36 \cdot 6521)$$

$$= 114.9 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(115; 724)$$

$$= 114.9 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(115; 414)$$

$$= 414.0 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(414; 697)$$

$$= 414.0 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 414 \geq 140 (130278 / (36 \cdot 6521))$$

$$:: 414.0 \geq 77.7 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 414 \geq (97 \cdot 130278 + \pi/4 \cdot 18 \cdot 1780^2) / (0.7 \cdot 6521 \cdot 36)$$

$$:: 414.0 \geq 349.5 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 414 \leq 276 (130278 / (36 \cdot 6521))$$

$$:: 414.0 \leq 153.2 \text{ MPa} \text{ Check Fails}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 414 \leq 697 (1/0.32)$$

$$:: 414.0 \leq 2158.6 \text{ MPa} \text{ Check Passes}$$



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Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 414 \cdot 0.2 \cdot 6521 \cdot 95.2$$

$$= 51447836.0 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 81 MPa
 Bolt Allowable Stress for: SA-540 B21 Sb 227 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 15.79E+09 / 1 \cdot 1 \cdot 0.27 / (1.18 \cdot 210987 \cdot 129^2 \cdot 477 \cdot 0.3)$$

$$= 0.372 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 11.98E+09 / 1 \cdot 1 \cdot 0.27 / (1.18 \cdot 186264 \cdot 129^2 \cdot 477 \cdot 0.3)$$

$$= 0.319 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 15.79E+09 / 1 \cdot 1 \cdot 0.27 / (1.2 \cdot 210987 \cdot 129^2 \cdot 477 \cdot 0.3)$$

$$= 0.365 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 11.98E+09 / 1 \cdot 1 \cdot 0.27 / (1.2 \cdot 186264 \cdot 129^2 \cdot 477 \cdot 0.3)$$

$$= 0.313 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	135.33	222.00	176.96 222.00
Radial Flange	82.39	148.00	107.74 148.00
Tangential Flange	57.68	148.00	75.42 148.00
Maximum Average	108.86	148.00	142.35 148.00
Bolting	219.39	228.00	14.27 228.00

Minimum Req'd Thickness, ambient, no Corrosion: 346.100 mm

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Internal Pressure Calculations: Step: 4 4:12pm Apr 10,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Channel head		14.28	64	3	1765	129.12	1
Channel shell		14.297	112	3	1756	129.12	2
Shell		18.001	363	3	1760	138.2	3
Shell		18.001	132	3	1756	138.2	4
Shell		18.001	164	2	1756	110.72	5
Cone-1		18.001	170	2	1651.7	110.72	6
Cone-2		18	106	2	1100	110.72	7
Forged pipe		18	88	2	914.4	110.72	8

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Channel head		14.28	14.7	16.7	54	52.5126	1
Channel shell		14.28	14.9	16.4	112	107.492	2
Shell		18	18.6	18.7	355	347.929	3
Shell		18	18.6	20.4	132	127.479	4
Shell		18	18.4	24.9	164	160.531	5
Cone-1		18	18.2	24.7	170	167.626	6
Cone-2		18	18.5	25.2	106	103.146	7
Forged pipe		18	18.6	25.5	88	84.7233	8

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 18.2 MPa
 Shell Side MAPnc = 18.7 MPa
 Channel Side MAWP = 14.7 MPa
 Channel Side MAPnc = 16.4 MPa

Elements Suitable for Design Internal Pressure.

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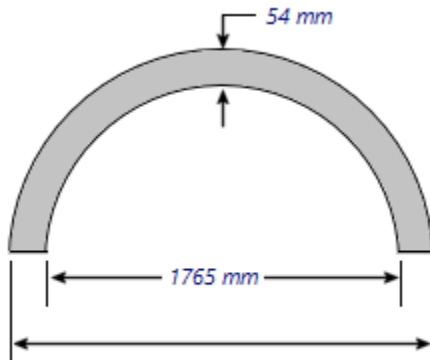
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Element: Channel head

Step: 5 4:12pm Apr 10,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel head

Spherical Head Design Information:			
Design Pressure P	14.280 MPa	Int. Design Temperature	343 C
Material	SA-516 70	Ext. Design Temperature	200 C
Length	112.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	129.1 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	1765.000 mm	Actual Stress	125.4 MPa
Spec. Min. Thk t _e , or s _a	54.0000 mm	Spec. Nominal Thk.	64.000 mm
Surface Area	6294936.0 mm ²	SG of Contents	0.000
Element Volume	1.7 m ³	Weight of Contents	0.0 N
Empty Weight	28700.0 N	Operating Weight	28700.0 N
		Comp. All. Ope. Stress	101.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.062 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.031 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (14.3 \cdot 886) / (2 \cdot 129 \cdot 1 - 0.2 \cdot 14.3)$$

$$= 49.5126 + 3.0000 = 52.5126 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 129 \cdot 1 \cdot 51) / (886 + 0.2 \cdot 51)$$

$$= 14.7 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 54) / (882 + 0.2 \cdot 54)$$



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= 16.7 MPa

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (14.3 \cdot (886 + 0.2 \cdot 51)) / (2 \cdot 1 \cdot 51)$$

$$= 125.398 \text{ MPa}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.249 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 54$, $t_r = 47.7$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.93$, Temp. Reduction = 4 C

Min Metal Temp. w/o impact per UCS-66, Curve B	18 C
Min Metal Temp. at Required thickness (UCS 66.1)	15 C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	14.704 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 C
Governing Thk.	54.000 mm	Basic MDMT	18 C
Stress/Thk. Ratio	0.935	Temperature Difference	4 C
UCS-66(c) red.	No	Computed Min. Temp.	15 C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 22.062 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 11.031 mm

EXTERNAL PRESSURE RESULTS for: Channel head

Spherical Head From 10 to 20 Ext. Chart: CS-2 at 200 C

Maximum External Pressure per Code Case 2286:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 192592 (51 / 936)$$

$$= 786.615 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 786.6 / 225.0$$

$$= 3.496$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr > 1.6 \text{ and } Fr < 6.25, F_{ic} = 1.31 \cdot F_y / (1.15 + S_y / F_{he})$$

$$= 1.31 \cdot 225 / (1.15 + 225 / 787)$$

$$= 205.259 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 205 / 225$$



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Element: Channel head Step: 5 4:12pm Apr 10,2026

= 1.731

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 1.6$ and $Fr < 6.25$, $Fha = 1.31 \cdot Fy / (FS \cdot (1.15 + Sy/Fhe))$
= $1.31 \cdot 225 / (1.73 \cdot (1.15 + 225/787))$
= 118.575 MPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha[t/Ro]$
= $2 \cdot 119 [51/936]$
= 12.915 MPa

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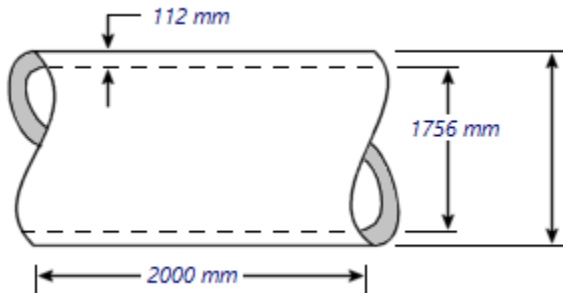
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Element: Channel shell

Step: 6 4:12pm Apr 10,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	14.297 MPa	Int. Design Temperature	343 C
Material	SA-516 70	Ext. Design Temperature	200 C
Length	2000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	129.1 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	1756.000 mm	Actual Stress	124.1 MPa
Spec. Min. Thk t,e, or sa	112.0000 mm	Spec. Nominal Thk.	112.000 mm
Surface Area	12440708.0 mm ²	SG of Contents	1.000
Element Volume	4.8 m ³	Weight of Contents	47475.9 N
Empty Weight	99907.2 N	Operating Weight	147383.1 N
		Comp. All. Ope. Stress	111.7 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (14.3 \cdot 881) / (129 \cdot 1 - 0.6 \cdot 14.3) \\
 &= 104.4920 + 3.0000 = 107.4920 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.017 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (129 \cdot 1 \cdot 109) / (881 + 0.6 \cdot 109) \\
 &= 14.9 - 0.017 = 14.9 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 112) / (878 + 0.6 \cdot 112) \\
 &= 16.4 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (14.3 \cdot (881 + 0.6 \cdot 109)) / (1 \cdot 109) \\
 &= 124.137 \text{ MPa}
 \end{aligned}$$



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 Element: Channel shell Step: 6 4:12pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.996 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:
 The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	14.704 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 C
Governing Thk.	112.000 mm	Basic MDMT	-17 C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 C
UCS-66(c) red.	No	Computed Min. Temp.	-17 C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Channel shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 200 C

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 2406 / \sqrt{990 \cdot 109}$$

$$= 7.325$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (7.32 - 0.579)$$

$$= 0.1364$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.14 \cdot 192592 (109 / 1980)$$

$$= 2313.529 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 2313.5 / 225.0$$

$$= 10.282$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 225.009 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [225] \geq F_y [225], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y / FS$$

$$= 225.009 / 1.667$$

$$= 134.978 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:



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Element: Channel shell

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= 2 • Fha(t / Do)

= 2 • 135 (109/1980)

= 14.861 MPa

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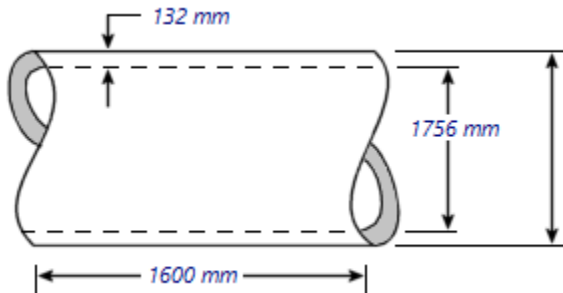
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Element: Shell

Step: 7 4:12pm Apr 10,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	18.001 MPa	Int. Design Temperature	370 C
Material	SA-387 22	Ext. Design Temperature	0 C
Length	1600.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.2 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 MPa
Inside Diameter Di	1756.000 mm	Actual Stress	133.7 MPa
Spec. Min. Thk t,e, or sa	132.0000 mm	Spec. Nominal Thk.	132.000 mm
Surface Area	10153627.0 mm ²	SG of Contents	0.029
Element Volume	2.9 m ³	Weight of Contents	833.2 N
Empty Weight	95206.7 N	Operating Weight	96039.9 N
		Comp. All. Ope. Stress	108.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 881) / (138 \cdot 1 - 0.6 \cdot 18) \\
 &= 124.4791 + 3.0000 = 127.4791 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 129) / (881 + 0.6 \cdot 129) \\
 &= 18.6 - 0.00051 = 18.6 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148 \cdot 1 \cdot 132) / (878 + 0.6 \cdot 132) \\
 &= 20.4 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (881 + 0.6 \cdot 129)) / (1 \cdot 129) \\
 &= 133.734 \text{ MPa}
 \end{aligned}$$



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Element: Shell

Step: 7 4:12pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.992 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	18.232 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 C
Governing Thk.	132.000 mm	Basic MDMT	-17 C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 C
UCS-66(c) red.	No	Computed Min. Temp.	-17 C
		Joint Efficiency E*	100 %

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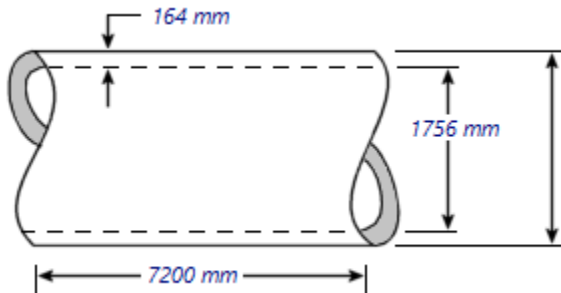
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Element: Shell

Step: 8 4:12pm Apr 10,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	18.001 MPa	Int. Design Temperature	480 C
Material	SA-336 F22	Ext. Design Temperature	0 C
Length	7200.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	110.7 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 MPa
Inside Diameter Di	1756.000 mm	Actual Stress	108.6 MPa
Spec. Min. Thk t,e, or sa	164.0000 mm	Spec. Nominal Thk.	164.000 mm
Surface Area	47138968.0 mm ²	SG of Contents	0.029
Element Volume	13.0 m ³	Weight of Contents	3750.9 N
Empty Weight	541313.7 N	Operating Weight	545064.5 N
		Comp. All. Ope. Stress	88.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 880) / (111 \cdot 1 - 0.6 \cdot 18) \\
 &= 158.5312 + 2.0000 = 160.5312 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (111 \cdot 1 \cdot 162) / (880 + 0.6 \cdot 162) \\
 &= 18.4 - 0.00051 = 18.4 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148 \cdot 1 \cdot 162) / (878 + 0.6 \cdot 162) \\
 &= 24.9 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (880 + 0.6 \cdot 162)) / (1 \cdot 162) \\
 &= 108.581 \text{ MPa}
 \end{aligned}$$



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Element: Shell

Step: 8 4:12pm Apr 10,2026

Minimum Design Metal Temperature Results:*Warning:*

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	18.232 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 C
Governing Thk.	132.000 mm	Basic MDMT	-17 C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 C
UCS-66(c) red.	No	Computed Min. Temp.	-17 C
		Joint Efficiency E*	100 %

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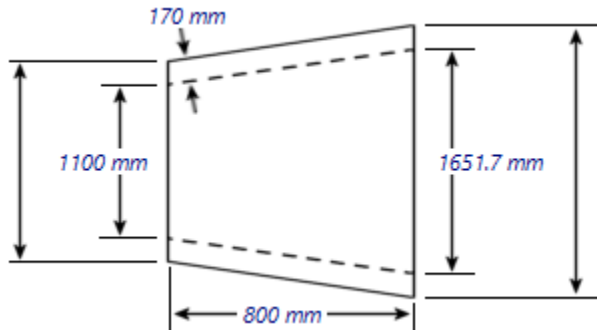
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Element: Cone-1

Step: 9 4:12pm Apr 10,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Cone-1

Conical Head/Section Design Information:			
Design Pressure P	18.001 MPa	Int. Design Temperature	480 C
Material	SA-336 F22	Ext. Design Temperature	0 C
Length	800.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	110.7 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 MPa
Inside Diameter Di	1651.729 mm	Actual Stress	109.3 MPa
Spec. Min. Thk t,e, or sa	170.0000 mm	Spec. Nominal Thk.	170.000 mm
Surface Area	3125696.8 mm ²	SG of Contents	0.029
Element Volume	1.3 m ³	Weight of Contents	368.4 N
Empty Weight	55975.8 N	Operating Weight	56344.2 N
		Comp. All. Ope. Stress	88.2 MPa
		To End Diameter	1100.000 mm
Ins. Knuckle Radius L	525.0000 mm	Small End Flare Radius r	... mm
Knuckle Thickness	175.0000 mm	Flare Thickness	... mm
Line of Support	Both Ends		

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P))$ per Appendix 1-4 (e)
 $= (18 \cdot 1656) / (2 \cdot 0.9 \cdot (111 \cdot 1 - 0.6 \cdot 18))$
 $= 165.6264 + 2.0000 = 167.6264 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 MPa
 $= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$ per App 1-4(e)
 $= (2 \cdot 111 \cdot 1 \cdot 168 \cdot 0.9) / (1656 + 1.2 \cdot 168 \cdot 0.9)$
 $= 18.2 - 0.00051 = 18.2 \text{ MPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$ per App 1-4(e)
 $= (2 \cdot 148 \cdot 1 \cdot 170 \cdot 0.9) / (1652 + 1.2 \cdot 170 \cdot 0.9)$
 $= 24.7 \text{ MPa}$

Actual stress at given pressure and thickness, corroded [Sact]:



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Element: Cone-1 Step: 9 4:12pm Apr 10,2026

$$\begin{aligned}
 &= (P \cdot (D + 1.2 \cdot t \cdot \cos(\alpha))) / (2 \cdot E \cdot t \cdot \cos(\alpha)) \\
 &= (18 \cdot (1656 + 1.2 \cdot 168 \cdot 0.9)) / (2 \cdot 1 \cdot 168 \cdot 0.9) \\
 &= 109.309 \text{ MPa}
 \end{aligned}$$

Thickness and Maximum Pressure for Large Knuckle:

Equivalent Crown Radius for the Cone Large End Knuckle [L]:

$$\begin{aligned}
 &= D_i / (2(\cos(\alpha)) + c_a) \text{ per UG-32(h)} \\
 &= 1652 / (2(\cos(25.8)) + 2) \\
 &= 918.920 \text{ mm}
 \end{aligned}$$

Stress Concentration Factor [M]:

$$\begin{aligned}
 &= (3 + \sqrt{(L / (r + c_a))}) / 4 \\
 &= (3 + \sqrt{(525 / (525 + 2))}) / 4 \\
 &= 1.080
 \end{aligned}$$

Required Knuckle Thickness due to Internal Pressure [TrKnuckle]:

$$\begin{aligned}
 &= (P(L + c_a)M) / (2 \cdot S \cdot E - 0.2 \cdot P) + c_a \text{ per App. 1-4(d)} \\
 &= (18 \cdot (919) \cdot 1.08) / (2 \cdot 111 \cdot 1 - 0.2 \cdot 18) + 2 \\
 &= 84.194 \text{ mm}
 \end{aligned}$$

Large End Knuckle Maximum Allowable Working Pressure [KnuckleMAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of 0.001 MPa} \\
 &= (2 \cdot S \cdot E \cdot (t - c_a)) / (M(L + c_a) + 0.2(t - c_a)) \text{ per Appendix 1-4 (d)} \\
 &= (2 \cdot 111 \cdot 1 \cdot (175 - 2)) / (1.08(919 + 2) + 0.2(175 - 2)) \\
 &= 37.3 - 0.00051 = 37.3 \text{ MPa}
 \end{aligned}$$

Minimum Knuckle Radius check per UF-13(b) [rL]:

$$\begin{aligned}
 &= 0.06(\text{Large End ID} + 2 \cdot c) \\
 &= 0.06(1756 + 2 \cdot 2) \\
 &= 105.6 \text{ mm}
 \end{aligned}$$

Warning:

The knuckle thickness should be the same as the attached cylinder thickness but they seem to be different. Please check the attached cylinder and knuckle thickness input. See UF-13 for more information.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	18.232 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 C
Governing Thk.	164.000 mm	Basic MDMT	-17 C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 C
UCS-66(c) red.	No	Computed Min. Temp.	-17 C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Cone-1

Cone From 60 to 70 Ext. Chart: CS-3 at 0 C

Maximum External Pressure per Code Case 2286:



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Element: Cone-1

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Determine Parameter [Mx]:

$$\begin{aligned} &= L / \sqrt{(R_o \cdot t)} \\ &= 888 / \sqrt{(953 \cdot 168)} \\ &= 2.219 \end{aligned}$$

Compute the Factor [Ch]:

$$\begin{aligned} &\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579) \\ &= 0.92 / (2.22 - 0.579) \\ &= 0.5608 \end{aligned}$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} &= 1.6 \cdot Ch \cdot E_y (t / D_o) \\ &= 1.6 \cdot 0.56 \cdot 212220 (168/1907) \\ &= 16779.791 \text{ MPa} \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} &= Fhe / F_y \\ &= 16779.8/310.0 \\ &= 54.127 \end{aligned}$$

Predicted Buckling Stress [Fic]:

$$\begin{aligned} &\text{Since } Fr \geq 2.439, Fic = F_y \\ &= 310.009 \text{ MPa} \end{aligned}$$

Factor of Safety [FS]:

$$\text{Since } Fic [310] \geq F_y [310], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\begin{aligned} &\text{Since } Fr > 2.439, Fha = F_y / FS \\ &= 310.009/1.667 \\ &= 185.968 \text{ MPa} \end{aligned}$$

Allowable External Pressure at the given Thickness [Pa]:

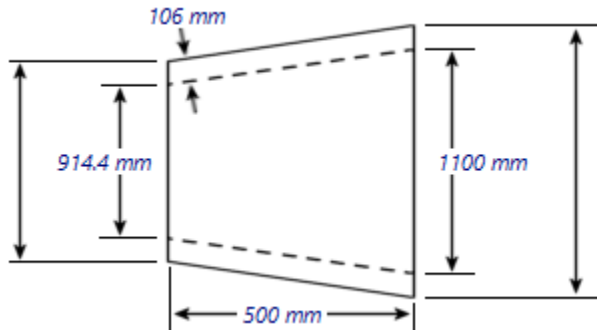
$$\begin{aligned} &= 2 \cdot Fha (t / D_o) \\ &= 2 \cdot 186 (168/1907) \\ &= 32.774 \text{ MPa} \end{aligned}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Cone-2

Conical Head/Section Design Information:			
Design Pressure P	18.000 MPa	Int. Design Temperature	480 C
Material	SA-336 F22	Ext. Design Temperature	0 C
Length	500.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	110.7 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 MPa
Inside Diameter Di	1100.000 mm	Actual Stress	108.0 MPa
Spec. Min. Thk t,e, or sa	106.0000 mm	Spec. Nominal Thk.	106.000 mm
Surface Area	1953605.0 mm2	SG of Contents	0.029
Element Volume	0.4 m3	Weight of Contents	115.1 N
Empty Weight	14350.9 N	Operating Weight	14466.1 N
		Comp. All. Ope. Stress	88.8 MPa
		To End Diameter	914.400 mm
Line of Support	Both Ends		

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P))$ per Appendix 1-4 (e)
 $= (18 \cdot 1104) / (2 \cdot 0.98 \cdot (111 \cdot 1 - 0.6 \cdot 18))$
 $= 101.1457 + 2.0000 = 103.1457 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 MPa

$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$ per App 1-4(e)
 $= (2 \cdot 111 \cdot 1 \cdot 104 \cdot 0.98) / (1104 + 1.2 \cdot 104 \cdot 0.98)$
 $= 18.5 - 0.00032 = 18.5 \text{ MPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$ per App 1-4(e)
 $= (2 \cdot 148 \cdot 1 \cdot 106 \cdot 0.98) / (1100 + 1.2 \cdot 106 \cdot 0.98)$
 $= 25.2 \text{ MPa}$

Actual stress at given pressure and thickness, corroded [Sact]:

$= (P \cdot (D + 1.2 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$
 $= (18 \cdot (1104 + 1.2 \cdot 104 \cdot 0.98)) / (2 \cdot 1 \cdot 104 \cdot 0.98)$



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Element: Cone-2

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= 107.978 MPa

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	18.232 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 C
Governing Thk.	106.000 mm	Basic MDMT	-17 C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 C
UCS-66(c) red.	No	Computed Min. Temp.	-17 C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Cone-2

Cone From 70 to 80 Ext. Chart: CS-3 at 0 C

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 509 / (622 \cdot 104)$$

$$= 2.000$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (2 - 0.579)$$

$$= 0.6476$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.65 \cdot 212220 (104 / 1244)$$

$$= 18386.938 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 18386.9 / 310.0$$

$$= 59.311$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 310.009 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [310] \geq F_y [310], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y / FS$$

$$= 310.009 / 1.667$$

$$= 185.968 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$



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Element: Cone-2

Step: 10 4:12pm Apr 10,2026

= 2 • 186 (104/1244)

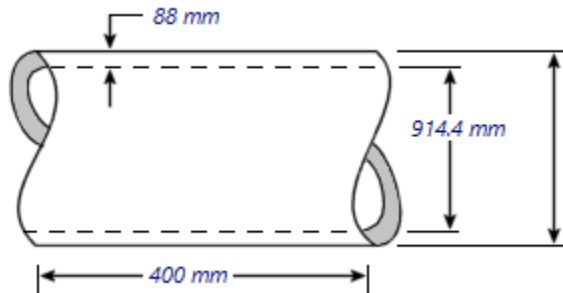
= 31.102 MPa

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Forged pipe

Cylindrical Shell Design Information:			
Design Pressure P	18.000 MPa	Int. Design Temperature	480 C
Material	SA-336 F22	Ext. Design Temperature	0 C
Length	400.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	110.7 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 MPa
Inside Diameter Di	914.400 mm	Actual Stress	106.9 MPa
Spec. Min. Thk t,e, or sa	88.0000 mm	Spec. Nominal Thk.	88.000 mm
Surface Area	1370237.1 mm ²	SG of Contents	0.029
Element Volume	0.3 m ³	Weight of Contents	75.7 N
Empty Weight	8424.7 N	Operating Weight	8500.4 N
		Comp. All. Ope. Stress	88.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 459) / (111 \cdot 1 - 0.6 \cdot 18) \\
 &= 82.7233 + 2.0000 = 84.7233 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (111 \cdot 1 \cdot 86) / (459 + 0.6 \cdot 86) \\
 &= 18.6 - 0.00026 = 18.6 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148 \cdot 1 \cdot 88) / (457 + 0.6 \cdot 88) \\
 &= 25.5 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (459 + 0.6 \cdot 86)) / (1 \cdot 86) \\
 &= 106.913 \text{ MPa}
 \end{aligned}$$



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 FileName : 123_C1_NVS Rev 02 Page 47 of 241
 Element: Forged pipe Step: 11 4:12pm Apr 10,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 88, tr = 27.6, c = 2 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.32, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve B 29 C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	18.232 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 C
Governing Thk.	88.000 mm	Basic MDMT	29 C
Stress/Thk. Ratio	0.321	Temperature Difference	133 C
UCS-66(c) red.	No	Computed Min. Temp.	-104 C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Forged pipe

Cylindrical Shell From 80 to 90 Ext. Chart: CS-3 at 0 C

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 400 / \sqrt{(545 \cdot 86)}$$

$$= 1.847$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (1.85 - 0.579)$$

$$= 0.7254$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.73 \cdot 212220 (86/1090)$$

$$= 19426.367 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 19426.4/310.0$$

$$= 62.664$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = Fy$$

$$= 310.009 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [310] \geq Fy[310], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = Fy/FS$$

$$= 310.009/1.667$$

$$= 185.968 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 186 (86/1090)$$



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Element: Forged pipe

Step: 11 4:12pm Apr 10,2026

= 29.335 MPa

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 Shop Pressure Test Results: Step: 12 4:12pm Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	25.383 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	25.059 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	24.301 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	21.478 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	25.740 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	25.740 MPa
App 27-4	= M.A.W.P.	18.232 MPa

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	20.312 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	19.726 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	21.240 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	17.187 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	20.420 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	20.420 MPa
App 27-4	= M.A.W.P.	14.704 MPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Channel head	175.8	234.0	0.751	21.26
Channel shell	179.4	234.0	0.767	21.26
Shell	176.3	279.0	0.632	24.32
Shell	144.8	279.0	0.519	24.32
Cone-1	145.8	279.0	0.522	24.32
Cone-2	142.9	279.0	0.512	24.31
Forged pipe	140.9	279.0	0.505	24.31

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	148.00	139.28	1.063
N1- Tube inlet	Nozzle	138.00	123.00	1.122
N2- Tube outlet	Nozzle	138.00	123.00	1.122
Shell Bypass No	Nozzle	148.00	111.00	1.333
Shell inlet-3	Nozzle	148.00	111.00	1.333
BD3	Nozzle	138.00	47.63	2.897

Minimum 1.063

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Channel head	138.00	129.12	1.069
Channel shell	138.00	129.12	1.069
Shell	148.00	138.20	1.071
Shell	148.00	138.20	1.071
Shell	148.00	110.72	1.337
Cone-1	148.00	110.72	1.337



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Shop Pressure Test Results:

Step: 12 4:12pm Apr 10, 2026

Cone-2	148.00	110.72	1.337
Forged pipe	148.00	110.72	1.337

Minimum 1.069

Stress ratios for Exchanger Materials (MPa):

Description	Ambient	Operating	Ratio
Tube Material	138.00	119.00	1.160
Tubesheet Material	148.00	138.20	1.071

Minimum 1.071

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
Manhole	140.80	192.40	0.732
N1- Tube inlet	156.74	179.00	0.876
N2- Tube outlet	147.74	179.00	0.825
Shell Bypass No	134.30	192.00	0.699
Shell inlet-3	113.81	192.00	0.593
BD3	37.53	179.40	0.209

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 13 4:12pm Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	25.383 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	25.059 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	24.301 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	21.478 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	25.740 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	25.740 MPa
App 27-4	= M.A.W.P.	18.232 MPa

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	20.312 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	19.726 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	21.240 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	17.187 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	20.420 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	20.420 MPa
App 27-4	= M.A.W.P.	14.704 MPa

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 18 • 1.07

= 25.059 MPa

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 14.3 • 1.06

= 19.726 MPa

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Channel head	173.4	234.0	0.741	19.74
Channel shell	171.4	234.0	0.733	19.74
Shell	186.3	279.0	0.668	25.08
Shell	151.3	279.0	0.542	25.08
Cone-1	152.3	279.0	0.546	25.08
Cone-2	150.4	279.0	0.539	25.07
Forged pipe	148.9	279.0	0.534	25.07

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	148.00	139.28	1.063
N1- Tube inlet	Nozzle	138.00	123.00	1.122
N2- Tube outlet	Nozzle	138.00	123.00	1.122
Shell Bypass No	Nozzle	148.00	111.00	1.333
Shell inlet-3	Nozzle	148.00	111.00	1.333
BD3	Nozzle	138.00	47.63	2.897

Minimum

1.063



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Field Pressure Test Results: Step: 13 4:12pm Apr 10,2026

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Channel head	138.00	129.12	1.069
Channel shell	138.00	129.12	1.069
Shell	148.00	138.20	1.071
Shell	148.00	138.20	1.071
Shell	148.00	110.72	1.337
Cone-1	148.00	110.72	1.337
Cone-2	148.00	110.72	1.337
Forged pipe	148.00	110.72	1.337

Minimum 1.069

Stress ratios for Exchanger Materials (MPa):

Description	Ambient	Operating	Ratio
Tube Material	138.00	119.00	1.160
Tubesheet Material	148.00	138.20	1.071

Minimum 1.071

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
Manhole	130.77	192.40	0.680
N1- Tube inlet	145.57	179.00	0.813
N2- Tube outlet	137.22	179.00	0.767
Shell Bypass No	138.48	192.00	0.721
Shell inlet-3	117.36	192.00	0.611
BD3	38.70	179.40	0.216

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 4:12p Apr 10,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		25.06 MPa
Shell Thickness	ts	132.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1756.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	148.00 MPa
Shell Allowable Stress at Ambient		148.00 MPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Test Pressure		19.73 MPa
Channel Thickness	tc	112.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1756.000 mm
Channel Design Temperature	TEMPC	21.11 C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 MPa
Channel Allowable Stress at Ambient		138.00 MPa

Tube Data:

Number of Tube Holes	Nt	2156
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	8500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8050.00 mm
Design Temperature of the Tubes		21.11 C
Tube Material		SA-213 TP321
Tube Material UNS Number		S32100
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		138.00 MPa
Tube Allowable Stress At Ambient		138.00 MPa
Tube Yield Stress At design Temperature	Syt	205.00 MPa
Tube Pitch (Center to Center Spacing)	P	28.5000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	858.460 mm
Largest Center-to-Center Tube Distance	U1	76.2000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	1695580.0 mm2
Perimeter of Tube Layout	Cp	5131.8 mm



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ASME TS Calc: Pressure Test

Case: 1 4:12p Apr 10,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	21.11 C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	148.00 MPa
Tubesheet Allowable Stress at Ambient	Tt	148.00 MPa
Thickness of Tubesheet	h	450.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2610.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	1847.00 mm
Step 1 Depth on the Shellside	hs1	9.00 mm
Step 2 Diameter on the Shellside	ds2	1751.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2610.00 mm

Note: Tubesheet diameter is now: 2610.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	192024.0 mm2
---------------------------	----	--------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2610.000 mm
Flange Inside Diameter	B	1760.000 mm
Flange Face Outside Diameter	Fod	1850.000 mm
Flange Face Inside Diameter	Fid	1760.000 mm
Gasket Outside Diameter	Go	1826.000 mm
Gasket Inside Diameter	Gi	1780.000 mm
Small end Hub thk.	g0	132.0000 mm
Large end Hub thk.	g1	206.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	69.00 MPa
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2410.000 mm
Nominal Bolt Diameter	dB	95.2500 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-540 B21
Bolt Allowable Stress At Temperature	Sb	228.00 MPa
Bolt Allowable Stress At Ambient	Sa	228.00 MPa
Weld between Flange and Shell/Channel		0.0000 mm

Tubesheet Integral with		Channel
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	413.0000 mm



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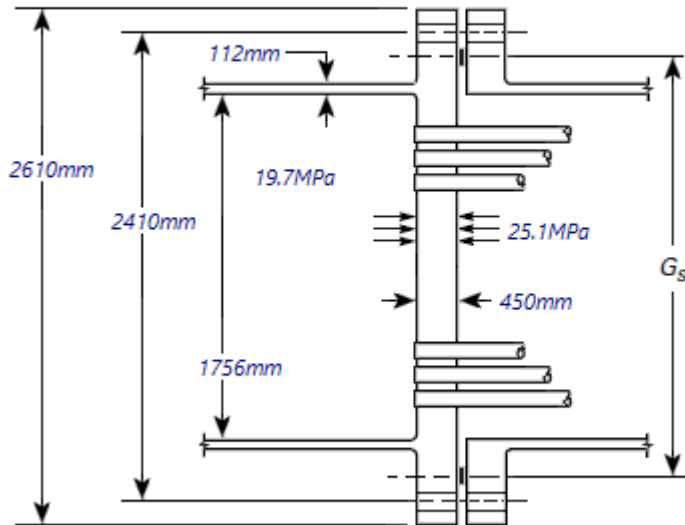
ASME TS Calc: Pressure Test

Case: 1 4:12p Apr 10, 2026

Is Bolt Load Transferred to the Tubesheet

Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group D

Elastic Mod. at Design Temperature	21.1 C	0.21099E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 C	0.20271E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.20271E+06 MPa

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesht. Design Temp.	21.1 C	0.19513E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.19513E+06 MPa

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	21.1 C	0.21099E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(19.7 \cdot (19/2 - 0)) / (138 \cdot 1 + 0.4 \cdot 19.7)$
= $1.2879 + 0.0000 = 1.2879$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:



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ASME TS Calc: Pressure Test

Case: 1 4:12p Apr 10, 2026

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 8908 / \sqrt{(9.52 \cdot 3.05)}$$

$$= 1653.342$$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)0.94$, $Ch = 0.55 / (Do/t)$

$$= 0.55 / (19/3.05)$$

$$= 0.0880$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey (t / Do)$$

$$= 1.6 \cdot 0.088 \cdot 195128 (3.05/19)$$

$$= 4395.855 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 4395.9/205.0$$

$$= 21.444$$

Predicted Buckling Stress [Fic]:

Since $Fr \geq 2.439$, $Fic = Fy$

$$= 204.996 \text{ MPa}$$

For non-Carbon and Low Alloy Steel Materials re-compute [Fic]:

Factor A [A]:

$$= F/Emod, \text{ see 2286-6 Step 1}$$

$$= 4396/195128$$

$$= 0.02253$$

From the External Pressure Chart, HA-2, determine the Stress [B]:

$$= 104.872 \text{ MPa}$$

The Tangent Modulus from 3-D.5.2, [Et]:

$$= 2B/A$$

$$= 2 \cdot 105/0.023$$

$$= 9310.4 \text{ MPa}$$

Predicted Buckling Stress [Fic]:

$$= (F \cdot Et) / (FS \cdot Emod)$$

$$= (4396 \cdot 9310) / (1 \cdot 195128)$$

$$= 209.744 \text{ MPa}$$

Factor of Safety [FS]:

Since $Fic [210] \geq Fy[205]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 2.439$, $Fha = Fy/FS$

$$= 204.996/1.667$$

$$= 122.973 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 123 (3.05/19)$$

$$= 39.351 \text{ MPa}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.9708 mm
Allowable Internal Pressure at Corroded thickness	50.64 MPa
Required Internal Design Pressure	19.73 MPa
Allowable External Pressure at Corroded thickness	39.35 MPa
Required External Design Pressure	25.06 MPa
Required Thickness due to Shell Side pressure	1.9708 mm



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ASME TS Calc: Pressure Test

Case: 1 4:12p Apr 10,2026

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	226.404 277.231		140 589	44 236	Ps+Pt D1	Ok
D2uc	316.416 ...		341 589	55 236	Ps+Pt D2	Ok
D3uc	165.373 ...		217 589	12 236	Ps+Pt D3	Ok
Max:	316.4162 277.231 mm		0.579	0.236 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc		...	...	...	370	Ok
D2uc		...	...	...	370	Ok
D3uc		...	...	...	370	Ok
Max		...	...	...	...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	316.416	450.000	Ok
Tubesheet thickness flanged extension :	277.231	413.000	Ok
Tube thickness :	1.971	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.000	Ok
Tube-tubesheet groove weld leg :	1.829	3.000	Ok

Given Shell Thickness: 132.0000 mm

Given Channel Thickness: 112.0000 mm

Min Channel length of thk, (112.000) adj. to tubesheet: 798.259 mm

Note: This is a full strength Tube to Tubesheet Joint.



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	1873	51	No Calc	No Calc
20	30	2406.17	1980	109	No Calc	No Calc
30	40	No Calc	...	352	No Calc	No Calc
40	50	8800	2020	129	No Calc	No Calc
50	60	8800	2084	162	No Calc	No Calc
60	70	800	1956.97	168	No Calc	No Calc
70	80	500	1315.62	104	No Calc	No Calc
80	90	400	1090.4	86	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	54	5.50079	0.103	12.9
20	30	112	9.53015	0.103	14.9
30	40	355	237.109	...	No Calc
40	50	132	No Calc	...	23.8
50	60	164	No Calc	...	28.9
60	70	170	6.05835	0.10343	32.8
70	80	106	4.50924	0.10343	31.1
80	90	88	4.10292	0.10343	29.3

Minimum

12.9

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2406.17	25399974	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	8800	No Calc	No Calc	No Calc
50	60	8800	No Calc	No Calc	No Calc
60	70	800	800	No Calc	No Calc
70	80	500	500	No Calc	No Calc
80	90	400	25399974	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 15

4:12pm Apr 10,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume m3	Corroded Metal Wgt. kg	Corroded ID Volume m3	Extra due Misc % kg
10	20	2926.78	1.71381	2798.15	1.73041	292.678
20	30	10188.4	4.84447	9931.39	4.87764	1018.84
30	40	10660.6	1.6412	10574	1.64634	1066.06
40	50	9709.03	2.89219	9503.44	2.91872	970.903
50	60	55202.3	13.0183	54585.9	13.0979	5520.23
60	70	5708.33	1.27872	5671.38	1.28348	570.833
70	80	1463.49	0.39957	1438.49	0.4028	146.349
80	90	859.139	0.26272	841.288	0.26503	85.9139
Total		96718	26.05	95344	26.22	9671



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Nozl	4431.82	-882.5	...	...	Manhole
10	Wght	200	-882.5	300	...	DAVIT WEIGHT
20	Liqd	4841.52	1000	...	...	Liquid: 20
20	Nozl	1603.48	900	1081.2	...	N1- Tube inlet
20	Nozl	983.84	900	-1030.4	...	N2- Tube outlet
20	Wght	450	800	...	...	Pass Partition
30	Liqd	48.217	332	-3.00003	...	Liquid: 30
30	Lini	...	...	...	...	cladding
40	Liqd	84.9699	800	...	...	Liquid: 40
40	Lini	211.728	800	...	...	Cladding
40	Nozl	1449.24	841.389	1081.2	...	Shell Bypass No
40	Nozl	2910.85	880	-1259	...	Shell inlet-3
50	Sadl	1925.06	1100	1408.5	...	Left Saddle
50	Sadl	1925.06	6500	1408.5	...	Sdl 2 Fr40
50	Liqd	382.506	3600	...	...	Liquid: 40
50	Lini	1586.11	3600	...	...	Caldding
50	Nozl	33.376	2330.98	-903.4	...	BD3
50	Wght	300	1100	2133	...	Fire prrofig w
50	Wght	300	6500.04	-2133	...	Fire prrofig w
50	Wght	200	2800	...	...	Bundle skid bar
60	Liqd	37.5675	392.237	-164	...	Liquid: 60
60	Lini	143.222	400	...	...	Cladding
70	Liqd	11.7391	250	-46.4	...	Liquid: 70
70	Lini	63.0438	250	...	...	cladding
70	Wght	200	225.039	...	...	weigt for incon
80	Liqd	7.71939	200	...	...	Liquid: 80
80	Lini	45.7647	200	...	...	cladding
20	FTsh	18168.5	2225	...	...	Tubesheet
20	Tube	28162.1	6250	...	...	Tubes
30	Bafl	249.042	666	...	...	Baffles
30	Bafl	249.042	1332	...	...	Baffles
30	Bafl	249.042	1998	...	...	Baffles
30	Bafl	249.042	2664	...	...	Baffles
30	Bafl	249.042	3330	...	...	Baffles
30	Bafl	249.042	3996	...	...	Baffles
30	Bafl	249.042	4662	...	...	Baffles



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Element and Detail Weights: Step: 15 4:12pm Apr 10,2026

30 Baf1	249.042	5328	...	...	Baffles
30 Baf1	249.042	5994	...	...	Baffles
30 Baf1	249.042	6660	...	...	Baffles
30 Baf1	249.042	7326	...	...	Baffles
30 Baf1	249.296	7992	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	3850.1
Liquid	5414.2
Lining	211.7
Cladding & Overlay	1838.1
Nozzles	11412.6
Weights	1650.0
Exchanger Components	49319.4
Liquid in Tubes	2814.2

Sum of the Detail Weights 76510.4 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	106389.8	106389.8	106389.8	106389.8	106389.8	106389.8
Saddles	3850.1	3850.1	3850.1	3850.1	3850.1	3850.1
Cladding	1838.1	1838.1	1838.1	1838.1	1838.1	1838.1
Nozzles	11412.6	11412.6	11412.6	11412.6	11412.6	11412.6
Wld Weights	200.0	200.0	200.0	200.0	200.0	200.0
Exchanger	49319.4	49319.4	49319.4	49319.4	49319.4	49319.4
Lining	211.7	211.7	211.7	211.7	211.7	211.7
Empty Weights	...	...	...	1450.0	1450.0	...
Test Weights	...	1450.0	...	...	...	...
Ope Weights	...	...	...	...	...	1450.0
Ope. Liquid	...	...	...	...	...	5414.2
Tube Ope Lqd	...	...	...	...	...	2814.2
Test Liquid	...	25779.3	...	...	...	...
Tube Tst Lqd	...	2814.2	...	...	...	...
Totals	173221.8	203265.3	173221.8	174671.8	174671.8	182900.2

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	173221.8 kg
Shop Test Wt.	- Fabricated weight + water (full)	203265.3 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	173221.8 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	174671.8 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	174671.8 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	182900.2 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	181388.9 kg
Field Test Wt.	- Empty weight + test medium (full)	203265.3 kg

Exchanger Tube Data

Volume of Exchanger tubes :	2.8 m3
Weight of Ope Liq in tubes :	2814.2 kg
Weight of Water in tubes :	2814.2 kg



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Element and Detail Weights:

Step: 15 4:12pm Apr 10,2026

Surface Areas of Elements:

From	To	Outside Surface Area mm2	Inside Surface Area mm2
10	20	6294936	5514414
20	30	12440708	11033274
30	40	8002275	3715624
40	50	10153627	8826619
50	60	47138968	39719784
60	70	3125696	2452751
70	80	1953605	1609124
80	90	1370237	1149068

Total	90480056.0	74020664.0 mm2
Total	90.5	74.0 m2

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Nozzle Flange MAWP:

Step: 16 4:12pm Apr 10,2026

Nozzle Flange MAWP Results: (MPa & C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Manhole	18.97	25.51	343	1500	GR 1.1	...	...	...	...
N1- Tube inlet	18.97	25.51	343	1500	GR 1.1	...	...	...	...

Shellside Flange Rating**Channelside Flange Rating**

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 18.967 MPa

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 25.511 MPa

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange.

See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation:

Step: 17 4:12pm Apr 10,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_z \bar{Q} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_z \bar{Q}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.98)/(1 + 1.7 \cdot 3.4 \cdot 0.23))) \\
 &= \min(0.85, 0.92) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.601
Structure Height to Diameter ratio	7.240

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 2133 + 300 = 2433 \text{ mm} \\
 &= 7.98 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (7.98 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Z_g)^{2/\alpha} \\
 &= 2.01(15/900)^{2/9.5} \\
 &= 0.85
 \end{aligned}$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 17 4:12pm Apr 10,2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [766] N/m2

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.6 • 17.5

= 143 lbs. [638] N

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz N/m2
Channel head	2433.0	0.000	0.000	0.000	0.849	1.000	766.064
Channel shell	2433.0	0.000	0.000	0.000	0.849	1.000	766.064
Shell	2433.0	0.000	0.000	0.000	0.849	1.000	766.064
Shell	2433.0	0.000	0.000	0.000	0.849	1.000	766.064
Shell	2433.0	0.000	0.000	0.000	0.849	1.000	766.064
Cone-1	2433.0	0.000	0.000	0.000	0.849	1.000	766.064
Cone-2	2433.0	0.000	0.000	0.000	0.849	1.000	766.064
Forged pipe	2433.0	0.000	0.000	0.000	0.849	1.000	766.064

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm2	Elevation mm	Pressure N/m2	Force N
DAVIT WEIGHT	0.00	2733.00	766.06	0.00
Pass Partitio	0.00	2433.00	766.06	0.00
Fire prrofig	0.00	4566.00	766.06	0.00
Fire prrofig	0.00	300.00	766.06	0.00
Bundle skid b	0.00	2433.00	766.06	0.00
weigt for inc	0.00	2433.00	766.06	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm2	Wind Pressure N/m2	Element Wind Load N
10	20	2433	2247.6	1629370	766.064	701.779
20	30	2433	2376	4752000	766.064	2046.71
30	40	2433	2112	1402368	766.064	604.008
40	50	2433	2424	3878400	766.064	1670.45
50	60	2433	2500.8	18005760	766.064	7755.19
60	70	2433	2121.6	1697280	766.064	731.029
70	80	2433	1463.04	731520	766.064	315.07
80	90	2433	1308.48	523392	766.064	225.428

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



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Wind Load Calculation:

Step: 17 4:12pm Apr 10, 2026

Be sure the wind speed is in accordance with the specified wind design code.

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 FileName : 123_C1_NVS Rev 02
 Earthquake Load Calculation: Step: 18 4:12pm Apr 10,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(9.890.75)$
 = 0.112 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.11$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.16 \cdot 1755764 \cdot 1.25$
 = 103238.984 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 1755764$
 = 55060.8 N

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	882.5	175576	11335.1



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Earthquake Load Calculation: Step: 18 4:12pm Apr 10,2026

20	30	878	175576	11277.3
30	40	880	175576	11303
40	50	878	175576	11277.3
50	Sad1	878	175576	11277.3
Sad1	60	878	175576	11277.3
50	60	878	175576	11277.3
60	70	878	175576	11277.3
70	80	550	175576	7064.39
80	90	457.2	175576	5872.43

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 19 4:12pm Apr 10,2026

Shop/Field Installation Options :

Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	8629.177 mm
Center of Gravity of the Liquid	1812.748 mm
Center of Gravity of the Lining	8511.056 mm
Center of Gravity of the Nozzles	1472.642 mm
Center of Gravity of the Added Weights (Operating)	5786.724 mm
Center of Gravity of the Added Weights (Empty)	5786.724 mm
Center of Gravity of the Tubesheet(s)	2337.000 mm
Center of Gravity of the Tubes	6362.000 mm
Center of Gravity of the Baffles	6894.486 mm
Center of Gravity of Bare Shell New and Cold	6689.606 mm
Center of Gravity of Bare Shell Corroded	6706.958 mm
Vessel CG in the Operating Condition	5790.829 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	5902.291 mm
Vessel CG in the Test Condition	5917.148 mm

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 Saddle Calcs: Bundle Pull New: Step: 20 4:12pm Apr 10,2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(288 + 1.56 \cdot \sqrt{960 \cdot 164}, 2 \cdot 2300)$
 $= 906.9879 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	960.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2300.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		110.72	MPa
Head Allowable Stress used in Calculation		0.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		480493.84	N
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		10349.47	N
Pressure used in Analysis	P	0.000	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-0.01	-278.99
Long. Stress at Top of Midspan	0.01	110.72
Long. Stress at Bottom of Midspan	0.01	110.72
Long. Stress at Top of Saddles	0.09	110.72
Long. Stress at Bottom of Saddles	-0.05	-278.99
Long. Stress at Bottom of Saddles	0.05	110.72
Tangential Shear in Shell	0.05	88.58
Circ. Stress at Horn of Saddle	0.05	166.08
Circ. Compressive Stress in Shell	0.01	110.72

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 $= \text{Bundle Pull Out Force} \cdot \text{Impact Factor} \cdot B / L_s$
 $= 480494 \cdot 1.5 \cdot 2133/5400$
 $= 284692.7 \text{ N}$



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Saddle Calcs: Bundle Pull New: Step: 20 4:12pm Apr 10, 2026

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= -274343 + max(284693, 0, 0, 0)

= 10349.5 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 29226.62 N

Transverse Shear Load 0.00 N

Longitudinal Shear Load 720740.75 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2) , 2•Rm)

= min(2(1756/2 + 164 + 30)sin(120/2) , 2•960)

= 1856.759 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -10349•2300[1-(1-2300/11164+(960²-0²)/

(2•2300•11164))/(1+(4•0)/(3•11164))]

= -4478675.5 N-mm

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 10349•11164/4((1+2(960²-0²)/(11164²))/(1+(4•0)/

(3•11164))-4•2300/11164)

= 5511001.5 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 960/(2•164) - 5511001/(π•960²•164)

= -0.01 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 960/(2 • 164) + 5511001/(π • 960² • 164)

= 0.01 MPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•960/(2•164)-4478675/(0.11•π•960²•164)

= 0.09 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•960/(2•164)+4478675/(0.19•π•960²•164)

= -0.05 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 10349 (11164 - 2 • 2300)/(11164 + (4 • 0/3))

= 6085.1 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ2]:

= K2 • T / (Rm • t)

= 1.17 • 6085/(960 • 164)



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= 0.05 MPa

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(960 \cdot 164)}$$

$$= 309.494 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 10349 \cdot 0.1 / (164(288 + 309 + 309))$$

$$= -0.01 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -10349 / (4 \cdot 164(288 + 309 + 309)) - 3 \cdot 0.053 \cdot 10349 / (2 \cdot 164^2)$$

$$= -0.05 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(288 + 1.56 \cdot \sqrt{(960 \cdot 164)}, 2 \cdot 2300)$$

$$= 906.99 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm
Web Thickness	Webtk	32.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	MPa
Height of Web at Center	Hw,c	1085.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(878 \cdot 164)} + 300)164^3) / 12$$

$$= 327865888.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 327865888 + 146282(306 - 82)^2$$

$$= 7656370176.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	892.0	164.0	82.0	146281.8	11995104.0	0.766E+10
Wearplate	300.0	30.0	179.0	9000.0	1611000.1	0.145E+09
Web	32.0	1006.0	697.0	32192.0	22437824.0	0.764E+10
BasePlate	420.0	55.0	1227.5	23100.0	28355246.0	0.196E+11
Totals	...	...	...	210573.7	64399172.0	0.351E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 64399172 / 210574$$

$$= 305.827 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$



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$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 10349$$

$$= 2106.3418 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.0328 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2133 - 878 \cdot \sin(120/2) / 1.05$$

$$= 1406.900 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2964613.7500 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.0258 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(10349 + 18877)420 / (4 \cdot 2616 \cdot 207)}$$

$$= 4.127 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30)$$

$$= 1856.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1857 / (5 - 1)$$

$$= 464.190 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 464 \cdot 420 / 2$$

$$= 97479.805 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 29227 / (2616 \cdot 420)$$

$$= 0.027 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 97480 \cdot 0.027$$

$$= 2593.022 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8192 + 7427$$

$$= 15619.032 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 2593 / 15619$$



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= 0.166 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:

= Saddle Width - Web Thickness

= 288 - 32

= 256.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/15619

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 288/2

= 144.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{63700928/15619}$

= 63.863 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 2133 - cos(120/2)(878 + 164 + 30) - 55

= 1542.000 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199955/310)}$

= 112.838

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1542/63.9

= 24.146

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 720741 / (2 • 2616) • 464 • 1542/2

= 49321604.000 N-mm

Compressive Allowable, KL/r < Cc (24.1 < 113) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (24.1)^2 / (2 \cdot 113^2)) 310 /$

$(5/3 + 3 \cdot (24.1) / (8 \cdot 113) - (24.1^3) / (8 \cdot 113^3))$

= 173.5 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 0.17/174 + (49321604 • 144/63700992) / 207

= 0.540

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



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Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288/2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2133 - \cos(1072/2)(55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(64881092/23046)}$$

$$= 53.059 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1006/53.1$$

$$= 18.960$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 720741/(2 \cdot 2616)$$

$$= 137.756 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 138 \cdot 464 \cdot 1006$$

$$= 64354780.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 64354780/450564$$

$$= 142.786 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($19 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 113^2)) 310 /$$

$$(5/3 + 3 \cdot (19) / (8 \cdot 113) - (19^3) / (8 \cdot 113^3))$$

$$= 176.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 0.2/177 + (64354780 \cdot 144/64881092) / 207$$

$$= 0.692$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1613.9999 mm2



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BasePlate Allowable Stress	S	148.00 MPa
Area Available in a Single Bolt	BltArea	1407.6318 mm2
Saddle Load Q0 (Weight)	Q0	29226.6 N
Saddle Load QL (Wind/Seismic contribution)	QL	720740.8 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	720740.8 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 720741 / (1408 \cdot 8)$$

$$= 64.0 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (1408 \cdot 8)$$

$$= 0.0 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - Q0) / (Stba \cdot Rnbolts)$$

$$= (720741 - 29227) / (172 \cdot 8)$$

$$= 502.6057 \text{ mm2}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 720741 / (103 \cdot 8)$$

$$= 873.0802 \text{ mm2}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2133 \cdot 0 + 0$$

$$= 0.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0 / 29227$$

$$= 0.00 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[503, 873, 0]$$

$$= 873.0802 \text{ mm2}$$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(288 + 1.56 \cdot \sqrt{961 \cdot 162}, 2 \cdot 2300)$
 $= 903.5223 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	961.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2300.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		110.72	MPa
Head Allowable Stress used in Calculation		0.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		480493.84	N
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		17759.46	N
Pressure used in Analysis	P	0.000	MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top of Midspan		-0.02	-278.99
Long. Stress at Top of Midspan		0.02	110.72
Long. Stress at Bottom of Midspan		0.02	110.72
Long. Stress at Top of Saddles		0.15	110.72
Long. Stress at Bottom of Saddles		-0.08	-278.99
Long. Stress at Bottom of Saddles		0.08	110.72
Tangential Shear in Shell		0.08	88.58
Circ. Stress at Horn of Saddle		0.08	166.08
Circ. Compressive Stress in Shell		0.01	110.72

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 $= \text{Bundle Pull Out Force} \cdot \text{Impact Factor} \cdot B / L_s$
 $= 480494 \cdot 1.5 \cdot 2133/5400$
 $= 284692.7 \text{ N}$



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Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= -266933 + max(284693, 0, 0, 0)

= 17759.5 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 36636.60 N

Transverse Shear Load 0.00 N

Longitudinal Shear Load 720740.75 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2) , 2•Rm)

= min(2(1756/2 + 164 + 30)sin(120/2) , 2•961)

= 1856.759 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -17759•2300[1-(1-2300/11164+(961²-0²)/

(2•2300•11164))/(1+(4•0)/(3•11164))]

= -7683777.5 N-mm

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 17759•11164/4((1+2(961²-0²)/(11164²))/(1+(4•0)/

(3•11164))-4•2300/11164)

= 9458284.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 961/(2•162) - 9458284/(π•961²•162)

= -0.02 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 961/(2 • 162) + 9458284/(π • 961² • 162)

= 0.02 MPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•961/(2•162)-7683777/(0.11•π•961²•162)

= 0.15 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•961/(2•162)+7683777/(0.19•π•961²•162)

= -0.08 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 17759 (11164 - 2 • 2300)/(11164 + (4 • 0/3))

= 10441.9 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ2]:

= K2 • T / (Rm • t)

= 1.17 • 10442/(961 • 162)



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= 0.08 MPa

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(961 \cdot 162)}$$

$$= 307.761 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 17759 \cdot 0.1 / (162(288 + 308 + 308))$$

$$= -0.01 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -17759 / (4 \cdot 162(288 + 308 + 308)) - 3 \cdot 0.053 \cdot 17759 / (2 \cdot 162^2)$$

$$= -0.08 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(288 + 1.56 \cdot \sqrt{(961 \cdot 162)}, 2 \cdot 2300)$$

$$= 903.52 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm
Web Thickness	Webtk	32.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	MPa
Height of Web at Center	Hw,c	1085.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(880 \cdot 162)} + 300)162^3) / 12$$

$$= 314971008.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 314971008 + 144020(307 - 81)^2$$

$$= 7667600384.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³ (3)	I + AD ² mm ⁴ (4)
Shell	889.0	162.0	81.0	144019.8	11665603.0	0.767E+10
Wearplate	300.0	30.0	177.0	9000.0	1593000.1	0.153E+09
Web	32.0	1006.0	695.0	32192.0	22373444.0	0.756E+10
BasePlate	420.0	55.0	1225.5	23100.0	28309048.0	0.195E+11
Totals	...	...	...	208311.8	63941092.0	0.349E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 63941092 / 208312$$

$$= 306.949 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$



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$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 17759$$

$$= 3614.4346 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.0562 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2133 - 880 \cdot \sin(120/2) / 1.05$$

$$= 1405.246 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5081229.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.0447 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(17759 + 18877)420) / (4 \cdot 2616 \cdot 207)}$$

$$= 4.620 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30)$$

$$= 1856.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1857 / (5 - 1)$$

$$= 464.190 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 464 \cdot 420 / 2$$

$$= 97479.805 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 36637 / (2616 \cdot 420)$$

$$= 0.033 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 97480 \cdot 0.033$$

$$= 3250.445 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8192 + 7427$$

$$= 15619.032 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 3250 / 15619$$



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= 0.208 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 288 - 32
 = 256.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/15619
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 288/2
 = 144.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{63700928/15619}$
 = 63.863 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 2133 - cos(120/2)(878 + 164 + 30) - 55
 = 1542.000 mm

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 199955/310)}$
 = 112.838

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1542/63.9
 = 24.146

Bending Moment [Rm]:
 $= FI / (2 \cdot Bpln) \cdot e \cdot L / 2$
 $= 720741 / (2 \cdot 2616) \cdot 464 \cdot 1542 / 2$
 = 49321604.000 N-mm

Compressive Allowable, $KL/r < Cc$ (24.1 < 113) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (24.1)^2 / (2 \cdot 113^2)) 310 /$
 $(5/3 + 3 \cdot (24.1) / (8 \cdot 113) - (24.1^3) / (8 \cdot 113^3))$
 = 173.5 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 0.21/174 + (49321604 \cdot 144/63700992) / 207$
 = 0.541

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



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Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288/2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2133 - \cos(1072/2)(55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(64881092/23046)}$$

$$= 53.059 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1006/53.1$$

$$= 18.960$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 720741/(2 \cdot 2616)$$

$$= 137.756 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 138 \cdot 464 \cdot 1006$$

$$= 64354780.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 64354780/450564$$

$$= 142.786 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($19 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 113^2)) 310 /$$

$$(5/3 + 3 \cdot (19) / (8 \cdot 113) - (19^3) / (8 \cdot 113^3))$$

$$= 176.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 0.25/177 + (64354780 \cdot 144/64881092) / 207$$

$$= 0.692$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1613.9999 mm2



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BasePlate Allowable Stress	S	148.00 MPa
Area Available in a Single Bolt	BltArea	1407.6318 mm2
Saddle Load Q0 (Weight)	Q0	36636.6 N
Saddle Load QL (Wind/Seismic contribution)	QL	720740.8 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	F1	720740.8 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 720741 / (1408 \cdot 8)$$

$$= 64.0 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (1408 \cdot 8)$$

$$= 0.0 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - Q0) / (Stba \cdot Rnbolts)$$

$$= (720741 - 36637) / (172 \cdot 8)$$

$$= 497.2200 \text{ mm2}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 720741 / (103 \cdot 8)$$

$$= 873.0802 \text{ mm2}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2133 \cdot 0 + 0$$

$$= 0.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0 / 36637$$

$$= 0.00 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[497, 873, 0]$$

$$= 873.0802 \text{ mm2}$$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

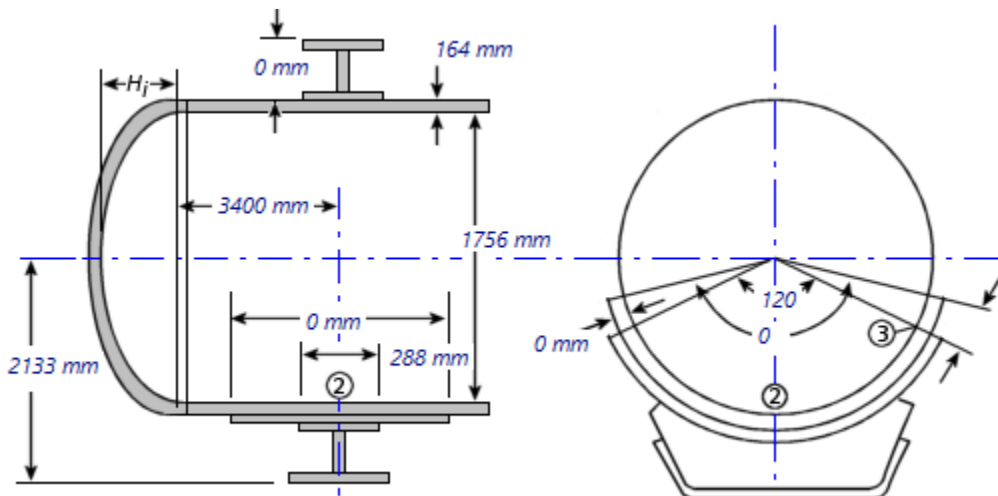
$$= \min(b + 1.56 \sqrt{R_m \cdot t}, 2a)$$

$$= \min(288 + 1.56 \sqrt{960 \cdot 164}, 2 \cdot 3400)$$

$$= 906.9879 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	960.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3400.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		110.72	MPa
Head Allowable Stress used in Calculation		138.20	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		1890503.88	N
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	MPa
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	km/hr



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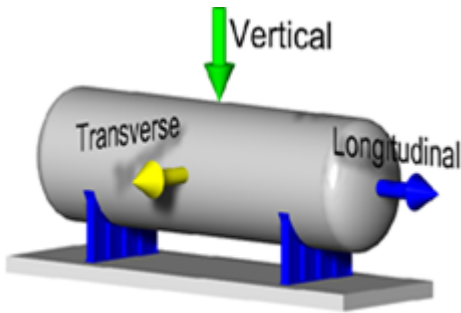
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Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	2.26	110.72
Long. Stress at Bottom of Midspan	-2.26	-278.99
Long. Stress at Bottom of Midspan	2.26	110.72
Long. Stress at Top of Saddles	37.13	110.72
Long. Stress at Bottom of Saddles	-20.58	-278.99
Long. Stress at Bottom of Saddles	20.58	110.72
Tangential Shear in Shell	5.50	88.58
Circ. Stress at Horn of Saddle	8.75	166.08
Circ. Compressive Stress in Shell	0.97	110.72

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 351 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 2982.5 \text{ lbf} \\
 &= 13266.164 \text{ N}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 6.84^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 374.3 \text{ lbf} \\
 &= 1664.7 \text{ N}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 1658781 \cdot 1 + 13266 \\
 &= 179144.3 \text{ N}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 1658781 \cdot 1 + 1665 \\
 &= 167542.8 \text{ N}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 179144/2 \cdot 2133/1857 \\
 &= 102898.4 \text{ N}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 167543 \cdot 2133/5400 \\
 &= 66179.4 \text{ N}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$= \text{Saddle Load} \cdot g_y$$



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$$= 1702481 \cdot 0.05$$

$$= 85124.078 \text{ N}$$

Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + \text{Frv} + \max(\text{Fwl}, \text{Fwt})$$

$$= 1702481 + 85124 + \max(66179, 102898)$$

$$= 1890503.9 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1909381.00	N
Transverse Shear Load	89572.17	N
Longitudinal Shear Load	167542.84	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1756/2 + 164 + 30)\sin(120/2), 2 \cdot 960)$$

$$= 1856.759 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -1890503 \cdot 3400 [1 - (1 - 3400/11164 + (960^2 - 0^2)/(2 \cdot 3400 \cdot 11164))/(1 + (4 \cdot 0)/(3 \cdot 11164))]$$

$$= -1880293376.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 1890503 \cdot 11164/4((1 + 2(960^2 - 0^2)/(11164^2))/(1 + (4 \cdot 0)/(3 \cdot 11164)) - 4 \cdot 3400/11164)$$

$$= -1073721472.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 960/(2 \cdot 164) - -1073721472/(\pi \cdot 960^2 \cdot 164)$$

$$= 2.26 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 960/(2 \cdot 164) + -1073721472/(\pi \cdot 960^2 \cdot 164)$$

$$= -2.26 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 960/(2 \cdot 164) - -1880293376/(0.11 \cdot \pi \cdot 960^2 \cdot 164)$$

$$= 37.13 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 960/(2 \cdot 164) + -1880293376/(0.19 \cdot \pi \cdot 960^2 \cdot 164)$$

$$= -20.58 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 1890503(11164 - 2 \cdot 3400)/(11164 + (4 \cdot 0/3))$$

$$= 738996.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:



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$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 738997 / (960 \cdot 164)$$

$$= 5.50 \text{ MPa}$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(960 \cdot 164)}$$

$$= 309.494 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 1890503 \cdot 0.1 / (164(288 + 309 + 309))$$

$$= -0.97 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.22) [σ7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -1890503 / (4 \cdot 164(288 + 309 + 309)) - 3 \cdot 0.053 \cdot 1890503 / (2 \cdot 164^2)$$

$$= -8.75 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(288 + 1.56 \cdot \sqrt{(960 \cdot 164)}, 2 \cdot 3400)$$

$$= 906.99 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm
Web Thickness	Webtk	32.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	MPa
Height of Web at Center	Hw,c	1085.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(878 \cdot 164)} + 300)164^3) / 12$$

$$= 327865888.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 327865888 + 146282(306 - 82)^2$$

$$= 7656370176.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	892.0	164.0	82.0	146281.8	11995104.0	0.766E+10
Wearplate	300.0	30.0	179.0	9000.0	1611000.1	0.145E+09
Web	32.0	1006.0	697.0	32192.0	22437824.0	0.764E+10
BasePlate	420.0	55.0	1227.5	23100.0	28355246.0	0.196E+11
Totals	...	...	...	210573.7	64399172.0	0.351E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 64399172 / 210574$$

$$= 305.827 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$



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$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1890503$$

$$= 384758.5312 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 5.9851 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2133 - 878 \cdot \sin(120/2) / 1.05$$

$$= 1406.900 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 541536192.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.7207 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(1890503 + 18877)420 / (4 \cdot 2616 \cdot 207)}$$

$$= 33.356 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30)$$

$$= 1856.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1857 / (5 - 1)$$

$$= 464.190 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 464 \cdot 420/2$$

$$= 97479.805 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1909380 / (2616 \cdot 420)$$

$$= 1.738 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 97480 \cdot 1.74$$

$$= 169402.672 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8192 + 7427$$

$$= 15619.032 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$



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= 169403/15619
= 10.847 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:
= Saddle Width - Web Thickness
= 288 - 32
= 256.000 mm

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/15619
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 288/2
= 144.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{63700928/15619}$
= 63.863 mm

Length of Outer Rib [L]:
= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 2133 - cos(120/2)(878 + 164 + 30) - 55
= 1542.000 mm

Intermediate Term [Cc]:
= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 176708/310)}$
= 106.076

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1542/63.9
= 24.146

Bending Moment [Rm]:
= FI / (2 • Bplen) • e • L / 2
= 167543 / (2 • 2616) • 464 • 1542/2
= 11465261.000 N-mm

Compressive Allowable, KL/r < Cc (24.1 < 106) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (24.1)^2 / (2 \cdot 106^2)) 310 /$
 $(5/3 + 3 \cdot (24.1) / (8 \cdot 106) - (24.1^3) / (8 \cdot 106^3))$
= 172.5 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)
= Sc/Sca + (Rm • C1 / I) / Sba
= 10.8/172 + (11465261 • 144/63700992) / 207
= 0.188

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction



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	B	D	Y	A	AY	Io
Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288/2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2133 - \cos(1072/2)(55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(64881092/23046)}$$

$$= 53.059 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1006/53.1$$

$$= 18.960$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 167543/(2 \cdot 2616)$$

$$= 32.023 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 32 \cdot 464 \cdot 1006$$

$$= 14959863.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 14959862/450564$$

$$= 33.192 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($19 < 106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 106^2)) 310 / (5/3 + 3 \cdot (19) / (8 \cdot 106) - (19^3) / (8 \cdot 106^3))$$

$$= 176 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 12.8/176 + (14959862 \cdot 144/64881092)/207$$

$$= 0.233$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$



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$$= \min(288 + 1.56\sqrt{960 \cdot 164}, 2 \cdot 2300)$$

$$= 906.9879 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	960.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2300.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		110.72	MPa
Head Allowable Stress used in Calculation		0.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		113765.32	N
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	MPa
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	km/hr

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-0.13	-278.99
Long. Stress at Top of Midspan	0.13	110.72
Long. Stress at Bottom of Midspan	0.13	110.72
Long. Stress at Top of Saddles	0.97	110.72
Long. Stress at Bottom of Saddles	-0.54	-278.99
Long. Stress at Bottom of Saddles	0.54	110.72
Tangential Shear in Shell	0.50	88.58
Circ. Stress at Horn of Saddle	0.53	166.08
Circ. Compressive Stress in Shell	0.06	110.72

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 351 \cdot 0.00256 \cdot 57.6^2 \cdot 1$$

$$= 2982.5 \text{ lbf}$$

$$= 13266.164 \text{ N}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 6.84^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2$$

$$= 374.3 \text{ lbf}$$

$$= 1664.7 \text{ N}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$= gz \cdot \text{Shipping Weight} \cdot gw + Ftw$$

$$= 0.1 \cdot 1658781 \cdot 1 + 13266$$

$$= 179144.3 \text{ N}$$



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Longitudinal Force due to Axial Acceleration [FI]:

$$= gx \cdot \text{Shipping Weight} \cdot gw + Flw$$

$$= 0.1 \cdot 1658781 \cdot 1 + 1665$$

$$= 167542.8 \text{ N}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$= Ft/\text{Num of Saddles} \cdot B / E$$

$$= 179144/2 \cdot 2133/1857$$

$$= 102898.4 \text{ N}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$= FI \cdot B / Ls$$

$$= 167543 \cdot 2133/5400$$

$$= 66179.4 \text{ N}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$= \text{Saddle Load} \cdot gy$$

$$= 10349 \cdot 0.05$$

$$= 517.474 \text{ N}$$

Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + Frv + \max(Fwl, Fwt)$$

$$= 10349 + 517 + \max(66179, 102898)$$

$$= 113765.3 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	132642.47	N
Transverse Shear Load	89572.17	N
Longitudinal Shear Load	167542.84	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1756/2 + 164 + 30)\sin(120/2), 2 \cdot 960)$$

$$= 1856.759 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -113765 \cdot 2300 [1 - (1 - 2300/11164 + (960^2 - 0^2)/(2 \cdot 2300 \cdot 11164))/(1 + (4 \cdot 0)/(3 \cdot 11164))]$$

$$= -49231304.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 113765 \cdot 11164/4((1 + 2(960^2 - 0^2)/(11164^2))/(1 + (4 \cdot 0)/(3 \cdot 11164)) - 4 \cdot 2300/11164)$$

$$= 60579020.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 960/(2 \cdot 164) - 60579020/(\pi \cdot 960^2 \cdot 164)$$

$$= -0.13 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 960/(2 \cdot 164) + 60579020/(\pi \cdot 960^2 \cdot 164)$$



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= 0.13 MPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

= $P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$

= $0.960 / (2 \cdot 164) - 49231304 / (0.11 \cdot \pi \cdot 960^2 \cdot 164)$

= 0.97 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

= $P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$

= $0.960 / (2 \cdot 164) + 49231304 / (0.19 \cdot \pi \cdot 960^2 \cdot 164)$

= -0.54 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L-2a) / (L + (4 \cdot h_m / 3))$

= $113765 (11164 - 2 \cdot 2300) / (11164 + (4 \cdot 0 / 3))$

= 66889.6 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= $K2 \cdot T / (R_m \cdot t)$

= $1.17 \cdot 66890 / (960 \cdot 164)$

= 0.50 MPa

Decay Length (4.15.20) [x_1, x_2]:

= $0.78 \cdot \sqrt{(R_m \cdot t)}$

= $0.78 \cdot \sqrt{(960 \cdot 164)}$

= 309.494 mm

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

= $-K5 \cdot Q \cdot k / (t(b + X1 + X2))$

= $-0.76 \cdot 113765 \cdot 0.1 / (164(288 + 309 + 309))$

= -0.06 MPa

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

= $-Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$

= $-113765 / (4 \cdot 164(288 + 309 + 309)) -$

$3 \cdot 0.053 \cdot 113765 / (2 \cdot 164^2)$

= -0.53 MPa

Effective reinforcing plate width (4.15.24) [B1]:

= $\min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$

= $\min(288 + 1.56 \cdot \sqrt{(960 \cdot 164)}, 2 \cdot 2300)$

= 906.99 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm
Web Thickness	Webtk	32.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	MPa
Height of Web at Center	Hw,c	1085.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

= $((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$

= $((1.56 \cdot \sqrt{(878 \cdot 164)} + 300)^3) / 12$

= 327865888.000 mm⁴

Resolved Inertia for the Shell [I,shell]:

= ShellInertia + AShell(C1 - Y)²



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$$= 327865888 + 146282 (306 - 82)^2$$

$$= 7656370176.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	892.0	164.0	82.0	146281.8	11995104.0	0.766E+10
Wearplate	300.0	30.0	179.0	9000.0	1611000.1	0.145E+09
Web	32.0	1006.0	697.0	32192.0	22437824.0	0.764E+10
BasePlate	420.0	55.0	1227.5	23100.0	28355246.0	0.196E+11
Totals	...	...	...	210573.7	64399172.0	0.351E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 64399172/210574$$

$$= 305.827 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 113765$$

$$= 23153.7090 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.3602 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2133 - 878 \cdot \sin(120/2) / 1.05$$

$$= 1406.900 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 32588158.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2841 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(113765 + 18877)420) / (4 \cdot 2616 \cdot 207)}$$

$$= 8.792 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30)$$

$$= 1856.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1857 / (5 - 1)$$

$$= 464.190 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 464 \cdot 420 / 2$$



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= 97479.805 mm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)

= 132642 / (2616 • 420)

= 0.121 N/mm²

Axial Load [P]:

= Ap • Bp

= 97480 • 0.12

= 11768.206 N

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 8192 + 7427

= 15619.032 mm²

Compressive Stress [Sc]:

= P/Ar

= 11768/15619

= 0.754 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:

= Saddle Width - Web Thickness

= 288 - 32

= 256.000 mm

Distance to Centroid from Datum [y_{tot}]:

= AY / A

= 0/15619

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 288/2

= 144.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{63700928 / 15619}$

= 63.863 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 2133 - cos(120/2)(878 + 164 + 30) - 55

= 1542.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199955 / 310}$

= 112.838

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1542/63.9

= 24.146



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Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 167543 / (2 \cdot 2616) \cdot 464 \cdot 1542 / 2$$

$$= 11465261.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($24.1 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (24.1)^2 / (2 \cdot 113^2)) 310 / (5/3 + 3 \cdot (24.1) / (8 \cdot 113) - (24.1)^3 / (8 \cdot 113^3))$$

$$= 173.5 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.75 / 174 + (11465261 \cdot 144 / 63700992) / 207$$

$$= 0.130$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2133 - \cos(1072/2) (55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{64881092 / 23046}$$

$$= 53.059 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1006 / 53.1$$

$$= 18.960$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 167543 / (2 \cdot 2616)$$

$$= 32.023 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 32 \cdot 464 \cdot 1006$$

$$= 14959863.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 14959862 / 450564$$

$$= 33.192 \text{ MPa}$$



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Compressive Allowable, $KL/r < C_c$ ($19 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 113^2)) 310 /$$

$$(5/3 + 3 \cdot (19) / (8 \cdot 113) - (19^3) / (8 \cdot 113^3))$$

$$= 176.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 0.89 / 177 + (14959862 \cdot 144 / 64881092) / 207$$

$$= 0.166$$

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Operating Case: Step: 23 4:12pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

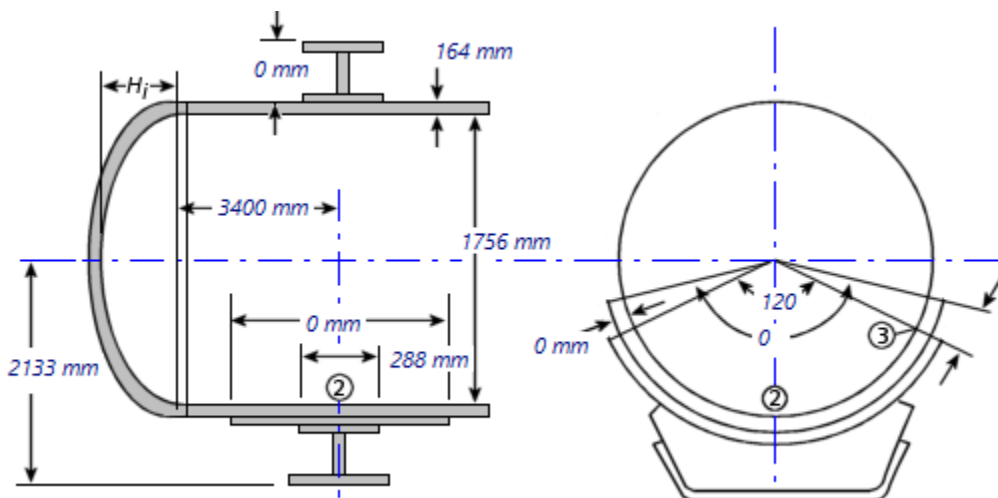
$$= \min(b + 1.56 \sqrt{R_m \cdot t}, 2a)$$

$$= \min(288 + 1.56 \sqrt{961 \cdot 162}, 2 \cdot 3400)$$

$$= 903.5223 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	961.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3400.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		110.72	MPa
Head Allowable Stress used in Calculation		138.20	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	2026858.00	N
Pressure used in Analysis	P	18.000 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	55.84	110.72
Long. Stress at Bottom of Midspan	50.94	110.72



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Saddle Calcs: Operating Case: Step: 23 4:12pm Apr 10, 2026

Long. Stress at Top of Saddles	93.60	110.72
Long. Stress at Bottom of Saddles	31.10	110.72
Tangential Shear in Shell	5.96	88.58
Circ. Stress at Horn of Saddle	9.59	138.40
Circ. Compressive Stress in Shell	1.05	110.72

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (14050/2 + 0) \cdot 2133/1857$$

$$= 24209.9 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4139, 0) \cdot 2133/5400$$

$$= 1634.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(103239, 0) \cdot 2133/5400$$

$$= 40779.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (103239/2 + 0) \cdot 2133/1857$$

$$= 177897.7 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1848960 + \max(1635, 24210, 40779, 177898)$$

$$= 2026858.0 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.08 \cdot 1.2)^2)$$

$$= 4139.213 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 2045735.12 N

Transverse Shear Load 51619.49 N

Longitudinal Shear Load 103238.98 N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1756/2 + 164 + 30)\sin(120/2), 2 \cdot 961)$$

$$= 1856.759 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -2026858 \cdot 3400 [1 - (1 - 3400/11164 + (961^2 - 0^2)/$$

$$(2 \cdot 3400 \cdot 11164))/(1 + (4 \cdot 0)/(3 \cdot 11164))]$$

$$= -2015736320.0 \text{ N-mm}$$



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Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\
 &= 2026858 \cdot 11164 / 4 \cdot ((1 + 2(961^2 - 0^2) / (11164^2)) / (1 + (4 \cdot 0) / (3 \cdot 11164)) - 4 \cdot 3400 / 11164) \\
 &= -1150989824.0 \text{ N-mm}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 18 \cdot 961 / (2 \cdot 162) - 1150989824 / (\pi \cdot 961^2 \cdot 162) \\
 &= 55.84 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 18 \cdot 961 / (2 \cdot 162) + 1150989824 / (\pi \cdot 961^2 \cdot 162) \\
 &= 50.94 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 18 \cdot 961 / (2 \cdot 162) - 2015736320 / (0.11 \cdot \pi \cdot 961^2 \cdot 162) \\
 &= 93.60 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 18 \cdot 961 / (2 \cdot 162) + 2015736320 / (0.19 \cdot \pi \cdot 961^2 \cdot 162) \\
 &= 31.10 \text{ MPa}
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 2026858 \cdot (11164 - 2 \cdot 3400) / (11164 + (4 \cdot 0 / 3)) \\
 &= 792297.3 \text{ N}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 792297 / (961 \cdot 162) \\
 &= 5.96 \text{ MPa}
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{Rm \cdot t} \\
 &= 0.78 \cdot \sqrt{961 \cdot 162} \\
 &= 307.761 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.76 \cdot 2026858 \cdot 0.1 / (162(288 + 308 + 308)) \\
 &= -1.05 \text{ MPa}
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned}
 &= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\
 &= -2026858 / (4 \cdot 162(288 + 308 + 308)) - 3 \cdot 0.053 \cdot 2026858 / (2 \cdot 162^2) \\
 &= -9.59 \text{ MPa}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a) \\
 &= \min(288 + 1.56 \cdot \sqrt{961 \cdot 162}, 2 \cdot 3400) \\
 &= 903.52 \text{ mm}
 \end{aligned}$$

Distance between Saddle Supports [Ls]:

$$= 5400.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$\begin{aligned}
 &= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature}) \\
 &= 0.14220\text{E-}04 \cdot 5400(480 - 21.1) \\
 &= 35.234 \text{ mm}
 \end{aligned}$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm
Web Thickness	Webtk	32.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		480.0	C
Saddle Yield Stress	Sy	223.6	MPa
Height of Web at Center	Hw,c	1085.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(880 \cdot 162)} + 300)162^3)/12$$

$$= 314971008.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 314971008 + 144020 (307 - 81)^2$$

$$= 7667600384.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	889.0	162.0	81.0	144019.8	11665603.0	0.767E+10
Wearplate	300.0	30.0	177.0	9000.0	1593000.1	0.153E+09
Web	32.0	1006.0	695.0	32192.0	22373444.0	0.756E+10
BasePlate	420.0	55.0	1225.5	23100.0	28309048.0	0.195E+11
Totals	...	...	...	208311.8	63941092.0	0.349E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 63941092/208312$$

$$= 306.949 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2026858$$

$$= 412509.5312 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 6.4167 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 134.1643 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 2133 - 880 \cdot \sin(120/2) / 1.05$$

$$= 1405.246 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 579912384.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.1018 \text{ MPa}$$



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Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 149.0714 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(2026858 + 18877) 420}{4 \cdot 2616 \cdot 149}}$$

$$= 40.652 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30)$$

$$= 1856.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1857 / (5 - 1)$$

$$= 464.190 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 464 \cdot 420 / 2$$

$$= 97479.805 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2045735 / (2616 \cdot 420)$$

$$= 1.862 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 97480 \cdot 1.86$$

$$= 181500.172 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8192 + 7427$$

$$= 15619.032 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 181500 / 15619$$

$$= 11.621 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 288 - 32$$

$$= 256.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 15619$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$



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= 144.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{63700928/15619}$

= 63.863 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 2133 - cos(120/2)(878 + 164 + 30) - 55

= 1542.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 176708/224}$

= 124.897

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1542/63.9

= 24.146

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 103239 / (2 • 2616) • 464 • 1542/2

= 7064831.000 N-mm

Compressive Allowable, KL/r < Cc (24.1 < 125) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (24.1)^2 / (2 \cdot 125^2)) 224 /$

$(5/3 + 3 \cdot (24.1) / (8 \cdot 125) - (24.1^3) / (8 \cdot 125^3))$

= 126.2 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 11.6/126 + (7064831 • 144/63700992) / 149

= 0.199

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/23046

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 288/2

= 144.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 2133 - cos(1072/2)(55 + 0 + 0) - 0

= 1006.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{64881092/23046}$



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= 53.059 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1006/53.1
 = 18.960

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 103239/(2 • 2616)
 = 19.732 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 19.7 • 464 • 1006
 = 9218185.000 N-mm

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus
 = 9218185/450564
 = 20.453 MPa

Compressive Allowable, KL/r < Cc (19 < 125) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³)
 = (1-(19)²/(2 • 125²))224/
 (5/3+3•(19)/(8• 125)-(19³)/(8•125³)
 = 128.3 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 13.7/128 + (9218185 • 144/64881092)/149
 = 0.244

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1613.9999 mm ²
BasePlate Allowable Stress	S	110.72 MPa
Area Available in a Single Bolt	BltArea	1407.6318 mm ²
Saddle Load QO (Weight)	QO	1867837.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	40779.4 N
Maximum Transverse Force	Ft	51619.5 N
Maximum Longitudinal Force	F1	103239.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

= FI / (Bolt Area • Number of Bolts)
 = 103239/(1408 • 8)
 = 9.2 MPa. Must be less than 82.8 MPa.

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 51619/(1408 • 8)
 = 4.6 MPa. Must be less than 82.8 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)



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Bolt Area due to Shear Load [Bltarears]:
 $= F / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
 $= 103239 / (82.8 \cdot 8)$
 $= 155.8692 \text{ mm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 2133 \cdot 51619 + 0$
 $= 110149016.00 \text{ N-mm}$

Eccentricity (e):
 $= R_{\text{mom}} / QO$
 $= 110149016 / 1867837$
 $= 58.95 \text{ mm} < B_{\text{plen}} / 6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[B_{\text{tarearl}}, B_{\text{tarears}}, B_{\text{tareart}}]$
 $= \max[0, 156, 0]$
 $= 155.8692 \text{ mm}^2$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 30$, $c = 2 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = -1$, Temp. Reduction = 0 C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B 4 C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(288 + 1.56 \cdot \sqrt{961 \cdot 162}, 2 \cdot 2300)$
 $= 903.5223 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	961.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2300.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		110.72	MPa
Head Allowable Stress used in Calculation		0.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.00	



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Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 233338.73 N
 Pressure used in Analysis P 18.000 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	53.13	110.72
Long. Stress at Bottom of Midspan	53.65	110.72
Long. Stress at Top of Saddles	55.40	110.72
Long. Stress at Bottom of Saddles	52.27	110.72
Tangential Shear in Shell	1.03	88.58
Circ. Stress at Horn of Saddle	1.10	138.40
Circ. Compressive Stress in Shell	0.12	110.72

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (14050/2 + 0) \cdot 2133/1857$
 $= 24209.9 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(4139, 0) \cdot 2133/5400$
 $= 1634.9 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl, eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(103239, 0) \cdot 2133/5400$
 $= 40779.4 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (103239/2 + 0) \cdot 2133/1857$
 $= 177897.7 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 55441 + \max(1635, 24210, 40779, 177898)$
 $= 233338.7 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform \text{ Area} + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.08 \cdot 1.2)^2)$
 $= 4139.213 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 252215.88 N
 Transverse Shear Load 51619.49 N
 Longitudinal Shear Load 103238.98 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1756/2 + 164 + 30) \sin(120/2), 2 \cdot 961)$
 $= 1856.759 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405



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$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1* = 0.1923$$

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\ &= -233339 \cdot 2300 [1 - (1 - 2300/11164 + (961^2 - 0^2)/(2 \cdot 2300 \cdot 11164))/(1 + (4 \cdot 0)/(3 \cdot 11164))] \\ &= -100955960.0 \text{ N-mm} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L] \\ &= 233339 \cdot 11164/4 [(1 + 2(961^2 - 0^2)/(11164^2))/(1 + (4 \cdot 0)/(3 \cdot 11164)) - 4 \cdot 2300/11164] \\ &= 124270920.0 \text{ N-mm} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\ &= 18 \cdot 961/(2 \cdot 162) - 124270920/(\pi \cdot 961^2 \cdot 162) \\ &= 53.13 \text{ MPa} \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\ &= 18 \cdot 961/(2 \cdot 162) + 124270920/(\pi \cdot 961^2 \cdot 162) \\ &= 53.65 \text{ MPa} \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\ &= 18 \cdot 961/(2 \cdot 162) - 100955960/(0.11 \cdot \pi \cdot 961^2 \cdot 162) \\ &= 55.40 \text{ MPa} \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t) \\ &= 18 \cdot 961/(2 \cdot 162) + 100955960/(0.19 \cdot \pi \cdot 961^2 \cdot 162) \\ &= 52.27 \text{ MPa} \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\ &= 233339 (11164 - 2 \cdot 2300)/(11164 + (4 \cdot 0/3)) \\ &= 137194.1 \text{ N} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned} &= K2 \cdot T / (R_m \cdot t) \\ &= 1.17 \cdot 137194/(961 \cdot 162) \\ &= 1.03 \text{ MPa} \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned} &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\ &= 0.78 \cdot \sqrt{(961 \cdot 162)} \\ &= 307.761 \text{ mm} \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned} &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ &= -0.76 \cdot 233339 \cdot 0.1/(162(288 + 308 + 308)) \\ &= -0.12 \text{ MPa} \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$\begin{aligned} &= -Q/(4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q/(2 \cdot t^2) \\ &= -233339/(4 \cdot 162(288 + 308 + 308)) - 3 \cdot 0.053 \cdot 233339/(2 \cdot 162^2) \\ &= -1.10 \text{ MPa} \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} &= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\ &= \min(288 + 1.56 \cdot \sqrt{(961 \cdot 162)}, 2 \cdot 2300) \end{aligned}$$



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= 903.52 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm
Web Thickness	Webtk	32.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	C
Saddle Yield Stress	Sy	223.6	MPa
Height of Web at Center	Hw,c	1085.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(880 \cdot 162)} + 300)162^3)/12$
 $= 314971008.000 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 314971008 + 144020(307 - 81)^2$
 $= 7667600384.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³ (3)	I + AD ² (2) mm ⁴ (4)
Shell	889.0	162.0	81.0	144019.8	11665603.0	0.767E+10
Wearplate	300.0	30.0	177.0	9000.0	1593000.1	0.153E+09
Web	32.0	1006.0	695.0	32192.0	22373444.0	0.756E+10
BasePlate	420.0	55.0	1225.5	23100.0	28309048.0	0.195E+11
Totals	...	...	...	208311.8	63941092.0	0.349E+11

Distance to Centroid [C1]:
 $= AY / A$
 $= 63941092/208312$
 $= 306.949 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 233339$
 $= 47489.4922 \text{ N}$

Tension Stress, St = (Fh/As) = 0.7387 MPa
 Allowed Stress, Sa = 0.6 * Yield Str = 134.1643 MPa

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 2133 - 880 \cdot \sin(120/2)/1.05$
 $= 1405.246 \text{ mm}$



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Bending Moment, $M = F_h \cdot d = 66761476.0000 \text{ N-mm}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 0.5873 \text{ MPa}$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 149.0714 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(233339 + 18877) \cdot 420) / (4 \cdot 2616 \cdot 149)}$

$= 14.274 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30)$

$= 1856.758 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 1857 / (5 - 1)$

$= 464.190 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 464 \cdot 420 / 2$

$= 97479.805 \text{ mm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 252216 / (2616 \cdot 420)$

$= 0.230 \text{ N/mm}^2$

Axial Load [P]:

$= A_p \cdot B_p$

$= 97480 \cdot 0.23$

$= 22376.908 \text{ N}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 8192 + 7427$

$= 15619.032 \text{ mm}^2$

Compressive Stress [Sc]:

$= P / A_r$

$= 22377 / 15619$

$= 1.433 \text{ MPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 288 - 32$

$= 256.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:

$= AY / A$

$= 0 / 15619$

$= 0.000 \text{ mm}$



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Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$

$$= 144.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{63700928 / 15619}$$

$$= 63.863 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2133 - \cos(120/2) (878 + 164 + 30) - 55$$

$$= 1542.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199955}{224}}$$

$$= 132.858$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1542 / 63.9$$

$$= 24.146$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 103239 / (2 \cdot 2616) \cdot 464 \cdot 1542 / 2$$

$$= 7064831.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($24.1 < 133$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (24.1)^2 / (2 \cdot 133^2)) 224 / (5/3 + 3 \cdot (24.1) / (8 \cdot 133) - (24.1)^3 / (8 \cdot 133^3))$$

$$= 126.8 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.43 / 127 + (7064831 \cdot 144 / 63700992) / 149$$

$$= 0.118$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2133 - \cos(1072/2) (55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$



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Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{64881092/23046}$
 $= 53.059 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 1006/53.1$
 $= 18.960$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 103239/(2 \cdot 2616)$
 $= 19.732 \text{ N/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 19.7 \cdot 464 \cdot 1006$
 $= 9218185.000 \text{ N-mm}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 9218185/450564$
 $= 20.453 \text{ MPa}$

Compressive Allowable, $KL/r < C_c$ ($19 < 133$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (19)^2 / (2 \cdot 133^2)) 224 /$
 $(5/3 + 3 \cdot (19) / (8 \cdot 133) - (19^3) / (8 \cdot 133^3))$
 $= 128.7 \text{ MPa}$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 1.69/129 + (9218185 \cdot 144/64881092) / 149$
 $= 0.150$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1613.9999 mm2
BasePlate Allowable Stress	S	110.72 MPa
Area Available in a Single Bolt	BltArea	1407.6318 mm2
Saddle Load Q0 (Weight)	Q0	74318.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	40779.4 N
Maximum Transverse Force	Ft	51619.5 N
Maximum Longitudinal Force	F1	103239.0 N
Saddle Bolted to Steel Foundation		Yes

Friction force due to thermal growth, [Ff]:
 $= 747134.9 \text{ N}$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= \max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 747135 / (1408 \cdot 8)$
 $= 66.4 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 51619 / (1408 \cdot 8)$



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= 4.6 MPa. Must be less than 82.8 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= FI / (BoltShearAllowable • Nbolts)

= 103239/(82.8 • 8)

= 155.8692 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 2133 • 51619 + 0

= 110149016.00 N-mm

Eccentricity (e):

= Rmom / QO

= 110149016/74318

= 1481.53 mm > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis

= 2616/2 - 70

= 1238.00 mm

K1 = 3 (e - 0.5 • Bplen)

= 3 (1482 - 0.5•2616)

= 520.58 mm

K2 = 6 • n1 • At / Bpwid • (f + e)

= 6 • 1 • 5631/420 • (1238 + 1482)

= 218747.98 mm²

K3 = -K2 • (0.5 • Bplen + f)

= -218748 • (0.5 • 2616 + 1238)

= -556931820.94 mm³

Iteratively Solving for the Effective Bearing Length:

$Y3 + K1 \cdot Y2 + K2 \cdot Y + K3 = 0$

$Y3 + 521 \cdot Y2 + 218748 \cdot Y + -556931776 = 0$

Y = 611.42 mm

Num = (Bplen / 2 - Y / 3 - e)

= (2616/2 - 611/3 - 1482)

= -377.33

Denom = (Bplen / 2 - Y / 3 + f)

= (2616/2 - 611/3 + 1238)

= 2342.19

Total Bolt Tension Force [Tforce]:

= - QO • Num / Denom

= - 74318 • -377/2342

= 11972.90 N

Bolt Area Required due to Transverse Load [Bltareart]:

= Tforce / (Stba • Nbt)

= 11973/(138 • 4)

= 21.6919 mm²

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]



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= max[0, 156, 21.7]
= 155.8692 mm2

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
= $2(QO + Tforce) / (Y \cdot Bpwid)$
= $2(74318 + 11973) / (611 \cdot 420)$
= 0.67 MPa

Distance from Baseplate Edge to the Web [ADIST]:
= $(Bplen - Weblength) / 2$
= $(2616 - 2565) / 2$
= 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
= $Tforce \cdot Adist$
= $11973 \cdot 25.4$
= 304234.94 N-mm

Equivalent Bearing Pressure (f1):
= $fc \cdot (Y - Adist) / Y$
= $0.67 \cdot (611 - 25.4) / 611$
= 0.64 MPa

Overturning Moment due to Bearing Pressure [Mc]:
= $(Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$
= $(25.4^2 \cdot 420 / 6) \cdot (0.64 + 2 \cdot 0.67)$
= 89828.12 N-mm

Baseplate Required Thickness [Treq]:
= $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
= $\sqrt{(6 \cdot \max(304235, 89828) / (420 \cdot 166))}$
= 5.1148 mm

MDMT Calculations for: Sdl 2 Fr40, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 30, c = 2 mm, E* = 1
Thickness Ratio = $tr \cdot E* / (tg - c) = -1$, Temp. Reduction = 0 C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	4 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

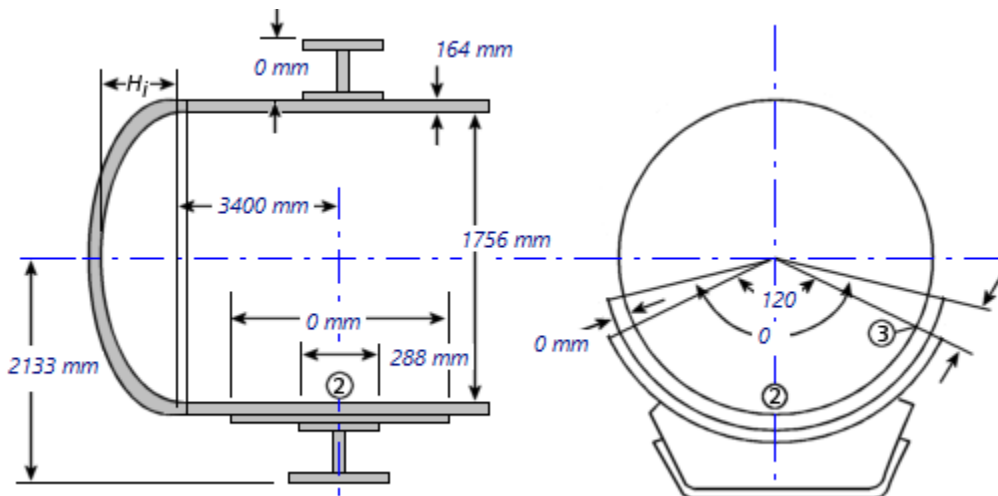
$$= \min(b + 1.56 \sqrt{R_m \cdot t}, 2a)$$

$$= \min(288 + 1.56 \sqrt{960 \cdot 164}, 2 \cdot 3400)$$

$$= 906.9879 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	960.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3400.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		278.99	MPa
Head Allowable Stress used in Calculation		278.99	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		-2240.43	N
Pressure used in Analysis	P	25.069	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	73.37	278.99
Long. Stress at Bottom of Midspan	73.37	278.99



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Long. Stress at Top of Saddles	73.33	278.99
Long. Stress at Bottom of Saddles	73.40	278.99
Tangential Shear in Shell	0.01	165.23
Circ. Stress at Horn of Saddle	0.01	418.49
Circ. Compressive Stress in Shell	0.00	278.99

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (7025/2 + 0) \cdot 2133/1857$$

$$= 12104.9 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(2069, 0) \cdot 2133/5400$$

$$= 817.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.08 \cdot 1.2)^2)$$

$$= 4139.213 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 16636.71 N

Transverse Shear Load 3512.42 N

Longitudinal Shear Load 2069.50 N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1756/2 + 164 + 30) \sin(120/2), 2 \cdot 960)$$

$$= 1856.759 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -2240 \cdot 3400 [1 - (1 - 3400/11164 + (960^2 - 0^2) / (2 \cdot 3400 \cdot 11164)) / (1 + (4 \cdot 0) / (3 \cdot 11164))]$$

$$= 2228332.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= -2240 \cdot 11164/4 ((1 + 2(960^2 - 0^2)/(11164^2)) / (1 + (4 \cdot 0) / (3 \cdot 11164)) - 4 \cdot 3400/11164)$$

$$= 1272465.1 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 25.1 \cdot 960 / (2 \cdot 164) - 1272465 / (\pi \cdot 960^2 \cdot 164)$$

$$= 73.37 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 t)$$

$$= 25.1 \cdot 960 / (2 \cdot 164) + 1272465 / (\pi \cdot 960^2 \cdot 164)$$

$$= 73.37 \text{ MPa}$$



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Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 25.1 \cdot 960 / (2 \cdot 164) - 2228332 / (0.11 \cdot \pi \cdot 960^2 \cdot 164)$$

$$= 73.33 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 25.1 \cdot 960 / (2 \cdot 164) + 2228332 / (0.19 \cdot \pi \cdot 960^2 \cdot 164)$$

$$= 73.40 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= -2240 (11164 - 2 \cdot 3400) / (11164 + (4 \cdot 0 / 3))$$

$$= -875.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot -876 / (960 \cdot 164)$$

$$= -0.01 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{960 \cdot 164}$$

$$= 309.494 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot -2240 \cdot 0.1 / (164 (288 + 309 + 309))$$

$$= 0.00 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -2240 / (4 \cdot 164 (288 + 309 + 309)) - 3 \cdot 0.053 \cdot -2240 / (2 \cdot 164^2)$$

$$= 0.01 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(288 + 1.56 \cdot \sqrt{960 \cdot 164}, 2 \cdot 3400)$$

$$= 906.99 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm
Web Thickness	Webtk	32.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	MPa
Height of Web at Center	Hw,c	1085.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{878 \cdot 164} + 300) 164^3) / 12$$

$$= 327865888.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 327865888 + 146282 (306 - 82)^2$$

$$= 7656370176.000 \text{ mm}^4$$



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	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	892.0	164.0	82.0	146281.8	11995104.0	0.766E+10
Wearplate	300.0	30.0	179.0	9000.0	1611000.1	0.145E+09
Web	32.0	1006.0	697.0	32192.0	22437824.0	0.764E+10
BasePlate	420.0	55.0	1227.5	23100.0	28355246.0	0.196E+11
Totals	...	...	...	210573.7	64399172.0	0.351E+11

Distance to Centroid [C1]:

= AY / A

= 64399172/210574

= 305.827 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.2 • -2240

= -455.9766 N

Tension Stress, St = (Fh/As) = -0.0071 MPa

Allowed Stress, Sa = 0.6 • Yield Str = 185.9966 MPa

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 2133 - 878 • sin(120/2)/1.05

= 1406.900 mm

Bending Moment, M = Fh • d = -641773.4375 N-mm

Bending Stress, Sb = (M • C1 / I) = -0.0056 MPa

Allowed Stress, Sa = 2/3 • Yield Str = 206.6629 MPa

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(-2240 + 18877)420/(4 • 2616 • 207))

= 3.114 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120/2)(878 + 164 + 30)

= 1856.758 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 1857/(5 - 1)

= 464.190 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 464 • 420/2

= 97479.805 mm²



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Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 16637 / (2616 \cdot 420)$$

$$= 0.015 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 97480 \cdot 0.015$$

$$= 1476.030 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8192 + 7427$$

$$= 15619.032 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1476 / 15619$$

$$= 0.095 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 288 - 32$$

$$= 256.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 15619$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$

$$= 144.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(63700928 / 15619)}$$

$$= 63.863 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2133 - \cos(120/2) (878 + 164 + 30) - 55$$

$$= 1542.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 176708 / 310)}$$

$$= 106.076$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1542 / 63.9$$

$$= 24.146$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$



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$$= 2069 / (2 \cdot 2616) \cdot 464 \cdot 1542 / 2$$

$$= 141619.609 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($24.1 < 106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (24.1)^2 / (2 \cdot 106^2)) 310 / (5/3 + 3 \cdot (24.1) / (8 \cdot 106) - (24.1)^3 / (8 \cdot 106^3))$$

$$= 172.5 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.095 / 172 + (141620 \cdot 144 / 63700992) / 207$$

$$= 0.002$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2133 - \cos(1072/2) (55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{64881092 / 23046}$$

$$= 53.059 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1006 / 53.1$$

$$= 18.960$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 2069 / (2 \cdot 2616)$$

$$= 0.396 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.4 \cdot 464 \cdot 1006$$

$$= 184785.125 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 184785 / 450564$$

$$= 0.410 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($19 < 106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$



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$$= (1 - (19)^2 / (2 \cdot 106^2)) \cdot 310 / (5/3 + 3 \cdot (19) / (8 \cdot 106) - (19^3) / (8 \cdot 106^3))$$

$$= 176 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 0.11/176 + (184785 \cdot 144/64881092) / 207$$

$$= 0.003$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1613.9999 mm ²
BasePlate Allowable Stress	S	148.00 MPa
Area Available in a Single Bolt	BltArea	1407.6318 mm ²
Saddle Load QO (Weight)	QO	4531.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	817.5 N
Maximum Transverse Force	Ft	3512.4 N
Maximum Longitudinal Force	F1	2069.5 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2069 / (1408 \cdot 8)$$

$$= 0.2 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3512 / (1408 \cdot 8)$$

$$= 0.3 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 2069 / (103 \cdot 8)$$

$$= 2.5069 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2133 \cdot 3512 + 0$$

$$= 7495021.50 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 7495021/4532$$

$$= 1653.21 \text{ mm} > Bplen/6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 2616/2 - 70$$

$$= 1238.00 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= ES / EC$$

$$= 203402/21526$$



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$$= 9.45$$

$$\begin{aligned} K1 &= 3 (e - 0.5 \cdot B_{plen}) \\ &= 3 (1653 - 0.5 \cdot 2616) \\ &= 1035.64 \text{ mm} \end{aligned}$$

$$\begin{aligned} K2 &= 6 \cdot n1 \cdot At / B_{pwid} \cdot (f + e) \\ &= 6 \cdot 9.45 \cdot 5631/420 \cdot (1238 + 1653) \\ &= 2197440.86 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} K3 &= -K2 \cdot (0.5 \cdot B_{plen} + f) \\ &= -2197440 \cdot (0.5 \cdot 2616 + 1238) \\ &= -5594679081.25 \text{ mm}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y3 + K1 \cdot Y2 + K2 \cdot Y + K3 &= 0 \\ Y3 + 1036 \cdot Y2 + 2197440 \cdot Y + -5.59E+09 &= 0 \\ Y &= 1170.50 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Num} &= (B_{plen} / 2 - Y / 3 - e) \\ &= (2616/2 - 1171/3 - 1653) \\ &= -735.38 \end{aligned}$$

$$\begin{aligned} \text{Denom} &= (B_{plen} / 2 - Y / 3 + f) \\ &= (2616/2 - 1171/3 + 1238) \\ &= 2155.83 \end{aligned}$$

Total Bolt Tension Force [Tforce]:

$$\begin{aligned} &= -QO \cdot \text{Num} / \text{Denom} \\ &= -4532 \cdot -735/2156 \\ &= 1545.84 \text{ N} \end{aligned}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= T_{force} / (Stba \cdot Nbt) \\ &= 1546 / (172 \cdot 4) \\ &= 2.2471 \text{ mm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0, 2.51, 2.25] \\ &= 2.5069 \text{ mm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned} &= 2(QO + T_{force}) / (Y \cdot B_{pwid}) \\ &= 2(4532 + 1546) / (1171 \cdot 420) \\ &= 0.02 \text{ MPa} \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$\begin{aligned} &= (B_{plen} - \text{WebLength}) / 2 \\ &= (2616 - 2565) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

Overturning Moment due To Bolt Tension [Mt]:

$$\begin{aligned} &= T_{force} \cdot \text{Adist} \\ &= 1546 \cdot 25.4 \\ &= 39280.31 \text{ N-mm} \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= f_c \cdot (Y - \text{Adist}) / Y \\ &= 0.025 \cdot (1171 - 25.4) / 1171 \\ &= 0.02 \text{ MPa} \end{aligned}$$



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Overturning Moment due to Bearing Pressure [Mc]:

$$= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$$

$$= (25.4^2 \cdot 420 / 6) \cdot (0.024 + 2 \cdot 0.025)$$

$$= 3326.99 \text{ N-mm}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$$

$$= \sqrt{(6 \cdot \max(39280, 3327) / (420 \cdot 222))}$$

$$= 1.5896 \text{ mm}$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(288 + 1.56 \cdot \sqrt{960 \cdot 164}, 2 \cdot 2300)$$

$$= 906.9879 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	960.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2300.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		278.99	MPa
Head Allowable Stress used in Calculation		278.99	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		-2326.78	N
Pressure used in Analysis	P	25.069	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	73.37	278.99
Long. Stress at Bottom of Midspan	73.37	278.99
Long. Stress at Top of Saddles	73.35	278.99
Long. Stress at Bottom of Saddles	73.38	278.99
Tangential Shear in Shell	0.01	165.23
Circ. Stress at Horn of Saddle	0.01	418.49
Circ. Compressive Stress in Shell	0.00	278.99

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (7025/2 + 0) \cdot 2133/1857$$

$$= 12104.9 \text{ N}$$



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Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(2069, 0) \cdot 2133/5400$$

$$= 817.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.08 \cdot 1.2)^2)$$

$$= 4139.213 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	16550.36	N
Transverse Shear Load	3512.42	N
Longitudinal Shear Load	2069.50	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1756/2 + 164 + 30) \sin(120/2), 2 \cdot 960)$$

$$= 1856.759 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -2327 \cdot 2300 [1 - (1 - 2300/11164 + (960^2 - 0^2)/(2 \cdot 2300 \cdot 11164)) / (1 + (4 \cdot 0)/(3 \cdot 11164))]$$

$$= 1006902.1 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= -2327 \cdot 11164/4 ((1 + 2(960^2 - 0^2)/(11164^2)) / (1 + (4 \cdot 0)/(3 \cdot 11164)) - 4 \cdot 2300/11164)$$

$$= -1238991.1 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 25.1 \cdot 960/(2 \cdot 164) - -1238991/(\pi \cdot 960^2 \cdot 164)$$

$$= 73.37 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 25.1 \cdot 960/(2 \cdot 164) + -1238991/(\pi \cdot 960^2 \cdot 164)$$

$$= 73.37 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 25.1 \cdot 960/(2 \cdot 164) - 1006902/(0.11 \cdot \pi \cdot 960^2 \cdot 164)$$

$$= 73.35 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 25.1 \cdot 960/(2 \cdot 164) + 1006902/(0.19 \cdot \pi \cdot 960^2 \cdot 164)$$

$$= 73.38 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= -2327 (11164 - 2 \cdot 2300) / (11164 + (4 \cdot 0/3))$$

$$= -1368.1 \text{ N}$$



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Saddle Calcs: Test Case:

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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot -1368 / (960 \cdot 164)$$

$$= -0.01 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(960 \cdot 164)}$$

$$= 309.494 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot -2327 \cdot 0.1 / (164(288 + 309 + 309))$$

$$= 0.00 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2327 / (4 \cdot 164(288 + 309 + 309)) - 3 \cdot 0.053 \cdot -2327 / (2 \cdot 164^2)$$

$$= 0.01 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(288 + 1.56 \cdot \sqrt{(960 \cdot 164)}, 2 \cdot 2300)$$

$$= 906.99 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm
Web Thickness	Webtk	32.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	MPa
Height of Web at Center	Hw,c	1085.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(878 \cdot 164)} + 300)164^3) / 12$$

$$= 327865888.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 327865888 + 146282(306 - 82)^2$$

$$= 7656370176.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	892.0	164.0	82.0	146281.8	11995104.0	0.766E+10
Wearplate	300.0	30.0	179.0	9000.0	1611000.1	0.145E+09
Web	32.0	1006.0	697.0	32192.0	22437824.0	0.764E+10
BasePlate	420.0	55.0	1227.5	23100.0	28355246.0	0.196E+11
Totals	...	...	...	210573.7	64399172.0	0.351E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 64399172 / 210574$$

$$= 305.827 \text{ mm}$$



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Saddle Calcs: Test Case:

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Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot -2327 \\
 &= -473.5508 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = -0.0074 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 185.9966 \text{ MPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 2133 - 878 \cdot \sin(120/2) / 1.05 \\
 &= 1406.900 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = -666508.6250 \text{ N-mm}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = -0.0058 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 206.6629 \text{ MPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(-2327 + 18877) \cdot 420) / (4 \cdot 2616 \cdot 207)} \\
 &= 3.105 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30) \\
 &= 1856.758 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1857 / (5 - 1) \\
 &= 464.190 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 464 \cdot 420 / 2 \\
 &= 97479.805 \text{ mm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 16550 / (2616 \cdot 420) \\
 &= 0.015 \text{ N/mm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 97480 \cdot 0.015 \\
 &= 1468.369 \text{ N}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 8192 + 7427 \\
 &= 15619.032 \text{ mm}^2
 \end{aligned}$$



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Saddle Calcs: Test Case:

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Compressive Stress [Sc]:

= P/A_r

= 1468/15619

= 0.094 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:

= Saddle Width - Web Thickness

= 288 - 32

= 256.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/15619

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 288/2

= 144.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{63700928/15619}$

= 63.863 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 2133 - $\cos(120/2)$ (878 + 164 + 30) - 55

= 1542.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199955/310}$

= 112.838

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1542/63.9

= 24.146

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

= 2069 / (2 • 2616) • 464 • 1542/2

= 141619.609 N-mm

Compressive Allowable, $KL/r < C_c$ (24.1 < 113) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (24.1)^2 / (2 \cdot 113^2)) 310 /$

$(5/3 + 3 \cdot (24.1) / (8 \cdot 113) - (24.1^3) / (8 \cdot 113^3))$

= 173.5 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (Rm \cdot C1 / I) / Sba$

= 0.094/174 + (141620 • 144/63700992)/207

= 0.002



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Saddle Calcs: Test Case:

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Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288/2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2133 - \cos(1072/2)(55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(64881092/23046)}$$

$$= 53.059 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1006/53.1$$

$$= 18.960$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2069 / (2 \cdot 2616)$$

$$= 0.396 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.4 \cdot 464 \cdot 1006$$

$$= 184785.125 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 184785/450564$$

$$= 0.410 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($19 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 113^2)) 310 / (5/3 + 3 \cdot (19) / (8 \cdot 113) - (19)^3 / (8 \cdot 113^3))$$

$$= 176.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 0.11/177 + (184785 \cdot 144/64881092) / 207$$

$$= 0.003$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7



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Saddle Calcs: Test Case:

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Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1613.9999 mm2
BasePlate Allowable Stress	S	148.00 MPa
Area Available in a Single Bolt	BlArea	1407.6318 mm2
Saddle Load Q0 (Weight)	Q0	4445.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	817.5 N
Maximum Transverse Force	Ft	3512.4 N
Maximum Longitudinal Force	Fl	2069.5 N
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2069 / (1408 \cdot 8)$$

$$= 0.2 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3512 / (1408 \cdot 8)$$

$$= 0.3 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolls})$$

$$= 2069 / (103 \cdot 8)$$

$$= 2.5069 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2133 \cdot 3512 + 0$$

$$= 7495021.50 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 7495021 / 4445$$

$$= 1685.33 \text{ mm} > Bplen / 6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 2616 / 2 - 70$$

$$= 1238.00 \text{ mm}$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (1685 - 0.5 \cdot 2616)$$

$$= 1131.98 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$= 6 \cdot 1 \cdot 5631 / 420 \cdot (1238 + 1685)$$

$$= 235140.73 \text{ mm}^2$$

K3 = -K2 • (0.5 • Bplen + f)

$$= -235141 \cdot (0.5 \cdot 2616 + 1238)$$

$$= -598667721.11 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y3 + K1 \cdot Y2 + K2 \cdot Y + K3 = 0$$

$$Y3 + 1132 \cdot Y2 + 235141 \cdot Y + -598667712 = 0$$

$$Y = 533.15 \text{ mm}$$



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Saddle Calcs: Test Case: Step: 24 4:12pm Apr 10,2026

$$\begin{aligned}\text{Num} &= (\text{Bplen} / 2 - Y / 3 - e) \\ &= (2616/2 - 533/3 - 1685) \\ &= -555.04\end{aligned}$$

$$\begin{aligned}\text{Denom} &= (\text{Bplen} / 2 - Y / 3 + f) \\ &= (2616/2 - 533/3 + 1238) \\ &= 2368.28\end{aligned}$$

$$\begin{aligned}\text{Total Bolt Tension Force [Tforce]:} \\ &= - \text{QO} \cdot \text{Num} / \text{Denom} \\ &= - 4445 \cdot -555/2368 \\ &= 1041.85 \text{ N}\end{aligned}$$

$$\begin{aligned}\text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= \text{Tforce} / (\text{Stba} \cdot \text{Nbt}) \\ &= 1042 / (172 \cdot 4) \\ &= 1.5145 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0, 2.51, 1.51] \\ &= 2.5069 \text{ mm}^2\end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned}\text{Bearing Pressure (fc)} \\ &= 2(\text{QO} + \text{Tforce}) / (Y \cdot \text{Bpwid}) \\ &= 2(4445 + 1042) / (533 \cdot 420) \\ &= 0.05 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (\text{Bplen} - \text{WebIngh}) / 2 \\ &= (2616 - 2565) / 2 \\ &= 25.4000 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= \text{Tforce} \cdot \text{Adist} \\ &= 1042 \cdot 25.4 \\ &= 26473.69 \text{ N-mm}\end{aligned}$$

$$\begin{aligned}\text{Equivalent Bearing Pressure (f1):} \\ &= \text{fc} \cdot (Y - \text{Adist}) / Y \\ &= 0.049 \cdot (533 - 25.4) / 533 \\ &= 0.05 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (f1 + 2 \cdot \text{fc}) \\ &= (25.4^2 \cdot 420 / 6) \cdot (0.047 + 2 \cdot 0.049) \\ &= 6537.34 \text{ N-mm}\end{aligned}$$

$$\begin{aligned}\text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))} \\ &= \sqrt{(6 \cdot \max(26474, 6537) / (420 \cdot 222))} \\ &= 1.3050 \text{ mm}\end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Ext. Pressure: Step: 25 4:12pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

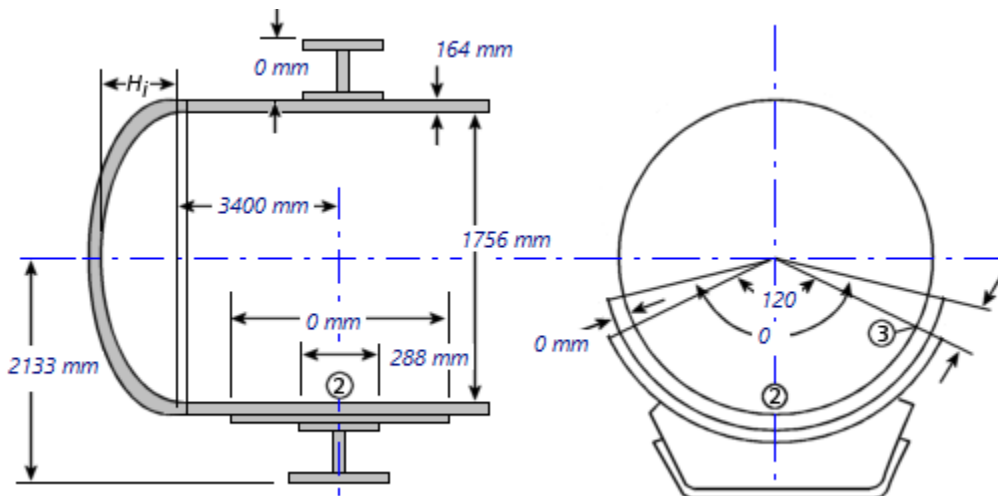
$$= \min(b + 1.56 \sqrt{R_m \cdot t}, 2a)$$

$$= \min(288 + 1.56 \sqrt{961 \cdot 162}, 2 \cdot 3400)$$

$$= 903.5223 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	961.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3400.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		110.72	MPa
Head Allowable Stress used in Calculation		138.20	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	2026858.00	N
Pressure used in Analysis	P	-0.000 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	2.45	110.72
Long. Stress at Bottom of Midspan	-2.45	-201.25



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Saddle Calcs: Ext. Pressure: Step: 25 4:12pm Apr 10, 2026

Long. Stress at Bottom of Midspan	2.45	110.72
Long. Stress at Top of Saddles	40.21	110.72
Long. Stress at Bottom of Saddles	-22.29	-201.25
Long. Stress at Bottom of Saddles	22.29	110.72
Tangential Shear in Shell	5.96	88.58
Circ. Stress at Horn of Saddle	9.59	138.40
Circ. Compressive Stress in Shell	1.05	110.72

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (14050/2 + 0) \cdot 2133/1857$$

$$= 24209.9 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl_{wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(4139, 0) \cdot 2133/5400$$

$$= 1634.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl_{eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(103239, 0) \cdot 2133/5400$$

$$= 40779.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (103239/2 + 0) \cdot 2133/1857$$

$$= 177897.7 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 1848960 + \max(1635, 24210, 40779, 177898)$$

$$= 2026858.0 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.08 \cdot 1.2)^2)$$

$$= 4139.213 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2045735.12	N
Transverse Shear Load	51619.49	N
Longitudinal Shear Load	103238.98	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1756/2 + 164 + 30) \sin(120/2), 2 \cdot 961)$$

$$= 1856.759 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -2026858 \cdot 3400 [1 - (1 - 3400/11164 + (961^2 - 0^2) / (2 \cdot 3400 \cdot 11164)) / (1 + (4 \cdot 0) / (3 \cdot 11164))]$$



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= -2015736320.0 N-mm

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L)$
 = 2026858 * 11164 / 4 * ((1 + 2(961^2 - 0^2) / (11164^2)) / (1 + (4 * 0) / (3 * 11164)) - 4 * 3400 / 11164)
 = -1150989824.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 = -0 * 961 / (2 * 162) - -1150989824 / (pi * 961^2 * 162)
 = 2.45 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$
 = -0 * 961 / (2 * 162) + -1150989824 / (pi * 961^2 * 162)
 = -2.45 MPa

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= $P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$
 = -0 * 961 / (2 * 162) - -2015736320 / (0.11 * pi * 961^2 * 162)
 = 40.21 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= $P \cdot Rm / (2t) + M1 / (K1 * \pi \cdot Rm^2 \cdot t)$
 = -0 * 961 / (2 * 162) + -2015736320 / (0.19 * pi * 961^2 * 162)
 = -22.29 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a) / (L + (4 \cdot hm / 3))$
 = 2026858 (11164 - 2 * 3400) / (11164 + (4 * 0 / 3))
 = 792297.3 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= $K2 \cdot T / (Rm \cdot t)$
 = 1.17 * 792297 / (961 * 162)
 = 5.96 MPa

Decay Length (4.15.20) [x1,x2]:

= $0.78 \cdot \sqrt{(Rm \cdot t)}$
 = $0.78 \cdot \sqrt{(961 \cdot 162)}$
 = 307.761 mm

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

= $-K5 \cdot Q \cdot k / (t (b + X1 + X2))$
 = -0.76 * 2026858 * 0.1 / (162 (288 + 308 + 308))
 = -1.05 MPa

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

= $-Q / (4 \cdot t (b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$
 = -2026858 / (4 * 162 (288 + 308 + 308)) -
 3 * 0.053 * 2026858 / (2 * 162^2)
 = -9.59 MPa

Effective reinforcing plate width (4.15.24) [B1]:

= $\min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$
 = $\min(288 + 1.56 \cdot \sqrt{(961 \cdot 162)}, 2 \cdot 3400)$
 = 903.52 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	32.0000	mm



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Web Thickness	Webtk	32.0000 mm
Web Location	Webloc	Center
Saddle Design Temperature		480.0 C
Saddle Yield Stress	Sy	223.6 MPa
Height of Web at Center	Hw,c	1085.0 mm
Friction Coefficient	mu	0.400

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(880 \cdot 162)} + 300)162^3)/12$$

$$= 314971008.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 314971008 + 144020(307 - 81)^2$$

$$= 7667600384.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	889.0	162.0	81.0	144019.8	11665603.0	0.767E+10
Wearplate	300.0	30.0	177.0	9000.0	1593000.1	0.153E+09
Web	32.0	1006.0	695.0	32192.0	22373444.0	0.756E+10
BasePlate	420.0	55.0	1225.5	23100.0	28309048.0	0.195E+11
Totals	...	...	...	208311.8	63941092.0	0.349E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 63941092/208312$$

$$= 306.949 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2026858$$

$$= 412509.5312 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 6.4167 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 134.1643 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 2133 - 880 \cdot \sin(120/2) / 1.05$$

$$= 1405.246 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 579912384.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.1018 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 149.0714 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2026858 + 18877)420 / (4 \cdot 2616 \cdot 149)}$$

$$= 40.652 \text{ mm}$$



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Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30)$$

$$= 1856.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1857 / (5 - 1)$$

$$= 464.190 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 464 \cdot 420 / 2$$

$$= 97479.805 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2045735 / (2616 \cdot 420)$$

$$= 1.862 \text{ N/mm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 97480 \cdot 1.86$$

$$= 181500.172 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8192 + 7427$$

$$= 15619.032 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 181500 / 15619$$

$$= 11.621 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 288 - 32$$

$$= 256.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 15619$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$

$$= 144.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(63700928 / 15619)}$$

$$= 63.863 \text{ mm}$$



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Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2133 - \cos(120/2) (878 + 164 + 30) - 55$$

$$= 1542.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 176708 / 224)}$$

$$= 124.897$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1542 / 63.9$$

$$= 24.146$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 103239 / (2 \cdot 2616) \cdot 464 \cdot 1542 / 2$$

$$= 7064831.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($24.1 < 125$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (24.1)^2 / (2 \cdot 125^2)) 224 / (5/3 + 3 \cdot (24.1) / (8 \cdot 125) - (24.1)^3 / (8 \cdot 125^3))$$

$$= 126.2 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 11.6 / 126 + (7064831 \cdot 144 / 63700992) / 149$$

$$= 0.199$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2133 - \cos(1072/2) (55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(64881092 / 23046)}$$

$$= 53.059 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1006 / 53.1$$

$$= 18.960$$



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Unit Force [Force,u]:
 $= F_l / (2 \cdot \text{Baseplate Length})$
 $= 103239 / (2 \cdot 2616)$
 $= 19.732 \text{ N/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 19.7 \cdot 464 \cdot 1006$
 $= 9218185.000 \text{ N-mm}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 9218185 / 450564$
 $= 20.453 \text{ MPa}$

Compressive Allowable, $KL/r < C_c$ ($19 < 125$) per AISC E2-1 [Sca]:
 $= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (19)^2 / (2 \cdot 125^2)) 224 /$
 $(5/3 + 3 \cdot (19) / (8 \cdot 125) - (19^3) / (8 \cdot 125^3))$
 $= 128.3 \text{ MPa}$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 13.7 / 128 + (9218185 \cdot 144 / 64881092) / 149$
 $= 0.244$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(288 + 1.56 \cdot \sqrt{961 \cdot 162}, 2 \cdot 2300)$
 $= 903.5223 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	961.00	mm
Shell Thickness used in this Case	t	164.000	mm
Stiffened Vessel Length per 4.15.6	L	11164.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2300.00	mm
Saddle Width	b1 or b	288.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		110.72	MPa
Head Allowable Stress used in Calculation		0.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2133.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 233338.73 N
 Pressure used in Analysis P -0.000 MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top	of Midspan	-0.26	-201.25



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Long. Stress at Top of Midspan	0.26	110.72
Long. Stress at Bottom of Midspan	0.26	110.72
Long. Stress at Top of Saddles	2.01	110.72
Long. Stress at Bottom of Saddles	-1.12	-201.25
Long. Stress at Bottom of Saddles	1.12	110.72
Tangential Shear in Shell	1.03	88.58
Circ. Stress at Horn of Saddle	1.10	138.40
Circ. Compressive Stress in Shell	0.12	110.72

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (14050/2 + 0) \cdot 2133/1857$$

$$= 24209.9 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4139, 0) \cdot 2133/5400$$

$$= 1634.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(103239, 0) \cdot 2133/5400$$

$$= 40779.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (103239/2 + 0) \cdot 2133/1857$$

$$= 177897.7 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 55441 + \max(1635, 24210, 40779, 177898)$$

$$= 233338.7 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.08 \cdot 1.2)^2)$$

$$= 4139.213 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	252215.88	N
Transverse Shear Load	51619.49	N
Longitudinal Shear Load	103238.98	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1756/2 + 164 + 30)\sin(120/2), 2 \cdot 961)$$

$$= 1856.759 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -233339 \cdot 2300 [1 - (1 - 2300/11164 + (961^2 - 0^2)/$$



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$$\begin{aligned} & (2 \cdot 2300 \cdot 11164) / ((1 + (4 \cdot 0) / (3 \cdot 11164))) \\ & = -100955960.0 \text{ N-mm} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} & = Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ & = 233339 \cdot 11164 / 4 \cdot ((1 + 2(961^2 - 0^2) / (11164^2)) / (1 + (4 \cdot 0) / (3 \cdot 11164)) - 4 \cdot 2300 / 11164) \\ & = 124270920.0 \text{ N-mm} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} & = P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\ & = -0 \cdot 961 / (2 \cdot 162) - 124270920 / (\pi \cdot 961^2 \cdot 162) \\ & = -0.26 \text{ MPa} \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} & = P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ & = -0 \cdot 961 / (2 \cdot 162) + 124270920 / (\pi \cdot 961^2 \cdot 162) \\ & = 0.26 \text{ MPa} \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned} & = P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\ & = -0 \cdot 961 / (2 \cdot 162) - 100955960 / (0.11 \cdot \pi \cdot 961^2 \cdot 162) \\ & = 2.01 \text{ MPa} \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned} & = P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ & = -0 \cdot 961 / (2 \cdot 162) + 100955960 / (0.19 \cdot \pi \cdot 961^2 \cdot 162) \\ & = -1.12 \text{ MPa} \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} & = Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ & = 233339 \cdot (11164 - 2 \cdot 2300) / (11164 + (4 \cdot 0 / 3)) \\ & = 137194.1 \text{ N} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned} & = K2 \cdot T / (Rm \cdot t) \\ & = 1.17 \cdot 137194 / (961 \cdot 162) \\ & = 1.03 \text{ MPa} \end{aligned}$$

Decay Length (4.15.20) [x1,x2]:

$$\begin{aligned} & = 0.78 \cdot \sqrt{(Rm \cdot t)} \\ & = 0.78 \cdot \sqrt{(961 \cdot 162)} \\ & = 307.761 \text{ mm} \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned} & = -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\ & = -0.76 \cdot 233339 \cdot 0.1 / (162(288 + 308 + 308)) \\ & = -0.12 \text{ MPa} \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned} & = -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\ & = -233339 / (4 \cdot 162(288 + 308 + 308)) - 3 \cdot 0.053 \cdot 233339 / (2 \cdot 162^2) \\ & = -1.10 \text{ MPa} \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} & = \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ & = \min(288 + 1.56 \cdot \sqrt{(961 \cdot 162)}, 2 \cdot 2300) \\ & = 903.52 \text{ mm} \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2616.0000	mm
Baseplate Thickness	Bpthk	55.0000	mm
Baseplate Width	Bpwid	420.0000	mm



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Saddle Calcs: Ext. Pressure: Step: 25 4:12pm Apr 10,2026

Number of Ribs (inc. outside ribs)	Nribs	5
Rib Thickness	Ribtk	32.0000 mm
Web Thickness	Webtk	32.0000 mm
Web Location	Webloc	Center
Saddle Design Temperature		21.1 C
Saddle Yield Stress	Sy	223.6 MPa
Height of Web at Center	Hw,c	1085.0 mm
Friction Coefficient	mu	0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(880 \cdot 162)} + 300)162^3)/12$$

$$= 314971008.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 314971008 + 144020(307 - 81)^2$$

$$= 7667600384.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	889.0	162.0	81.0	144019.8	11665603.0	0.767E+10
Wearplate	300.0	30.0	177.0	9000.0	1593000.1	0.153E+09
Web	32.0	1006.0	695.0	32192.0	22373444.0	0.756E+10
BasePlate	420.0	55.0	1225.5	23100.0	28309048.0	0.195E+11
Totals	...	...	...	208311.8	63941092.0	0.349E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 63941092/208312$$

$$= 306.949 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 233339$$

$$= 47489.4922 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.7387 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 134.1643 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 2133 - 880 \cdot \sin(120/2) / 1.05$$

$$= 1405.246 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 66761476.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.5873 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 149.0714 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$



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Saddle Calcs: Ext. Pressure: Step: 25 4:12pm Apr 10,2026

$$= \sqrt{3(233339 + 18877)420 / (4 \cdot 2616 \cdot 149)}$$

$$= 14.274 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (878 + 164 + 30)$$

$$= 1856.758 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1857 / (5 - 1)$$

$$= 464.190 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 464 \cdot 420 / 2$$

$$= 97479.805 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 252216 / (2616 \cdot 420)$$

$$= 0.230 \text{ N/mm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 97480 \cdot 0.23$$

$$= 22376.908 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 8192 + 7427$$

$$= 15619.032 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 22377 / 15619$$

$$= 1.433 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	32.0	288.0	...	9216.0	...	0.637E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 288 - 32$$

$$= 256.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 15619$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288 / 2$$

$$= 144.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$



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Saddle Calcs: Ext. Pressure: Step: 25 4:12pm Apr 10, 2026

$$= \sqrt{(63700928/15619)}$$

$$= 63.863 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2133 - \cos(120/2) (878 + 164 + 30) - 55$$

$$= 1542.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/224)}$$

$$= 132.858$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1542/63.9$$

$$= 24.146$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 103239 / (2 \cdot 2616) \cdot 464 \cdot 1542/2$$

$$= 7064831.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($24.1 < 133$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (24.1)^2 / (2 \cdot 133^2)) 224 /$$

$$(5/3 + 3 \cdot (24.1) / (8 \cdot 133) - (24.1)^3 / (8 \cdot 133^3))$$

$$= 126.8 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.43/127 + (7064831 \cdot 144/63700992) / 149$$

$$= 0.118$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	32.0	256.0	0.0	8192.0	0.0	0.636E+08
Web	464.2	32.0	0.0	14854.1	0.0	0.127E+07
Totals	...	...	...	23046.1	...	0.649E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/23046$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 288/2$$

$$= 144.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2133 - \cos(1072/2) (55 + 0 + 0) - 0$$

$$= 1006.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(64881092/23046)}$$

$$= 53.059 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



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Saddle Calcs: Ext. Pressure: Step: 25 4:12pm Apr 10, 2026

$$= 1 \cdot 1006/53.1$$

$$= 18.960$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 103239 / (2 \cdot 2616)$$

$$= 19.732 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 19.7 \cdot 464 \cdot 1006$$

$$= 9218185.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9218185/450564$$

$$= 20.453 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($19 < 133$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19)^2 / (2 \cdot 133^2)) 224 / (5/3 + 3 \cdot (19) / (8 \cdot 133) - (19^3) / (8 \cdot 133^3))$$

$$= 128.7 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 1.69/129 + (9218185 \cdot 144/64881092) / 149$$

$$= 0.150$$

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Section: Cone-1

Cone: 13 4:12pm Apr 10,2026

Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

Cone-1

Design Internal Pressure, Small End	18.001	MPa
Design Internal Pressure, Large End	18.001	MPa
Temperature for Internal Pressure	480.0	C
Design External Pressure	0.103	MPa
Temperature for External Pressure	-0.0	C

CONE Properties:

Cone Material		SA-336 F22
Cone Allowable Stress at Temperature	Ss	110.72 MPa
Cone Allowable Stress At Ambient		148.00 MPa
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00
Circumferential Joint Efficiency of Cone		1.00
Actual Thickness of Cone	Tc	170.0000 mm
Corrosion Allowance for Cone	Cac	2.0000 mm
Diameter Basis for Cone and Cylinders		ID
Diameter of Small End of Cone	Ds	1100.000 mm
Diameter of Large End of Cone	Dl	1756.000 mm
Half Apex Angle for Cone		25.75 degrees
Axial Length of Cone	Lc	800.000 mm

SMALL-END Properties:

Attached Small End Material Specification		SA-336 F22
Operating Allowable Stress	Ss	110.72 MPa
Ambient Allowable Stress		148.00 MPa
Longitudinal Joint Efficiency	Es,E1	1.00
Actual Thickness	Ts	170.0000 mm
Corrosion Allowance	Cas	2.0000 mm
Axial Length of Attached Element	Ls	500.000 mm

The element type attached to the small end of the cone was not a cylinder. Therefore, the material properties printed above reflect those of the cone.

LARGE-END Properties:

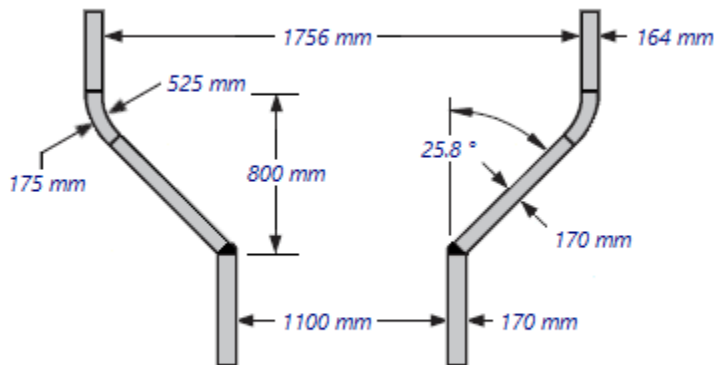
Attached Large End Material Specification		SA-336 F22
Operating Allowable Stress	Sl	110.72 MPa
Ambient Allowable Stress		148.00 MPa
Longitudinal Joint Efficiency	El,E1	1.00
Actual Thickness	Tl	164.0000 mm
Corrosion Allowance	Cal	2.0000 mm
Axial Length of Attached Element	Ll	8800.000 mm

Type of Reinforcement at Large End of Cone: Knuckle

Large End Reinforcing/Knuckle Material		None
Large Reinforcing/Knuckle Allowable, Operating		110.72 MPa
Large Reinforcing/Knuckle Allowable, Ambient		148.00 MPa
Bend Radius for Knuckle at Large End	Rblk	525.0000 mm
Thickness for Knuckle at Large End	Rtlk	175.0000 mm

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange	No
Force Arl & Ars calculation, cone -> cylinder	No



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 0 C

Elastic Modulus of Cone Material	0.212E+06 MPa at 0 C
Elastic Modulus of Small Cylinder Material	0.212E+06 MPa at 0 C
Elastic Modulus of Large Cylinder Material	0.212E+06 MPa at 0 C
Elastic Modulus of Small End Reinforcement	0.212E+06 MPa at 0 C

Elastic Modulus Data from ASME Section II Part D at 480 C

Elastic Modulus of Cone Material	0.177E+06 MPa at 479 C
Elastic Modulus of Small Cylinder Material	0.177E+06 MPa at 479 C
Elastic Modulus of Large Cylinder Material	0.177E+06 MPa at 479 C
Elastic Modulus of Small End Reinforcement	0.177E+06 MPa at 479 C

Axial forces and moments are not computed for Horizontal geometries.
Compute them manually and enter the loads using the Force/Moment dialog.
Place the loads at the ends of the cone as needed.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	25.4000 mm
Maximum Centroid Reinforcement Distance Small End	25.4000 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Performing discontinuity stress calculation for cone to cone junction (top or right end).

Junction Calculations per Code Case 2286-1 Section 4.2

Calculations for the Large End Cone/Shell Junction - With Knuckle

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure MPa	Reason for Failure at this Pressure
MAWP	18.238	Thickness due to internal pressure, Cone Large End
MAPnc	24.693	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.



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Conical Section: Cone-1

Cone: 13 4:12pm Apr 10, 2026

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Conical Section: Cone-2

Cone: 14 4:12pm Apr 10,2026

Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

Cone-2

Design Internal Pressure, Small End	18.000	MPa
Design Internal Pressure, Large End	18.000	MPa
Temperature for Internal Pressure	480.0	C
Design External Pressure	0.103	MPa
Temperature for External Pressure	-0.0	C

CONE Properties:

Cone Material	SA-336 F22	
Cone Allowable Stress at Temperature	Ss	110.72 MPa
Cone Allowable Stress At Ambient		148.00 MPa
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00
Circumferential Joint Efficiency of Cone		1.00
Actual Thickness of Cone	Tc	106.0000 mm
Corrosion Allowance for Cone	Cac	2.0000 mm
Diameter Basis for Cone and Cylinders	ID	
Diameter of Small End of Cone	Ds	914.400 mm
Diameter of Large End of Cone	Dl	1100.000 mm
Half Apex Angle for Cone		10.51 degrees
Axial Length of Cone	Lc	500.000 mm

SMALL-END Properties:

Attached Small End Material Specification	SA-336 F22	
Operating Allowable Stress	Ss	110.72 MPa
Ambient Allowable Stress		148.00 MPa
Longitudinal Joint Efficiency	Es,E1	1.00
Actual Thickness	Ts	88.0000 mm
Corrosion Allowance	Cas	2.0000 mm
Axial Length of Attached Element	Ls	400.000 mm

LARGE-END Properties:

Attached Large End Material Specification	SA-336 F22	
Operating Allowable Stress	Sl	110.72 MPa
Ambient Allowable Stress		148.00 MPa
Longitudinal Joint Efficiency	El,E1	1.00
Actual Thickness	Tl	106.0000 mm
Corrosion Allowance	Cal	2.0000 mm
Axial Length of Attached Element	Ll	800.000 mm

The element type attached to the Large end of the cone was not a cylinder. Therefore, the material properties printed above reflect those of the cone.

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange Yes

Force Arl & Ars calculation, cone -> cylinder Yes



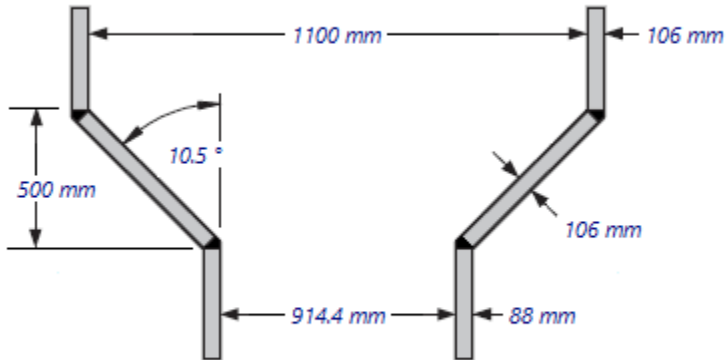
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Conical Section: Cone-2

Cone: 14 4:12pm Apr 10,2026

**Cone Geometry****Calculated Values:**

Elastic Modulus Data from ASME Section II Part D at 0 C

Elastic Modulus of Cone Material	0.212E+06 MPa at 0 C
Elastic Modulus of Small Cylinder Material	0.212E+06 MPa at 0 C
Elastic Modulus of Large Cylinder Material	0.212E+06 MPa at 0 C
Elastic Modulus of Large End Reinforcement	0.212E+06 MPa at 0 C
Elastic Modulus of Small End Reinforcement	0.212E+06 MPa at 0 C

Elastic Modulus Data from ASME Section II Part D at 480 C

Elastic Modulus of Cone Material	0.177E+06 MPa at 479 C
Elastic Modulus of Small Cylinder Material	0.177E+06 MPa at 479 C
Elastic Modulus of Large Cylinder Material	0.177E+06 MPa at 479 C
Elastic Modulus of Large End Reinforcement	0.177E+06 MPa at 479 C
Elastic Modulus of Small End Reinforcement	0.177E+06 MPa at 479 C

Axial forces and moments are not computed for Horizontal geometries.
Compute them manually and enter the loads using the Force/Moment dialog.
Place the loads at the ends of the cone as needed.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	25.4000 mm
Maximum Centroid Reinforcement Distance Small End	25.4000 mm

Note: No ring was found close enough to the large end to be considered.**Note:** No ring was found close enough to the small end to be considered.**Performing discontinuity stress calculation for cone to cone junction (left or bottom end).**

Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.



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Conical Section: Cone-2

Cone: 14 4:12pm Apr 10, 2026

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	93.74	447.21	Small Cyl. Long.
Compres. Stress	0.00	-447.21	Small Cyl. Long. *
Membrane Stress	124.34	-166.08	Small End Tang.
Tensile Stress	73.20	447.21	Cone Longitudinal
Compres. Stress	0.00	-447.21	Cone Longitudinal *
Compres Stress	109.31	-166.08	Cone Tangential

Calculations for the Small End Cone/Shell Junction

Ch per 4.4.13.2 [Ch]:

$$= 0.55 \cdot t / D$$

$$= 0.55 \cdot 104 / 1090$$

$$= 0.051577$$

Hoop Stress per 4.4.133 [fh]:

$$= P \cdot D_o / (2 \cdot t_c \cdot \cos(\alpha))$$

$$= 0.1 \cdot 1090 / (2 \cdot 104 \cdot 0.98)$$

$$= 0.551 \text{ MPa}$$

Fhe per 4.1.3 or 4.4.19 [Fhe]:

$$= 1.6 \cdot Ch \cdot E \cdot t_c / D$$

$$= 1.6 \cdot 0.052 \cdot 212220 \cdot 104 / 1090$$

$$= 1670.4 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [224] \geq F_y [224], \text{ FS} = 1.667$$

Available Moment of Inertia, Small End, External Pressure:

	Area	Centroid	Ar*Ce	Dist	I	Ar*Di ²
Shl	0.145E+05	0.000E+00	0.000E+00	0.981E+01	0.893E+07	0.139E+07
Con	0.193E+05	0.172E+02	0.331E+06	-0.738E+01	0.202E+08	0.105E+07
Sec	0.000E+00	0.430E+02	0.000E+00	-0.332E+02	0.000E+00	0.000E+00
TOT	0.337E+05		0.331E+06		0.291E+08	0.244E+07

Centroid of Section 0.981E+01 Moment of Inertia 0.316E+08

$$f_h = 0.551 \text{ MPa per 4.1.3}$$

$$F_{ha} = 134.137 \text{ MPa per 4.1.3}$$

Since $f_h < F_{ha}$, a junction ring is not required on this end due to external pressure buckling per 4.1.3 or 4.4.13, but may be required for Uniform Axial Compression and Bending.

Localized Bending Stress in the Cylinder, 4.2.2 [f_b]:

$$= 0.6 \cdot t \cdot \sqrt{(D \cdot (t + t_c))} \cdot (f_x + f_b) \cdot \tan(\alpha) / t^2$$

$$= 0.6 \cdot 86 \cdot \sqrt{(1090 \cdot (86 + 104))} \cdot (0.36 + 0) \cdot 0.19 / 86^2$$

$$= 0.210 \text{ MPa per eq 4-2}$$

Localized Bending Stress in the Cone, 4.2.2 [f_{bc}]:

$$= 0.6 \cdot t \cdot \sqrt{(D \cdot (t + t_c))} \cdot (f_x + f_b) \cdot \tan(\alpha) / t_c^2$$

$$= 0.6 \cdot 86 \cdot \sqrt{(1090 \cdot (86 + 104))} \cdot (0.36 + 0) \cdot 0.19 / 104^2$$

$$= 0.143 \text{ MPa per eq 4-2}$$

Tensile Hoop Stress from the unbalanced Radial Line Load [f_{ht}]:

$$= 0.45 \cdot \sqrt{(D/t)} \cdot (f_x + f_b) \cdot \tan(\alpha) \text{ per eq 4-3}$$

$$= 0.45 \cdot \sqrt{(1090/86)} \cdot (0.36 + 0) \cdot 0.19$$

$$= 0.106 \text{ MPa}$$

Compare Localized Stresses against defined Allowables



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Conical Section: Cone-2

Cone: 14 4:12pm Apr 10, 2026

2: $f_x + f_b + f'b = [\text{Total}]$ must be $\leq 3 * S$
 2: $0.36 + 0 + 0.21 = [0.57]$ must be ≤ 332 MPa

3: $f_x + f_b = [\text{Total}]$ must be $\leq S$
 3: $0.36 + 0 = [0.36]$ must be ≤ 111 MPa

4: f'_{ht} must be $\leq 1.5 * S * E$
 4: 0.11 MPa must be ≤ 166 MPa

Reqd Moment of Inertia of the Junction for Axial Compression and Bending [I_r]:

$= t \cdot D \cdot (Dc)^2 / (8 \cdot E) \cdot (f_x + f_b) \cdot \tan(\alpha)$ per 4.2.3 eq(4-5)
 $= 86 \cdot 1090 \cdot (1090)^2 / (8 \cdot 212220) \cdot (0.36 + 0) \cdot 0.19$
 $= 4337.998 \text{ mm}^4$

Required Area of the Junction [A_r]:

$= t \cdot D / F_y \cdot (f_x + f_b) \cdot \tan(\alpha)$
 $= 86 \cdot 1090 / 224 \cdot (0.36 + 0) \cdot 0.19$
 $= 27.702 \text{ mm}^2$

Summary of Results for Cone/Shell Junction per Code Case 2286

Area of reinforcement required per 4.2.3 Comp + Bend	27.70 mm ²
Area of reinforcement in the Cone/Shell/Ring Junction	33746.68 mm ²
Reqd Inertia of the Cone/Shell/Ring Junction per 4.1.3	0.00 mm ⁴
Reqd Inertia of the Cone/Shell/Ring Junction per 4.2.3	4338.00 mm ⁴
Available Moment of Inertia of Junction	31565294.00 mm ⁴



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	80.75	447.21	Small Cyl. Long.
Compres. Stress	0.00	-447.21	Small Cyl. Long. *
Membrane Stress	118.28	-166.08	Small End Tang.
Tensile Stress	63.48	447.21	Cone Longitudinal
Compres. Stress	0.00	-447.21	Cone Longitudinal *
Compres Stress	101.58	-166.08	Cone Tangential

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure MPa	Reason for Failure at this Pressure
MAWP	18.458	Thickness due to internal pressure, Cone Large End
MAPnc	25.182	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Nozzle Calcs.: Manhole

Nozl: 13 4:12pm Apr 10,2026

Input, Nozzle Desc: Manhole

From: 10

Pressure for Reinforcement Calculations	P	14.7041	MPa
Temperature for Internal Pressure	Temp	343	C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	129.12	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	1765.00	mm
Head Finished (Minimum) Thickness	t	54.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-12.00	C
---	--	--------	---

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22 +IT	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	139.28	MPa
Allowable Stress At Ambient	Sna	148.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	50.6000	mm

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		56.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



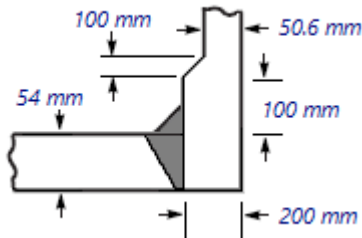
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Nozzle Calcs.: Manhole

Nozl: 13 4:12pm Apr 10, 2026



Reinforcement CALCULATION, Description: Manhole

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.992 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 14.7 \cdot 886 / (2 \cdot 129 \cdot 1 - 0.2 \cdot 14.7)$
 $= 51.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.7 \cdot 308 / (139 \cdot 1 - 0.6 \cdot 14.7)$
 $= 34.6931 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.7201 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 149 / 0.5773$
 $= 258.7685 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001	mm
Parallel to Vessel Wall, opening length	d	615.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	127.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 139 / 129)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 139 / 129)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$



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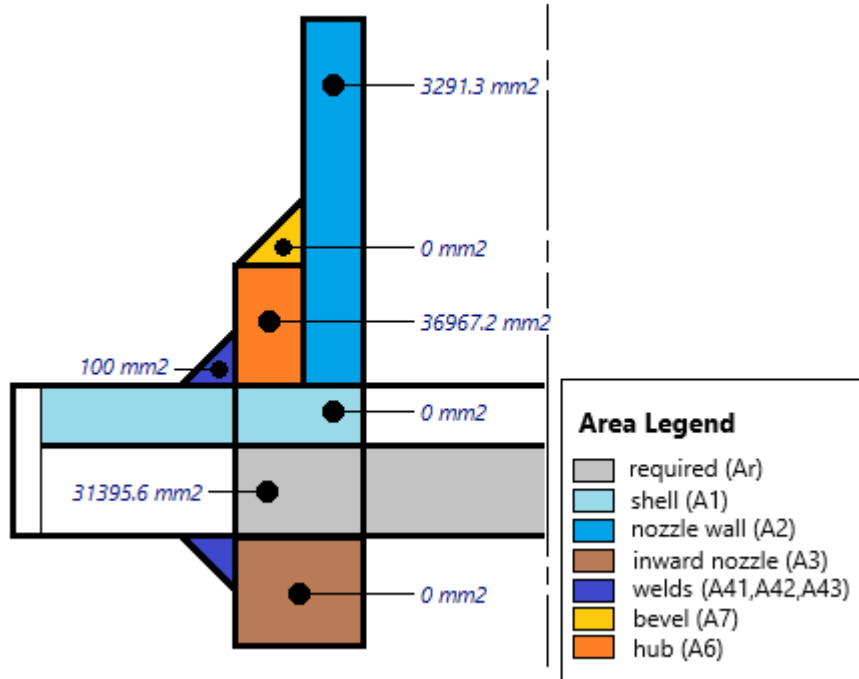
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Nozzle Calcs.: Manhole

Nozl: 13 4:12pm Apr 10,2026

= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	31395.602	769.742	NA
Area in Shell	A1	0.000	29856.115	NA
Area in Nozzle Wall	A2	3291.265	11699.368	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	36967.168	36967.168	NA
TOTAL AREA AVAILABLE	Atot	40358.434	78622.648	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (616 \cdot 51 \cdot 1 + 2 \cdot 197 \cdot 51 \cdot 1 \cdot (1 - 1))$
 $= 31395.602 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 616(1 \cdot 51 - 1 \cdot 51) - 2 \cdot 197$
 $(1 \cdot 51 - 1 \cdot 51) \cdot (1 - 1)$
 $= 0.000 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$



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Nozzle Calcs.: Manhole

Nozl: 13 4:12pm Apr 10, 2026

$$= (2 \cdot 127)(47.6 - 34.7)1$$

$$= 3291.265 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_n / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(127, 350, 100)) \cdot (200 - 50.6) \cdot 1$$

$$= 36967.168 \text{ mm}^2 \text{ Includes Bevel Area of } 7087.168 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 37.6931 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 54.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 54.0000 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3532 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(54, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(37.7, 11.3)$$

$$= 37.6931 \text{ mm}$$

Available Nozzle Neck Thickness = 50.6000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 50.6, tr = 32.5, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.68, Temp. Reduction = 18 C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 C
Min Metal Temp. at Required thickness (UCS 66.1)	19 C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 54, tr = 47.7, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.93, Temp. Reduction = 4 C

Min Metal Temp. w/o impact per UCS-66, Curve B	18 C
Min Metal Temp. at Required thickness (UCS 66.1)	15 C

Gov. MDMT of the nozzle to shell joint welded assembly : 19 C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 14.70/25.51 = 0.576$$

Weld Size Calculations, Description: Manhole

Intermediate Calc. for nozzle/shell welds	Tmin	19.0000 mm
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Nozzle Calcs.: Manhole

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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (31396 - 0 + 2 \cdot 197 \cdot 1 \cdot \\
 &\quad (1 \cdot 51 - 51)) \cdot 129) \\
 &= 4053526.75 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (40258 + 100 - 0 \cdot 1) \cdot 129 \\
 &= 5210730.00 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (40258 + 0 + 100 + (20094)) \cdot 129 \\
 &= 7805092.50 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (40258 + 0 + 100 + 0 + (20094)) \cdot 129 \\
 &= 7805092.50 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 1010 \cdot 10 \cdot 0.49 \cdot 129 \\
 &= 1003297. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 406) \cdot (200 - 3) \cdot 0.7 \cdot 139 \\
 &= 24513682. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 1010 \cdot (54 - 3) \cdot 0.74 \cdot 139 \\
 &= 8335251. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (1003297 + 24513682) = 25516980 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (1003297 + 0 + 8335251 + 0) = 9338548 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (1003297 + 8335251 + 0) = 9338548 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:



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Nozzle Calcs.: Manhole

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Path 1-1 = 25516980N , must exceed W = 4053526 N or W1 = 5210730 N

Path 2-2 = 9338548 N , must exceed W = 4053526 N or W2 = 7805092 N

Path 3-3 = 9338548 N , must exceed W = 4053526 N or W3 = 7805092 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.7 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 158.6331 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 562.6331 mm

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Nozzle Calcs.: N1- Tube inlet

Nozl: 14 4:12pm Apr 10,2026

Input, Nozzle Desc: N1- Tube inlet

From: 20

Pressure for Reinforcement Calculations	P	14.7214	MPa
Temperature for Internal Pressure	Temp	343	C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	129.12	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1756.00	mm
Design Length of Section	L	2406.1667	mm
Shell Finished (Minimum) Thickness	t	112.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1012.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	C
---	--	--------	---

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	123.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		16.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	105.0000	mm
Height of Beveled Transition	L'	105.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		210.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	750.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	64.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



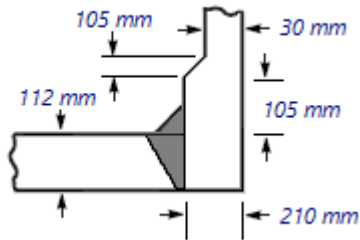
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Nozzle Calcs.: N1- Tube inlet

Nozl: 14 4:12pm Apr 10, 2026



Reinforcement CALCULATION, Description: N1- Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	1.181 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.7 \cdot 881 / (129 \cdot 1 - 0.6 \cdot 14.7)$
 $= 107.8192 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.7 \cdot 206 / (123 \cdot 1 - 0.6 \cdot 14.7)$
 $= 26.5886 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8463 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 179 / 0.5773$
 $= 310.3836 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	824.8000 mm
Parallel to Vessel Wall, opening length	d	412.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	272.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 123 / 129)$
 $= 0.953$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 123 / 129)$
 $= 0.953$



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Nozzle Calcs.: N1- Tube inlet

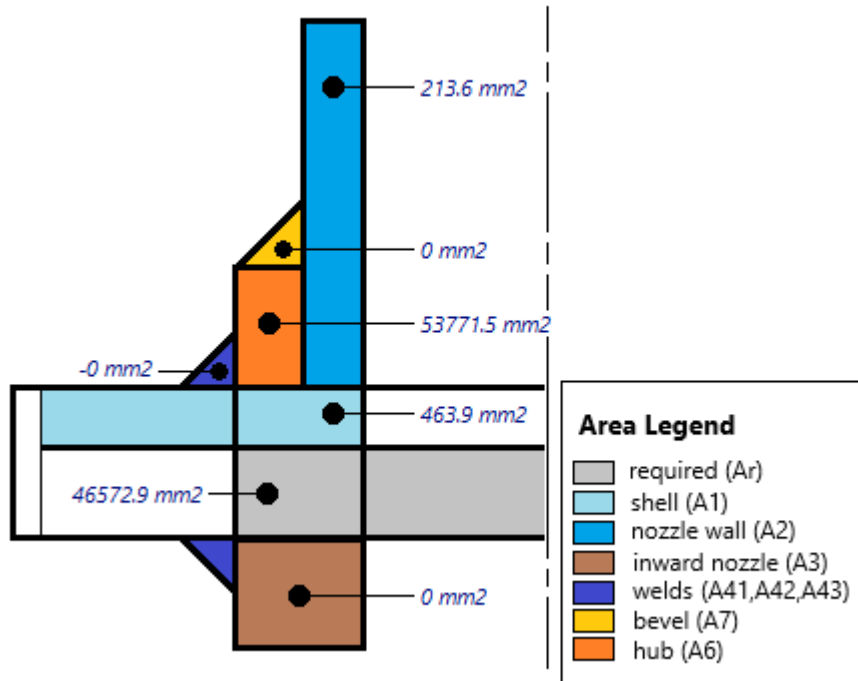
Nozl: 14 4:12pm Apr 10,2026

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.95, 1)

= 0.953



Results of Nozzle Reinforcement Area Calculations: (mm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	46572.902	1410.361	NA
Area in Shell	A1	463.879	40254.898	NA
Area in Nozzle Wall	A2	213.591	13058.795	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		-0.000	-0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	53771.547	53771.547	NA
TOTAL AREA AVAILABLE	Atot	54449.016	107085.234	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (412•108•1+2•206•108•1•(1-0.95))

= 46572.902 mm2

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)

= 412 (1 • 109 - 1 • 108) - 2 • 206

(1 • 109 - 1 • 108) • (1 - 0.95)

= 463.879 mm2



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 4:12pm Apr 10, 2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_r^2$$

$$= (2 \cdot 273) (27 - 26.6) 0.95$$

$$= 213.591 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_r^2 + (W_i - \text{can}/0.707)^2 - \text{Area Lost} \cdot f_r^2$$

$$= (15^2 - 225) \cdot 0.95 + (0^2 - 0) \cdot 0.95$$

$$= -0.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_r^2$$

$$= (2 \cdot \min(273, 750, 105)) \cdot (209 - 30) \cdot 0.95$$

$$= 53771.547 \text{ mm}^2 \text{ Includes Bevel Area of } 17923.848 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 29.5886 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 110.8192 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 110.8192 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.7031 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(111, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(29.6, 11.3)$$

$$= 29.5886 \text{ mm}$$

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.9,	Allowable	: 123.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 257.3 MPa	Passed
Occasional	: 52.8,	Allowable	: 163.6 MPa	Passed
Shear	: 11.0,	Allowable	: 86.1 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:C, Curve:A)

Govrn. thk, tg = 30, tr = 23.5, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.87, Temp. Reduction = 7 C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 C
Min Metal Temp. at Required thickness (UCS 66.1)	16 C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:C, Curve:B)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 16 C



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 Nozzle Calcs.: N1- Tube inlet Nozl: 14 4:12pm Apr 10, 2026

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 14.72/25.51 = 0.577

Weld Size Calculations, Description: N1- Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (46573 - 464 + 2 \cdot 206 \cdot 0.95 \cdot (1 \cdot 109 - 108)) \cdot 129)$
 $= 6013088.00 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_r2) \cdot S_v$
 $= (53985 + -0.00013 - 0 \cdot 0.95) \cdot 129$
 $= 6970091.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (53985 + 0 + -0.00013 + (42820)) \cdot 129$
 $= 12498669.00 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (53985 + 0 + -0.00013 + 0 + (42820)) \cdot 129$
 $= 12498669.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 825 \cdot 15 \cdot 0.49 \cdot 123$
 $= 1171181. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 309) \cdot (209 - 3) \cdot 0.7 \cdot 123$
 $= 17249826. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 825 \cdot (64 - 0) \cdot 0.74 \cdot 129$
 $= 7922176. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



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 Nozzle Calcs.: N1- Tube inlet Nozl: 14 4:12pm Apr 10,2026

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (1171181 + 17249826) = 18421008 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (1171181 + 0 + 7922176 + 0) = 9093357 \text{ N}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (1171181 + 7922176 + 0) = 9093357 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 18421008N , must exceed W = 6013088 N or W1 = 6970091 N
 Path 2-2 = 9093357 N , must exceed W = 6013088 N or W2 = 12498669 N
 Path 3-3 = 9093357 N , must exceed W = 6013088 N or W3 = 12498669 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.9 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 102.8805 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 964.8806 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 412.40 mm

Input Echo, WRC297 Item 1, Description: N1- Tube inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1756.000	mm
Shell Thickness	112.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 129.12	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	406.400	mm
Nozzle Thickness	210.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 123.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp 14.72	MPa
Radial Load	P -49999.97	N
Circumferential Shear	Vc 40000.02	N
Longitudinal Shear	Vl 40000.02	N



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 4:12pm Apr 10, 2026

Circumferential Moment	Mc	0.480E+08	N-mm
Longitudinal Moment	Ml	0.480E+08	N-mm
Torsional Moment	Mt	0.560E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, D_v/T_v (17.679) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_n (3.935) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 826.4 + 2 \cdot 1.65 \cdot \sqrt{935.5(112.0 - 3.0)}$$

$$= 1880.178 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1871.000 mm
Nozzle Outside Diameter	(d) =	826.400 mm
Vessel Thickness used	(T) =	109.000 mm
Nozzle Thickness used	(t) =	207.000 mm
T / t	=	0.527
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.830$$

Nr/P	=	0.073
Mr/P	=	0.149
M0/P	=	0.044
N0/P	=	0.102
MrD/Mc	=	0.286
NrDL/Mc	=	0.102
M0d/Mc	=	0.090
N0DL/Mc	=	0.065
MrD/Ml	=	0.121
NrDL/Ml	=	0.070
M0D/Ml	=	0.040
N0DL/Ml	=	0.127

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-1	1	1	-1
Normal Pressure Stress	112	126	112	126
Outplane Stress Summary	111	126	115	125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 4:12pm Apr 10,2026

Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	56	55	63	49

Vessel Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		0	0	0	0
Shear Stress Summary		0	0	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		111	126	115	125

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		0	0	0	0
Outplane Bending (P)		1	-1	1	-1
Outplane Membrane (Mc)		0	0	0	0
Outplane Bending (Mc)		-2	2	2	-2
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		56	56	56	56
Outplane Stress Summary		54	57	60	53

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		0	0	0	0
Inplane Bending (P)		3	-3	3	-3
Inplane Membrane (Mc)		0	0	0	0
Inplane Bending (Mc)		-8	8	8	-8
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		112	126	112	126
Inplane Stress Summary		107	131	124	115

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		0	0	0	0



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Nozzle Calcs.: N1- Tube inlet

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Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	107	131	124	115

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	29	11	29
Hoop Stress Summary	11	29	12	30

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	5	4	8	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	11	29	12	30

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 4:12pm Apr 10,2026

Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	29	11	29

Hoop Stress Summary	11	29	12	30
---------------------	----	----	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	3	6	9	4
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	11	29	12	30
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	126	112	126	112	126	112	126
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	2	-2	-5	5	11	-11
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	6	-6	-1	1	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	112.0	126.0	112.0	126.0	112.0	126.0	112.0	126.0
Pm+Pl (SUS)	112.0	126.0	112.0	126.0	112.0	126.0	112.0	126.0
Pm+Pl+Q (Total)	112.0	126.0	114.0	124.0	107.0	131.0	123.0	115.0

Vessel Stress Summation Comparison (MPa):



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 Nozzle Calcs.: N1- Tube inlet Nozl: 14 4:12pm Apr 10,2026

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	126.00	129.12	Passed
Pm+Pl (SUS)	126.00	193.68	Passed
Pm+Pl+Q (TOTAL)	131.00	387.37	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	29	11	29	11	29	11	29
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	-1	1	1	-1
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	11.0	29.0	11.0	29.0	11.0	29.0	11.0	29.0
Pm+Pl (SUS)	11.0	29.0	11.0	29.0	11.0	29.0	11.0	29.0
Pm+Pl+Q (Total)	11.0	29.0	11.0	29.0	11.0	29.0	11.0	29.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.00	123.00	Passed
Pm+Pl (SUS)	29.00	184.50	Passed
Pm+Pl+Q (TOTAL)	29.00	369.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2- Tube outlet

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Input, Nozzle Desc: N2- Tube outlet

From: 20

Pressure for Reinforcement Calculations	P	14.7214	MPa
Temperature for Internal Pressure	Temp	343	C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	129.12	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1756.00	mm
Design Length of Section	L	2406.1667	mm
Shell Finished (Minimum) Thickness	t	112.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1012.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-12.00	C
---	--	--------	---

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	123.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		12.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	105.0000	mm
Height of Beveled Transition	L'	105.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		210.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	450.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	64.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



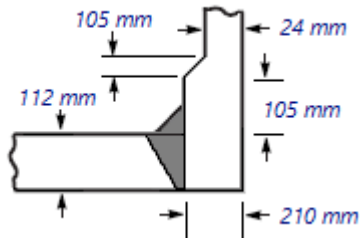
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Nozzle Calcs.: N2- Tube outlet

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Reinforcement CALCULATION, Description: N2- Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	12.000 in.
Actual Thickness Used in Calculation	0.945 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $14.7 \cdot 881 / (129 \cdot 1 - 0.6 \cdot 14.7)$

= 107.8192 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $14.7 \cdot 155 / (123 \cdot 1 - 0.6 \cdot 14.7)$

= 20.0381 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.2658 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

= (Hub Thickness - Neck Thickness) / $\tan(30)$

= $134 / 0.5773$

= 232.7877 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	621.6000 mm
--	----	-------------

Parallel to Vessel Wall, opening length	d	310.8000 mm
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	T1np	262.5000 mm
---	------	-------------

Weld Strength Reduction Factor [fr1]:

= $\min(1, S_n / S_v)$

= $\min(1, 123 / 129)$

= 0.953

Weld Strength Reduction Factor [fr2]:

= $\min(1, S_n / S_v)$

= $\min(1, 123 / 129)$

= 0.953



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Nozzle Calcs.: N2- Tube outlet

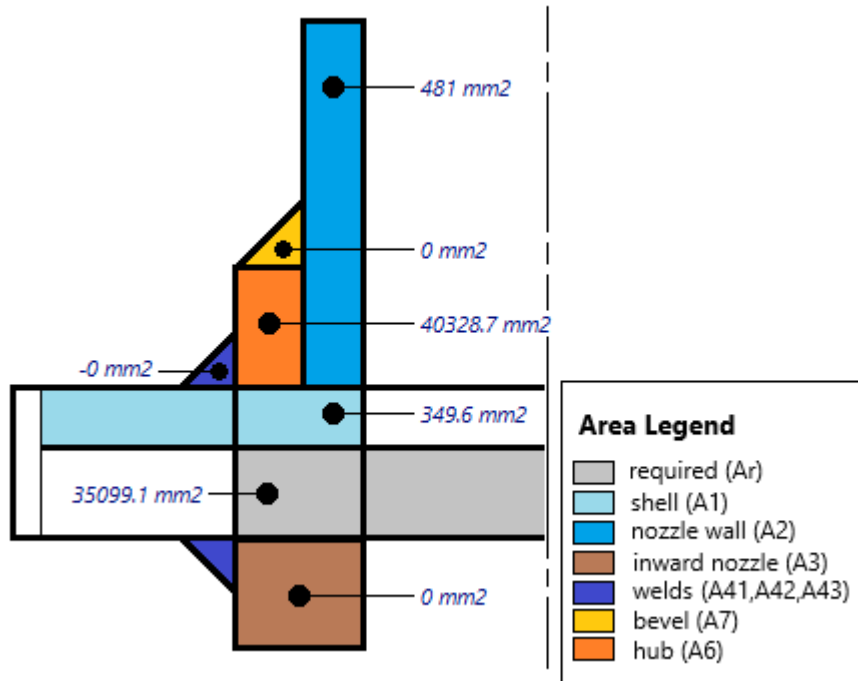
Noz1: 15 4:12pm Apr 10, 2026

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.95, 1)

= 0.953



Results of Nozzle Reinforcement Area Calculations: (mm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	35099.074	1062.901	NA
Area in Shell	A1	349.596	30337.590	NA
Area in Nozzle Wall	A2	481.032	9869.228	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		-0.000	-0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	40328.652	40328.652	NA
TOTAL AREA AVAILABLE	Atot	41159.281	80535.469	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (311•108•1+2•155•108•1•(1-0.95))

= 35099.074 mm2

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)

= 311 (1 • 109 - 1 • 108) - 2 • 155

(1 • 109 - 1 • 108) • (1 - 0.95)

= 349.596 mm2



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Nozzle Calcs.: N2- Tube outlet Nozl: 15 4:12pm Apr 10, 2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_r^2$$

$$= (2 \cdot 262) (21 - 20) 0.95$$

$$= 481.032 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_r^2 + (W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot f_r^2$$

$$= (15^2 - 225) \cdot 0.95 + (0^2 - 0) \cdot 0.95$$

$$= -0.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_r^2$$

$$= (2 \cdot \min(262, 450, 105)) \cdot (158 - 24) \cdot 0.95$$

$$= 40328.652 \text{ mm}^2 \text{ Includes Bevel Area of } 13442.885 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 23.0381 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 110.8192 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 110.8192 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.7031 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(111, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(23, 11.3)$$

$$= 23.0381 \text{ mm}$$

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 71.6,	Allowable	: 123.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 254.7 MPa	Passed
Occasional	: 51.0,	Allowable	: 163.6 MPa	Passed
Shear	: 14.0,	Allowable	: 86.1 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:C, Curve:A)

Govrn. thk, tg = 24, tr = 17.7, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.84, Temp. Reduction = 9 C

Min Metal Temp. w/o impact per UCS-66, Curve A	18 C
Min Metal Temp. at Required thickness (UCS 66.1)	9 C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:C, Curve:B)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 9 C



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 Nozzle Calcs.: N2- Tube outlet Nozl: 15 4:12pm Apr 10, 2026

Weld Size Calculations, Description: N2- Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (35099 - 350 + 2 \cdot 155 \cdot 0.95 \cdot (1 \cdot 109 - 108)) \cdot 129)$
 $= 4531687.00 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (40810 + -0.00013 - 0 \cdot 0.95) \cdot 129$
 $= 5268992.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (40810 + 0 + -0.00013 + (32271)) \cdot 129$
 $= 9435533.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (40810 + 0 + -0.00013 + 0 + (32271)) \cdot 129$
 $= 9435533.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 622 \cdot 15 \cdot 0.49 \cdot 123$
 $= 882646. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 233) \cdot (158 - 3) \cdot 0.7 \cdot 123$
 $= 9797367. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 622 \cdot (64 - 0) \cdot 0.74 \cdot 129$
 $= 5970446. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (882646 + 9797367) = 10680013 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (882646 + 0 + 5970446 + 0) = 6853092 \text{ N}$

Failure Path [Path 3-3]:



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$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (882646 + 5970446 + 0) = 6853092 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 10680013N , must exceed W = 4531687 N or W1 = 5268992 N
 Path 2-2 = 6853092 N , must exceed W = 4531687 N or W2 = 9435533 N
 Path 3-3 = 6853092 N , must exceed W = 4531687 N or W3 = 9435533 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.9 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 56.8500 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 618.8500 mm

Input Echo, WRC297 Item 1, Description: N2- Tube outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1756.000	mm
Shell Thickness	112.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 129.12	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	304.800	mm
Nozzle Thickness	210.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 123.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	14.72	MPa
Radial Load	P	-37499.98	N
Circumferential Shear	Vc	29999.98	N
Longitudinal Shear	Vl	29999.98	N
Circumferential Moment	Mc	0.270E+08	N-mm
Longitudinal Moment	Ml	0.270E+08	N-mm
Torsional Moment	Mt	0.315E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dv/Tv (17.679) is < 20
 check the limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (3.451) is < 10
 check the limitations of WRC bulletin 297.



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Nozzle Calcs.: N2- Tube outlet

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Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 724.8 + 2 \cdot 1.65 \cdot \sqrt{935.5(112.0 - 3.0)}$$

$$= 1778.578 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1871.000 mm

Nozzle Outside Diameter (d) = 724.800 mm

Vessel Thickness used (T) = 109.000 mm

Nozzle Thickness used (t) = 207.000 mm

T / t = 0.527

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.605$$

Nr/P = 0.079

Mr/P = 0.164

M0/P = 0.048

N0/P = 0.108

MrD/Mc = 0.292

NrDL/Mc = 0.101

M0d/Mc = 0.092

N0DL/Mc = 0.060

MrD/Ml = 0.136

NrDL/Ml = 0.075

M0D/Ml = 0.044

N0DL/Ml = 0.130

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	112	126	112	126
Outplane Stress Summary	112	126	114	125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-2	2	2	-2
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	56	55	62	50

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom



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Nozzle Calcs.: N2- Tube outlet

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longitudinal plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stress Intensities			Top	Top	Bottom	Bottom
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			112	126	114	125

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Outplane Membrane (P)			0	0	0	0
Outplane Bending (P)			0	0	0	0
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-1	1	1	-1
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			56	56	56	56
Outplane Stress Summary			55	56	59	53

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Inplane Membrane (P)			0	0	0	0
Inplane Bending (P)			3	-3	3	-3
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			-5	5	5	-5
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			112	126	112	126
Inplane Stress Summary			109	129	121	118

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	0	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stress Intensities			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			109	129	121	118

Nozzle Stresses (MPa):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
--------------------	--	--	----	----	----	----



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Nozzle Calcs.: N2- Tube outlet Nozl: 15 4:12pm Apr 10,2026

(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	25	7	25
Hoop Stress Summary	7	25	8	26

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	3	3	5	3

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	7	25	8	26

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	25	7	25
Hoop Stress Summary	8	25	8	26

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0



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Nozzle Calcs.: N2- Tube outlet Nozl: 15 4:12pm Apr 10,2026

Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	2	3	6	2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	8	25	8	26

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	126	112	126	112	126	112	126
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	-2	2	8	-8
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	1	-1	5	-5	-1	1	1	-1
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	112.0	126.0	112.0	126.0	112.0	126.0	112.0	126.0
Pm+Pl (SUS)	112.0	126.0	112.0	126.0	112.0	126.0	112.0	126.0
Pm+Pl+Q (Total)	112.0	126.0	112.0	126.0	110.0	128.0	120.0	118.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	126.00	129.12	Passed
Pm+Pl (SUS)	126.00	193.68	Passed
Pm+Pl+Q (TOTAL)	128.00	387.37	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



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Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	25	7	25	7	25	7	25
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	-1	1	1	-1
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	7.0	25.0	7.0	25.0	7.0	25.0	7.0	25.0
Pm+Pl (SUS)	7.0	25.0	7.0	25.0	7.0	25.0	7.0	25.0
Pm+Pl+Q (Total)	7.0	25.0	7.0	25.0	7.0	25.0	7.0	25.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.00	123.00	Passed
Pm+Pl (SUS)	25.00	184.50	Passed
Pm+Pl+Q (TOTAL)	25.00	369.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell Bypass No

Nozl: 16

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Input, Nozzle Desc: Shell Bypass No

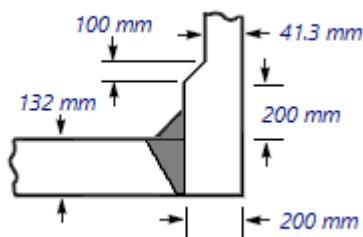
From: 40

Pressure for Reinforcement Calculations	P	18.2330	MPa
Temperature for Internal Pressure	Temp	370	C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	138.20	MPa
Parent Allowable Stress At Ambient	Sva	148.00	MPa
Inside Diameter of Cylindrical Shell	D	1756.00	mm
Shell Finished (Minimum) Thickness	t	132.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4070.56	mm
User Entered Minimum Design Metal Temperature		-12.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	111.00	MPa
Allowable Stress At Ambient	Sna	148.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	41.2800	mm
Hub Height of Integral Nozzle	h	200.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	155.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: Shell Bypass No



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Nozzle Calcs.: Shell Bypass No Nozl: 16 4:12pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 16.000 in.
Actual Thickness Used in Calculation 1.625 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $18.2 \cdot 881 / (138 \cdot 1 - 0.6 \cdot 18.2)$
= 126.2249 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $18.2 \cdot 206 / (111 \cdot 1 - 0.6 \cdot 18.2)$
= 37.5738 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 159/0.5773
= 274.9113 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	824.8000	mm
Parallel to Vessel Wall, opening length	d	412.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	322.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 111/138)
= 0.803

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 111/138)
= 0.803

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.8, 1)
= 0.803



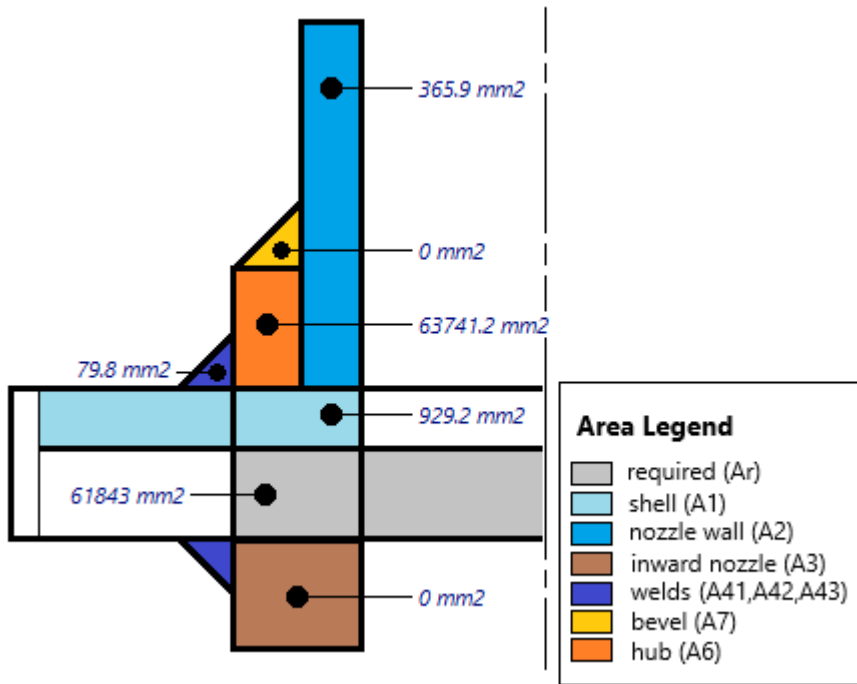
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Nozzle Calcs.: Shell Bypass No

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Results of Nozzle Reinforcement Area Calculations: (mm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	61842.992	NA	NA
Area in Shell	A1	929.245	NA	NA
Area in Nozzle Wall	A2	365.863	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	79.805	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	63741.242	NA	NA
TOTAL AREA AVAILABLE	Atot	65116.152	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (412 \cdot 126 \cdot 1 + 2 \cdot 197 \cdot 126 \cdot 1 \cdot (1 - 0.8))$$

$$= 61842.992 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 412(1 \cdot 129 - 1 \cdot 126) - 2 \cdot 197$$

$$(1 \cdot 129 - 1 \cdot 126) \cdot (1 - 0.8)$$

$$= 929.245 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 322)(38.3 - 37.6)0.8$$

$$= 365.863 \text{ mm}^2$$



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Nozzle Calcs.: Shell Bypass No Nozl: 16 4:12pm Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot fr_2 + ((W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr_2$$

$$= (10^2 - 0.64) \cdot 0.8 + (0^2 - 0) \cdot 0.8$$

$$= 79.805 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(322, 500, 200)) \cdot (200 - 41.3) \cdot 0.8$$

$$= 63741.242 \text{ mm}^2 \text{ Includes Bevel Area of } 12748.248 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 40.5738 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 129.2249 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 129.2249 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(129, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(40.6, 11.3)$$

$$= 40.5738 \text{ mm}$$

Available Nozzle Neck Thickness = 41.2800 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 61.2,	Allowable	: 111.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 262.6 MPa	Passed
Occasional	: 44.9,	Allowable	: 147.6 MPa	Passed
Shear	: 10.6,	Allowable	: 77.7 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: Shell Bypass No

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$$

$$= \max(0, (61843 - 929 + 2 \cdot 197 \cdot 0.8 \cdot (1 \cdot 129 - 126)) 138)$$



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Nozzle Calcs.: Shell Bypass No Nozl: 16 4:12pm Apr 10, 2026

= 8538842.00 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an}/1.707)^2 \cdot fr2) \cdot S_v$$

$$= (64107 + 79.8 - 0 \cdot 0.8) \cdot 138$$

$$= 8869797.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (64107 + 0 + 79.8 + (40823)) \cdot 138$$

$$= 14511003.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (64107 + 0 + 79.8 + 0 + (40823)) \cdot 138$$

$$= 14511003.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 806 \cdot 10 \cdot 0.49 \cdot 111$$

$$= 688894. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 305) \cdot (200 - 3) \cdot 0.7 \cdot 111$$

$$= 14651181. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 806 \cdot (132 - 3) \cdot 0.74 \cdot 138$$

$$= 16709330. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (688894 + 14651181) = 15340075 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (688894 + 0 + 16709330 + 0) = 17398224 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (688894 + 16709330 + 0) = 17398224 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 15340075 N , must exceed W = 8538842 N or W1 = 8869797 N
 Path 2-2 = 17398224 N , must exceed W = 8538842 N or W2 = 14511003 N
 Path 3-3 = 17398224 N , must exceed W = 8538842 N or W3 = 14511003 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 18.5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 98.0553 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 730.0553 mm



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 Nozzle Calcs.: Shell Bypass No Nozl: 16 4:12pm Apr 10,2026

Input Echo, WRC297 Item 1, Description: Shell Bypass N:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1756.000	mm
Shell Thickness	132.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 22	
Vessel Cold S.I. Allowable	Smc 148.00	MPa
Vessel Hot S.I. Allowable	Smh 138.20	MPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	406.400	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F22	
Nozzle Cold S.I. Allowable	SNmc 148.00	MPa
Nozzle Hot S.I. Allowable	SNmh 111.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	18.23	MPa
Radial Load	P	58000.14	N
Circumferential Shear	Vc	48000.15	N
Longitudinal Shear	Vl	48000.15	N
Circumferential Moment	Mc	0.608E+08	N-mm
Longitudinal Moment	Ml	0.608E+08	N-mm
Torsional Moment	Mt	0.800E+08	N-mm
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, Dv/Tv (15.303) is < 20
check the limitations of WRC bulletin 297.
Warning - The ratio, Dn/Tn (4.032) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 806.4 + 2 \cdot 1.65 \cdot \sqrt{(945.5(132.0 - 3.0))}$
 $= 1958.896 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1891.000	mm
Nozzle Outside Diameter	(d) =	806.400	mm
Vessel Thickness used	(T) =	129.000	mm
Nozzle Thickness used	(t) =	197.000	mm
T / t	=	0.655	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.633$$

$$Nr/P = 0.083$$



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Nozzle Calcs.: Shell Bypass No Nozl: 16 4:12pm Apr 10,2026

Mr/P = 0.154
MØ/P = 0.047
NØ/P = 0.108
MrD/Mc = 0.282
NrDL/Mc = 0.113
MØd/Mc = 0.092
NØDL/Mc = 0.059
MrD/Ml = 0.135
NrDL/Ml = 0.077
MØD/Ml = 0.045
NØDL/Ml = 0.132

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-1	1	1	-1
Normal Pressure Stress	116	134	116	134
Outplane Stress Summary	112	135	116	134

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-3	3	-3	3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	50	64	58	57

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	112	135	116	134

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: Shell Bypass No Nozl: 16 4:12pm Apr 10,2026

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-2	2	2	-2
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58
Outplane Stress Summary	53	60	59	56

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-3	3	-3	3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-7	7	7	-7
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	134	116	134
Inplane Stress Summary	104	144	120	130

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	104	144	120	130

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	37	15	37
Hoop Stress Summary	14	36	15	37

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-1	1	-1	1



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Nozzle Calcs.: Shell Bypass No Nozl: 16 4:12pm Apr 10,2026

Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	4	8	8	8
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	14	36	15	37

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	37	15	37
Hoop Stress Summary	14	36	15	37

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-1	1	-1	1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-2	2	2	-2
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	2	9	10	7

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):



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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	14	36	15	37

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	134	116	134	116	134	116	134
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-1	1	1	-1	-10	10	4	-4
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-6	6	0	0	-2	2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	116.0	134.0	116.0	134.0	116.0	134.0	116.0	134.0
Pm+Pl (SUS)	116.0	134.0	116.0	134.0	116.0	134.0	116.0	134.0
Pm+Pl+Q (Total)	115.0	135.0	117.0	133.0	106.0	144.0	120.0	130.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	134.00	138.20	Passed
Pm+Pl (SUS)	134.00	207.30	Passed
Pm+Pl+Q (TOTAL)	144.00	414.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	37	15	37	15	37	15	37
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-2	2	0	0	-3	3	1	-1
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	15.0	37.0	15.0	37.0	15.0	37.0	15.0	37.0
Pm+Pl (SUS)	15.0	37.0	15.0	37.0	15.0	37.0	15.0	37.0



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Nozzle Calcs.: Shell Bypass No

Nozl: 16

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Pm+Pl+Q (Total)	15.0	37.0	15.0	37.0	15.0	37.0	15.0	37.0
-----------------	------	------	------	------	------	------	------	------

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	37.00	111.00	Passed
Pm+Pl (SUS)	37.00	166.50	Passed
Pm+Pl+Q (TOTAL)	37.00	333.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell inlet-3

Nozl: 17 4:12pm Apr 10,2026

Input, Nozzle Desc: Shell inlet-3

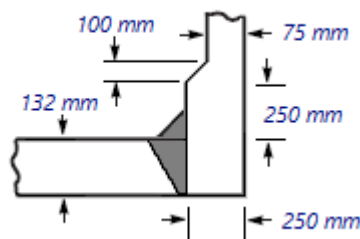
From: 40

Pressure for Reinforcement Calculations	P	18.2330	MPa
Temperature for Internal Pressure	Temp	370	C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	138.20	MPa
Parent Allowable Stress At Ambient	Sva	148.00	MPa
Inside Diameter of Cylindrical Shell	D	1756.00	mm
Shell Finished (Minimum) Thickness	t	132.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4109.18	mm
User Entered Minimum Design Metal Temperature		-12.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	111.00	MPa
Allowable Stress At Ambient	Sna	148.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	75.0000	mm
Hub Height of Integral Nozzle	h	250.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		250.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1085.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	132.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: Shell inlet-3



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Nozzle Calcs.: Shell inlet-3 Nozl: 17 4:12pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 30.000 in.
Actual Thickness Used in Calculation 2.953 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $18.2 \cdot 881 / (138 \cdot 1 - 0.6 \cdot 18.2)$
= 126.2249 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $18.2 \cdot 381 / (111 \cdot 1 + 0.4 \cdot 18.2)$
= 58.7250 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 175/0.5773
= 303.1090 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1236.0000	mm
Parallel to Vessel Wall, opening length	d	618.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	322.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 111/138)
= 0.803

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 111/138)
= 0.803

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.8, 1)
= 0.803



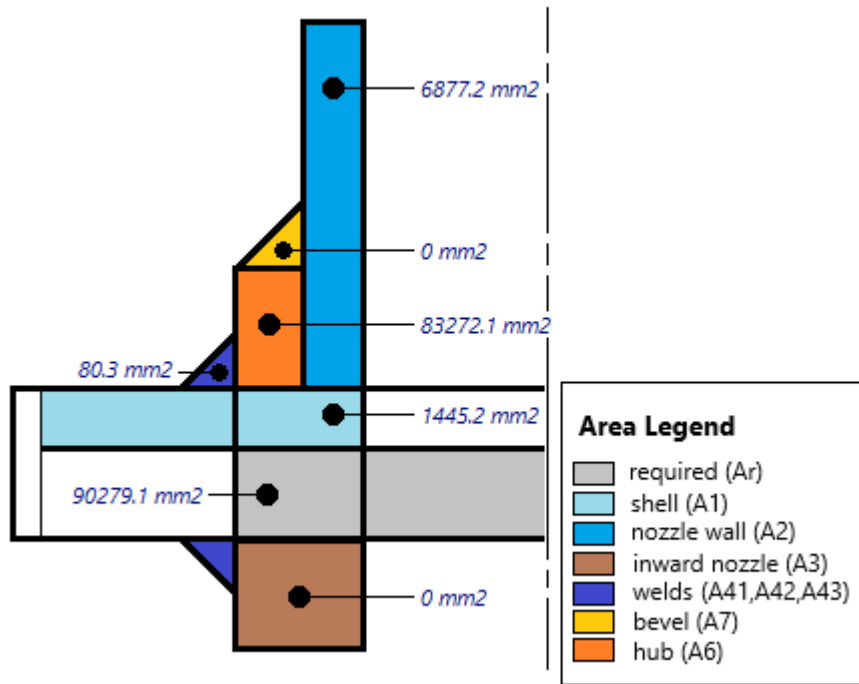
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Nozzle Calcs.: Shell inlet-3

Nozl: 17 4:12pm Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	90279.062	NA	NA
Area in Shell	A1	1445.180	NA	NA
Area in Nozzle Wall	A2	6877.232	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		80.319	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	83272.086	NA	NA
TOTAL AREA AVAILABLE	Atot	91674.812	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (618 \cdot 126 \cdot 1 + 2 \cdot 247 \cdot 126 \cdot 1 \cdot (1 - 0.8)) \\
 &= 90279.062 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 618(1 \cdot 129 - 1 \cdot 126) - 2 \cdot 247 \\
 &\quad (1 \cdot 129 - 1 \cdot 126) \cdot (1 - 0.8) \\
 &= 1445.180 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tnp)(tn - trn)fr2 \\
 &= (2 \cdot 322)(72 - 58.7)0.8 \\
 &= 6877.232 \text{ mm}^2
 \end{aligned}$$



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Nozzle Calcs.: Shell inlet-3

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.8 + (0)^2 \cdot 0.8$$

$$= 80.319 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(322, 1085, 250)) \cdot (250 - 75) \cdot 0.8$$

$$= 83272.086 \text{ mm}^2 \text{ Includes Bevel Area of } 12992.867 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 61.7250 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 129.2249 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 129.2249 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(129, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(61.7, 11.3)$$

$$= 61.7250 \text{ mm}$$

Available Nozzle Neck Thickness = 75.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 35.3,	Allowable	: 111.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 288.4 MPa	Passed
Occasional	: 35.0,	Allowable	: 147.6 MPa	Passed
Shear	: 0.5,	Allowable	: 77.7 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:C, Curve:A)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 129 / 0.5773$$

$$= 223.4347 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]



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Nozzle Calcs.: Shell inlet-3 Nozl: 17 4:12pm Apr 10,2026

Parallel to Vessel Wall (Diameter Limit) D1 1020.0001 mm
 Parallel to Vessel Wall Rn+tn+t 510.0000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 322.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	58665.320	58665.320
Area Available in Shell	A1	896.016	896.016
Area Available in Nozzle Wall	A2	6877.232	6877.232
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	-0.000	-0.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	61383.434	61383.434
TOTAL AREA AVAILABLE	Atot	69156.680	69156.680

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Shell inlet-3

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (90279 - 1445 + 2 \cdot 247 \cdot 0.8 \cdot (1 \cdot 129 - 126)) \cdot 138)$
 $= 12427841.00 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (68261 + 80.3 - 0 \cdot 0.8) \cdot 138$
 $= 9443836.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (68261 + 0 + 80.3 + (51184)) \cdot 138$
 $= 16516821.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (68261 + 0 + 80.3 + 0 + (51184)) \cdot 138$
 $= 16516821.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 762 \cdot 10 \cdot 0.49 \cdot 111$
 $= 650964. \text{ N}$



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Nozzle Calcs.: Shell inlet-3 Nozl: 17 4:12pm Apr 10,2026

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 345) \cdot (250 - 3) \cdot 0.7 \cdot 111$$

$$= 20799362. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 762 \cdot (132 - 3) \cdot 0.74 \cdot 138$$

$$= 15789320. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (650964 + 20799362) = 21450326 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (650964 + 0 + 15789320 + 0) = 16440284 N$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (650964 + 15789320 + 0) = 16440284 N$$

Summary of Failure Path Calculations:

Path 1-1 = 21450326N , must exceed W = 12427841 N or W1 = 9443836 N
 Path 2-2 = 16440284N , must exceed W = 12427841 N or W2 = 16516821 N
 Path 3-3 = 16440284N , must exceed W = 12427841 N or W3 = 16516821 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 18.4 MPa

The Drop for this Nozzle is : 86.9735 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1303.9735 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Shell inlet-3 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1756.000	mm
Shell Thickness	132.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 22	
Vessel Cold S.I. Allowable	Smc	148.00 MPa
Vessel Hot S.I. Allowable	Smh	138.20 MPa

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	1112.000	mm
Nozzle Thickness	250.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F22	
Nozzle Cold S.I. Allowable	SNmc	148.00 MPa
Nozzle Hot S.I. Allowable	SNmh	111.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:



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Nozzle Calcs.: Shell inlet-3

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External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	18.23	MPa
Radial Load	P	-18750.05	N
Circumferential Shear	Vc	15000.04	N
Longitudinal Shear	Vl	15000.04	N
Circumferential Moment	Mc	0.675E+07	N-mm
Longitudinal Moment	Ml	0.675E+07	N-mm
Torsional Moment	Mt	0.785E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (15.303) is < 20
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Dv (.550) is > 0.5
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (4.448) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 1112.0 + 2 \cdot 1.65 \cdot \sqrt{(945.5(132.0 - 3.0))}$$

$$= 2264.496 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1891.000 mm
Nozzle Outside Diameter	(d) =	1112.000 mm
Vessel Thickness used	(T) =	129.000 mm
Nozzle Thickness used	(t) =	247.000 mm
T / t	=	0.522
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.251$$

Nr/P	=	0.066
Mr/P	=	0.129
M0/P	=	0.039
N0/P	=	0.094
MrD/Mc	=	0.277
NrDL/Mc	=	0.103
M0d/Mc	=	0.086
N0DL/Mc	=	0.074
MrD/Ml	=	0.107
NrDL/Ml	=	0.060
M0D/Ml	=	0.033
N0DL/Ml	=	0.116

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	116	134	116	134



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Nozzle Calcs.: Shell inlet-3

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Outplane Stress Summary	116	134	116	134
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Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	58	58	58

Inplane Stress Summary	58	57	59	57
------------------------	----	----	----	----

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	116	134	116	134
------------------------	-----	-----	-----	-----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary	58	58	58	57
-------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	134	116	134

Inplane Stress Summary	116	134	117	132
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Nozzle Calcs.: Shell inlet-3

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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	0	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			116	134	117	132

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)			0	0	0	0
Hoop Bending (P)			0	0	0	0
Hoop Membrane (Mc)			0	0	0	0
Hoop Bending (Mc)			0	0	0	0
Hoop Membrane (ML)			0	0	0	0
Hoop Bending (ML)			0	0	0	0
Hoop Pressure Stress			19	41	19	41
Hoop Stress Summary			19	41	19	41

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)			0	0	0	0
Axial Bending (P)			0	0	0	0
Axial Membrane (Mc)			0	0	0	0
Axial Bending (Mc)			0	0	0	0
Axial Membrane (ML)			0	0	0	0
Axial Bending (ML)			0	0	0	0
Axial Pressure Stress			9	9	9	9
Axial Stress Summary			9	9	10	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)			Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)			0	0	0	0
Shear due to (Vl)			0	0	0	0
Shear due to Torsion			0	0	0	0
Shear Stress Summary			0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)			Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress			19	41	19	41



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Nozzle Calcs.: Shell inlet-3

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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	41	19	41
Hoop Stress Summary	19	41	19	41

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	9	9	10	9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	19	41	19	41

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	134	116	134	116	134	116	134
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0



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Pm (SUS)	116.0	134.0	116.0	134.0	116.0	134.0	116.0	134.0
Pm+P1 (SUS)	116.0	134.0	116.0	134.0	116.0	134.0	116.0	134.0
Pm+P1+Q (Total)	116.0	134.0	116.0	134.0	116.0	134.0	116.0	134.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	134.00	138.20	Passed
Pm+P1 (SUS)	134.00	207.30	Passed
Pm+P1+Q (TOTAL)	134.00	414.60	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	41	19	41	19	41	19	41
Circ. P1 (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. P1 (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	19.0	41.0	19.0	41.0	19.0	41.0	19.0	41.0
Pm+P1 (SUS)	19.0	41.0	19.0	41.0	19.0	41.0	19.0	41.0
Pm+P1+Q (Total)	19.0	41.0	19.0	41.0	19.0	41.0	19.0	41.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	111.00	Passed
Pm+P1 (SUS)	41.00	166.50	Passed
Pm+P1+Q (TOTAL)	41.00	333.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Nozzle Calcs.: BD3

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Input, Nozzle Desc: BD3

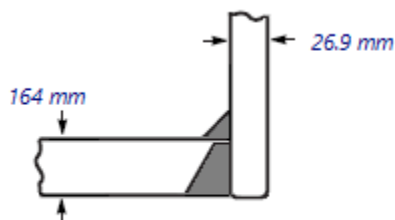
From: 50

Pressure for Reinforcement Calculations	P	18.2330	MPa
Temperature for Internal Pressure	Temp	480	C
Parent Material		SA-336 F22	
Parent Allowable Stress at Temperature	Sv	110.72	MPa
Parent Allowable Stress At Ambient	Sva	148.00	MPa
Inside Diameter of Cylindrical Shell	D	1756.00	mm
Shell Finished (Minimum) Thickness	t	164.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7160.16	mm
User Entered Minimum Design Metal Temperature		-12.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	47.63	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	190.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	155.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: BD3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



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 FileName : 123_C1_NVS Rev 02 Page 200 of 241
 Nozzle Calcs.: BD3 Nozl: 18 4:12pm Apr 10,2026

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.2 \cdot 880 / (111 \cdot 1 - 0.6 \cdot 18.2)$
 $= 160.8031 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([18.2 / (47.6 \cdot 1)] - 1))$
 $= 13.2465 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	428.6480	mm
Parallel to Vessel Wall	Rn+tn+tr	214.3240	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: BD3.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 16.2465 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 162.8031 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 162.8031 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 8.2578 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(163, 5.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(16.2, 8.26)$
 $= 16.2465 \text{ mm}$

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 26.9, tr = 4.01, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.17$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve A	21 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: BD3

Intermediate Calc. for nozzle/shell welds	Tmin	19.0000	mm
---	------	---------	----

Results Per UW-16.1:



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Nozzle Calcs.: BD3

Nozl: 18 4:12pm Apr 10,2026

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.5606 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 356.0605 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 55.31 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 214.32 mm

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ASME TS Calc: DESIGN 1

Case: 1 4:12p Apr 10,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	18.00 MPa
Shell Maximum Operating Pressure	Psox,max	15.93 MPa
Shell Minimum Operating Pressure	Psox,min	0.00 MPa
Shell Thickness	ts	132.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1756.000 mm
Shell Temperature for Internal Pressure	Ts	479.96 C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	110.72 MPa
Shell Allowable Stress at Ambient		148.00 MPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	14.28 MPa
Channel Maximum Operating Pressure	Ptox,max	12.55 MPa
Channel Minimum Operating Pressure	Ptox,min	0.00 MPa
Channel Thickness	tc	112.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1756.000 mm
Channel Design Temperature	TEMPC	342.97 C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	129.12 MPa
Channel Allowable Stress at Ambient		138.00 MPa

Tube Data:

Number of Tube Holes	Nt	2156
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	8500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8050.00 mm
Design Temperature of the Tubes		399.97 C
Tube Material		SA-213 TP321
Tube Material UNS Number		S32100
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		119.00 MPa
Tube Allowable Stress At Ambient		138.00 MPa
Tube Yield Stress At design Temperature	Syt	132.00 MPa
Tube Pitch (Center to Center Spacing)	P	28.5000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	858.460 mm
Largest Center-to-Center Tube Distance	Ul	76.2000 mm



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ASME TS Calc: DESIGN 1

Case: 1 4:12p Apr 10,2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	1695580.0 mm2
Perimeter of Tube Layout	Cp	5131.8 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	369.97 C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	138.20 MPa
Tubesheet Allowable Stress at Ambient	Tt	148.00 MPa
Thickness of Tubesheet	h	450.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2610.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	1847.00 mm
Step 1 Depth on the Shellside	hs1	9.00 mm
Step 2 Diameter on the Shellside	ds2	1751.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2610.00 mm

Note: Tubesheet diameter is now: 2610.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	192024.0 mm2
---------------------------	----	--------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2610.000 mm
Flange Inside Diameter	B	1760.000 mm
Flange Face Outside Diameter	Fod	1850.000 mm
Flange Face Inside Diameter	Fid	1760.000 mm
Gasket Outside Diameter	Go	1826.000 mm
Gasket Inside Diameter	Gi	1780.000 mm
Small end Hub thk.	g0	132.0000 mm
Large end Hub thk.	g1	206.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	69.00 MPa
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option	Program Selects	

Bolting Information:

Diameter of Bolt Circle	C	2410.000 mm
Nominal Bolt Diameter	dB	95.2500 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-540 B21
Bolt Allowable Stress At Temperature	Sb	228.00 MPa
Bolt Allowable Stress At Ambient	Sa	228.00 MPa
Weld between Flange and Shell/Channel		0.0000 mm



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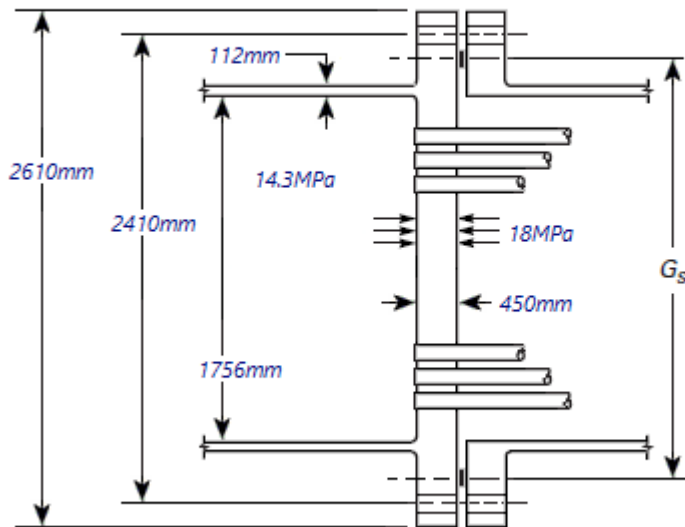
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ASME TS Calc: DESIGN 1

Case: 1 4:12p Apr 10,2026

Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 413.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group D

Elastic Mod. at Design Temperature	480.0 C	0.17671E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	343.0 C	0.17931E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.20271E+06 MPa

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesheet Design Temp.	370.0 C	0.17107E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.19513E+06 MPa

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	370.0 C	0.18626E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (14.3 \cdot (19/2 - 0)) / (119 \cdot 1 + 0.4 \cdot 14.3)$
 $= 1.0907 + 0.0000 = 1.0907 \text{ mm}$



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ASME TS Calc: DESIGN 1

Case: 1 4:12p Apr 10, 2026

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 8908 / \sqrt{(9.52 \cdot 3.05)}$$

$$= 1653.342$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (19/3.05)$$

$$= 0.0880$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.088 \cdot 168489 (3.05/19)$$

$$= 3795.723 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3795.7/132.0$$

$$= 28.756$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 131.999 \text{ MPa}$$

For non-Carbon and Low Alloy Steel Materials re-compute [Fic]:

Factor A [A]:

$$= F/E_{mod}, \text{ see 2286-6 Step 1}$$

$$= 3796/168489$$

$$= 0.02253$$

From the External Pressure Chart, HA-2, determine the Stress [B]:

$$= 83.760 \text{ MPa}$$

The Tangent Modulus from 3-D.5.2, [Et]:

$$= 2B/A$$

$$= 2 \cdot 83.8/0.023$$

$$= 7436.1 \text{ MPa}$$

Predicted Buckling Stress [Fic]:

$$= (F \cdot Et) / (FS \cdot E_{mod})$$

$$= (3796 \cdot 7436) / (1 \cdot 168489)$$

$$= 167.519 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [168] \geq F_y [132], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y/FS$$

$$= 131.999/1.667$$

$$= 79.184 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$

$$= 2 \cdot 79.2 (3.05/19)$$

$$= 25.339 \text{ MPa}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.1653 mm
Allowable Internal Pressure at Corroded thickness	43.67 MPa
Required Internal Design Pressure	14.28 MPa
Allowable External Pressure at Corroded thickness	25.34 MPa



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ASME TS Calc: DESIGN 1

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Required External Design Pressure	18.00 MPa
Required Thickness due to Shell Side pressure	2.1653 mm

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	23.000 mm
Basic Gasket Width,	$b_0 = N / 2$	11.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) * 2.5$	8.542 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	1808.915 mm

Required Tubesheet Flanged Extension thk. per 4.18.5 (operating) [hropr]:

$$= \sqrt{(1.9 \cdot W_{M1} \cdot h_g / (S \cdot G))}$$

$$= \sqrt{(1.9 \cdot 51500748 \cdot 301 / (138 \cdot 1809))}$$

$$= 343.000 \text{ mm}$$

Required Tubesheet Flanged Extension thk. per 4.18.5 (seating) [hrseat]:

$$= \sqrt{(1.9 \cdot W \cdot h_g / (S_a \cdot G))}$$

$$= \sqrt{(1.9 \cdot 52510936 \cdot 301 / (148 \cdot 1809))}$$

$$= 334.686 \text{ mm}$$

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + d_t$$

$$= 2 \cdot 858 + 19$$

$$= 1735.970 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$

$$= (28.5 - 19) / 28.5$$

$$= 0.332$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 1809 / 1736$$

$$= 1.0420$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1762 / 1736$$

$$= 1.0150$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 434$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(d_t - 2t_t (E_t/E) \cdot (S_t/S) \cdot (\rho); d_t - 2t_t)$$

$$= \max(19 - 2 \cdot 3.05 (171067 / 186264) \cdot$$



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ASME TS Calc: DESIGN 1

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$$\begin{aligned} & (121/138) \cdot (0); 19 - 2 \cdot 3.05 \\ & = 19.0500 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} & = p / \sqrt{(1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))} \\ & = 28.5 / \sqrt{(1 - 4 \cdot \min(192024; 4 \cdot 1736 \cdot 28.5) / (\pi \cdot 1736^2))} \\ & = 29.7316 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ*]:

$$\begin{aligned} & = (p^* - d^*) / p^* \\ & = (29.7 - 19) / 29.7 \\ & = 0.3593 \end{aligned}$$

h/p = 15.228069 ; μ* = 0.359267 ; E* = 66954 MPa

E*/E = 0.359461 ; ν* = 0.322844

Note: As h/p (15.228) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

βs, ks, λs, δs and ωs are all = 0

Channel coefficient [βc]:

$$\begin{aligned} & = ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(Dc + tc)tc)} \\ & = ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1762 + 109)109)} \\ & = 0.0040 \text{ 1/mm} \end{aligned}$$

Channel coefficient [kc]:

$$\begin{aligned} & = \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_s^2)) \\ & = 0.004 \cdot 179315 \cdot 109^3 / (6(1 - 0.3^2)) \\ & = 171202064.0000 \text{ MPa} \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

CqDelta = CqTheta = CmTheta = CmDelta = 1

Channel coefficient [λc]:

$$\begin{aligned} & = 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (Cm\theta + h \cdot \beta_c \cdot (Cm\Delta + Cq\theta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta) \\ & = 6(1762 + 109/\cos(0)) \cdot 171202064/434^3 \cdot (1 + 434 \cdot 0.004 \cdot (1 + 1)/2 + (434 \cdot 0.004)^2/2 \cdot 1) \\ & = 100461.8 \text{ MPa} \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned} & = Dc^2 / (4 \cdot E_c \cdot tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc))) \\ & = 1762^2 / (4 \cdot 179315 \cdot 109) \cdot (1 - 1762 \cdot 0.3 / (2(1762 + 109))) \\ & = 0.34101E-01 \text{ mm/MPa} \end{aligned}$$

Channel coefficient [zetac]:

zetac = 0

Parameter [ωc]:

$$\begin{aligned} & = pc \cdot kc [\delta_c (Cm\theta + \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (Cm\theta + h \cdot Cq\theta \cdot h \cdot \beta_c / 2)] \\ & = 1.01 \cdot 171202064 [0.034101211 (1 + 0.004 \cdot 1 \cdot 434) - 0.000 (1/434 + 1 \cdot 0.004/2)] \\ & = 65525.1562 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} & = A / Do \\ & = 2610/1736 \\ & = 1.5035 \end{aligned}$$



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ASME TS Calc: DESIGN 1

Case: 1 4:12p Apr 10, 2026

Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu) / E \cdot (\lambda c + E \cdot \ln(K)) \\
 &= (1 - 0.32) / 66955 \cdot (100462 + 186264 \cdot \ln(1.5)) \\
 &= 1.7842
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= Do^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt] \\
 &= 1736^2 / 16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 0 - \\
 &\quad (1.01 - 1) \cdot (1.01^2 + 1) \cdot 14.3] \\
 &= -81882.6172 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= Mts + (\omega c \cdot Pt) + W \cdot (C - Gs) / (2 \cdot \pi \cdot Do) \\
 &= -81883 + (935754) + 0 \cdot (2410 - 1809) / (2 \cdot \pi \cdot 1736) \\
 &= 853871.0000 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M^*) - Do^2 / 32 \cdot F (Ps - Pt)) / (1 + F) \\
 &= (853871) - 1736^2 / 32 \cdot 1.78 (0 - 14.3) / (1 + 1.78) \\
 &= 1168534.3750 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2 / 64 \cdot (3 + \nu) \cdot (Ps - Pt) \\
 &= 1168534 + 1736^2 / 64 \cdot (3 + 0.32) \cdot (0 - 14.3) \\
 &= -1065901.0000 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(|Mp|; |Mo|) \\
 &= \max(|1168534|; |-1065901|) \\
 &= 1168534.3750 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - hg')^2) \\
 &= 6 \cdot 1168534 / (0.36 (434 - 0)^2) \\
 &= 103.6087 \text{ MPa}
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 138 \\
 &= 276.40 \text{ MPa}
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - hg')^2) \\
 &= 6 \cdot 1183796 / (0.36 (268 - 0)^2) \\
 &= 276.1915 \text{ MPa}
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 268 + 3 + 13 \\
 &= 283.5472 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(284; 145) \\
 &= 283.5472 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|Ps - Pt|$]:



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ASME TS Calc: DESIGN 1

Case: 1 4:12p Apr 10, 2026

$$= |P_s - P_t|$$

$$= |0 - 14.3|$$

$$= 14.281 \text{ MPa}$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.33 \cdot 434 / 1736 \cdot \min(0.8 \cdot 138; 0.533 \cdot 246)$$

$$= 36.660 \text{ MPa}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.33 \cdot 111) \cdot (4 \cdot 1695580 / 5132) \cdot |0 - 14.3| + 16$$

$$= 144.7138 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 109.000 mm

Min. Channel len. adjacent to the tubesheet is: 788.840mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1762^2 / (4 \cdot 109 (1762 + 109) \cdot \cos(0)) \cdot 14.3$$

$$= 54.351 \text{ MPa}$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 38486400 / 109^2 \cdot [(0.034101211 \cdot 1 \cdot 0.004 - 0.000 \cdot 1) \cdot 14.3 - 6((1 - 0.32) / 66955) \cdot 1736 / 434^3 \cdot (1 + 0.004 \cdot 434 / 2 \cdot 1) \cdot (1168534 + 1736^2 / 32 \cdot (0 - 14.3))]$$

$$= 206.299 \text{ MPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 260.650 \text{ MPa (Should be } \leq 193.68)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation (option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 179315 \cdot \sqrt{(1.5 \cdot 129 / 261)}$$

$$= 154573.2 \text{ MPa}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 1159183 / (0.36 (434 - 0)^2)$$

$$= 102.8 \text{ MPa (Should be } \leq 276.4)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 268 + 3 + 13 = 284 \text{ mm}$$



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Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:
 $= \max(\text{HreqB}; \text{HreqS}) = \max(284; 145) = 284 \text{ mm}$

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (19 - 3.05) 3.05$
 $= 153.228 \text{ mm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 3.05 (19 - 3.05) 119$
 $= 18232.701 \text{ N}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$
 $= 0.85 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 119$
 $= 18232.7012 \text{ N}$

Weld Strength Factor [fw]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 119/119$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$
 $= 0.55 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 119$
 $= 12989.8496 \text{ N}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o} \right)$
 $= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) 1 \cdot 1) - 0.75 \cdot 19} \right)$
 $= 3.445 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm
 Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 18233 is $\geq$ tube strength 18233 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	102.8 <=	276.4 MPa	Ok
Tubesheet shear stress	32.8 <=	110.6 MPa	Ok
Stress in Channel at Tubesheet	260.6 <=	389.4 MPa	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	283.547	450.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok



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Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	23.000 mm
Basic Gasket Width,	$b0 = N / 2$	11.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	8.542 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	1808.915 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 858 + 19$$

$$= 1735.970 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (28.5 - 19) / 28.5$$

$$= 0.332$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= Gs / Do \text{ (Configurations d, e, f)}$$

$$= 1809 / 1736$$

$$= 1.0420$$

Ratio [pc]:

$$= Dc / Do \text{ (Configurations a, e, f)}$$

$$= 1762 / 1736$$

$$= 1.0150$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= ltx / h$$

$$= 0 / 434$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2tt)$$

$$= \max(19 - 2 \cdot 3.05 (171067 / 186264) \cdot (121 / 138) \cdot (0); 19 - 2 \cdot 3.05)$$

$$= 19.0500 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 28.5 / \sqrt{ (1 - 4 \cdot \min(192024; 4 \cdot 1736 \cdot 28.5) / (\pi \cdot 1736^2)) }$$

$$= 29.7316 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (29.7 - 19) / 29.7$$

$$= 0.3593$$

$h/p = 15.228069$; $\mu^* = 0.359267$; $E^* = 66954 \text{ MPa}$



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$$E^*/E = 0.359461 ; v^* = 0.322844$$

Note: As h/p (15.228) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 β_s , k_s , λ_s , δ_s and ω_s are all = 0

$$\begin{aligned} \text{Channel coefficient } [\beta_c]: \\ &= \left(\left(12(1 - v_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right) \\ &= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(1762 + 109)109} \right) \\ &= 0.0040 \text{ 1/mm} \end{aligned}$$

$$\begin{aligned} \text{Channel coefficient } [k_c]: \\ &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2)) \\ &= 0.004 \cdot 179315 \cdot 109^3 / (6(1 - 0.3^2)) \\ &= 171202064.0000 \text{ MPa} \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:
 $CqDelta = CqTheta = CmTheta = CmDelta = 1$

$$\begin{aligned} \text{Channel coefficient } [\lambda_c]: \\ &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\ &\quad h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ &= 6(1762 + 109 / \cos(0)) \cdot 171202064 / 434^3 \cdot (1 + \\ &\quad 434 \cdot 0.004 \cdot (1 + 1) / 2 + (434 \cdot 0.004)^2 / 2 \cdot 1) \\ &= 100461.8 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Channel coefficient } [\delta_c]: \\ &= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c))) \\ &= 1762^2 / (4 \cdot 179315 \cdot 109) \cdot (1 - 1762 \cdot 0.3 / (2(1762 + 109))) \\ &= 0.34101E-01 \text{ mm/MPa} \end{aligned}$$

Channel coefficient [zeta_c]:
 $zeta_c = 0$

$$\begin{aligned} \text{Parameter } [\omega_c]: \\ &= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \\ &\quad \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ &= 1.01 \cdot 171202064 [0.034101211 (1 + 0.004 \cdot 1 \cdot 434) - \\ &\quad 0.000 (1/434 + 1 \cdot 0.004/2)] \\ &= 65525.1562 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:
 $= A / D_o$
 $= 2610 / 1736$
 $= 1.5035$

$$\begin{aligned} \text{Coefficient } [F]: \\ &= (1 - v^*) / E^* \cdot (\lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.32) / 66955 \cdot (100462 + 186264 \cdot \ln(1.5)) \\ &= 1.7842 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

$$\begin{aligned} \text{Moment on tubesheet due to pressures (Ps, Pt) [Mts]:} \\ &= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t] \\ &= 1736^2 / 16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 18 - \\ &\quad (1.01 - 1) \cdot (1.01^2 + 1) \cdot 0] \\ &= 297160.8750 \text{ MPa} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]



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$$\begin{aligned}
 &= Mts + (\omega c \cdot Pt) + W \star (C - Gs) / (2 \cdot \pi \cdot Do) \\
 &= 297161 + (0) + 51500748 \cdot (2410 - 1809) / (2 \cdot \pi \cdot 1736) \\
 &= 3135498.0000 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= (M \star) - Do^2 / 32 \cdot F (Ps - Pt) / (1 + F) \\
 &= (3135498) - 1736^2 / 32 \cdot 1.78 (18 - 0) / (1 + 1.78) \\
 &= 39797.9414 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2 / 64 \cdot (3 + \nu \star) \cdot (Ps - Pt) \\
 &= 39798 + 1736^2 / 64 \cdot (3 + 0.32) \cdot (18 - 0) \\
 &= 2856313.0000 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(|Mp|; |Mo|) \\
 &= \max(|39798|; |2856313|) \\
 &= 2856313.0000 \text{ MPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \star (h - hg')^2) \\
 &= 6 \cdot 2856313 / (0.36 (434 - 0)^2) \\
 &= 253.2563 \text{ MPa}
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 138 \\
 &= 276.40 \text{ MPa}
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \star (h - hg')^2) \\
 &= 6 \cdot 2791924 / (0.36 (412 - 0)^2) \\
 &= 275.2613 \text{ MPa}
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 412 + 3 + 13 \\
 &= 427.5724 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(428; 178) \\
 &= 427.5724 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$\begin{aligned}
 &= |Ps - Pt| \\
 &= |18 - 0| \\
 &= 18.001 \text{ MPa}
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\
 &= 4 \cdot 0.33 \cdot 434 / 1736 \cdot \min(0.8 \cdot 138; 0.533 \cdot 246) \\
 &= 36.660 \text{ MPa}
 \end{aligned}$$

|Ps - Pt| is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc$$



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$$= 1 / (4 \cdot 0.33 \cdot 111) \cdot (4 \cdot 1695580 / 5132) \cdot | 18 - 0 | + 16$$

$$= 178.2443 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 109.000 mm
Min. Channel len. adjacent to the tubesheet is: 788.840mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1762^2 / (4 \cdot 109 (1762 + 109) \cdot \cos(0)) \cdot 0$$

$$= 0.000 \text{ MPa}$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6 ((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 38486400 / 109^2 \cdot [(0.034101211 \cdot 1 \cdot 0.004 - 0.000 \cdot 1) \cdot 0 -$$

$$6 ((1 - 0.32) / 66955) \cdot 1736 / 434^3 \cdot (1 +$$

$$0.004 \cdot 434 / 2 \cdot 1) \cdot (39798 + 1736^2 / 32 \cdot (18 - 0))]$$

$$= -362.163 \text{ MPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 362.163 \text{ MPa (Should be } \leq 193.68)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$= E_c \cdot \sqrt{ (1.5 \cdot S_c / \sigma_c) }$$

$$= 179315 \cdot \sqrt{ (1.5 \cdot 129 / 362) }$$

$$= 131132.6 \text{ MPa}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 3044941 / (0.36 (434 - 0)^2)$$

$$= 270.0 \text{ MPa (Should be } \leq 276.4)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 429 + 3 + 13 = 445 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS}) = \max(445 ; 178) = 445 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 3.05) 3.05$$

$$= 153.228 \text{ mm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (19 - 3.05) 119$$



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= 18232.701 N

Groove Weld Strength [Fg]:

= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)

= $0.85 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 119$

= 18232.7012 N

Weld Strength Factor [fw]:

= S / S_w where S_w = min(S and S_t)

= 119/119

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)

= $0.55 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 119$

= 12989.8496 N

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$

= $2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) 1 \cdot 1)} - 0.75 \cdot 19 \right)$

= 3.445 mm

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm

Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 18233 is >= tube strength 18233 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	270.0 <=	276.4 MPa	Ok
Tubesheet shear stress	41.3 <=	110.6 MPa	Ok
Stress in Channel at Tubesheet	362.2 <=	389.4 MPa	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	444.933	450.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width, N = (Goc-Gic) / 2 23.000 mm

Basic Gasket Width, b0 = N / 2 11.500 mm

Effective Gasket Width, b = sqrt(b0) * 2.5 8.542 mm

Gasket Reaction Diameter, G = Go-2.0*b 1808.915 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:



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Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 858 + 19 \\ &= 1735.970 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (28.5 - 19) / 28.5 \\ &= 0.332 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= G_s / D_o \text{ (Configurations d, e, f)} \\ &= 1809 / 1736 \\ &= 1.0420 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1762 / 1736 \\ &= 1.0150 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 434 \\ &= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt) \\ &= \max(19 - 2 \cdot 3.05 (171067/186264) \cdot \\ &\quad (121/138) \cdot (0); 19 - 2 \cdot 3.05) \\ &= 19.0500 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 28.5 / \sqrt{ (1 - 4 \cdot \min(192024; 4 \cdot 1736 \cdot 28.5) / (\pi \cdot 1736^2)) } \\ &= 29.7316 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (29.7 - 19) / 29.7 \\ &= 0.3593 \end{aligned}$$

$h/p = 15.228069$; $\mu^* = 0.359267$; $E^* = 66954 \text{ MPa}$

$E^*/E = 0.359461$; $v^* = 0.322844$

Note: As h/p (15.228) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s and w_s are all = 0

Channel coefficient [β_c]:

$$\begin{aligned} &= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1762 + 109) 109 } \\ &= 0.0040 \text{ 1/mm} \end{aligned}$$



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Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.004 \cdot 179315 \cdot 109^3 / (6(1 - 0.3^2))$$

$$= 171202064.0000 \text{ MPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1762 + 109 / \cos(0)) \cdot 171202064 / 434^3 \cdot (1 + 434 \cdot 0.004 \cdot (1 + 1) / 2 + (434 \cdot 0.004)^2 / 2 \cdot 1)$$

$$= 100461.8 \text{ MPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1762^2 / (4 \cdot 179315 \cdot 109) \cdot (1 - 1762 \cdot 0.3 / (2(1762 + 109)))$$

$$= 0.34101\text{E-}01 \text{ mm/MPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 171202064 [0.034101211(1 + 0.004 \cdot 1 \cdot 434) - 0.000(1/434 + 1 \cdot 0.004/2)]$$

$$= 65525.1562 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2610 / 1736$$

$$= 1.5035$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 66955 \cdot (100462 + 186264 \cdot \ln(1.5))$$

$$= 1.7842$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$$

$$= 1736^2 / 16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 18 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 14.3]$$

$$= 215278.2656 \text{ MPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + (\omega_c \cdot Pt) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= 215278 + (935754) + 51500748 \cdot (2410 - 1809) / (2 \cdot \pi \cdot 1736)$$

$$= 3989368.7500 \text{ MPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F(Ps - Pt) / (1 + F)$$

$$= (3989368) - 1736^2 / 32 \cdot 1.78(18 - 14.3) / (1 + 1.78)$$

$$= 1208332.3750 \text{ MPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= 1208332 + 1736^2 / 64 \cdot (3 + 0.32) \cdot (18 - 14.3)$$



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$$= 1790412.0000 \text{ MPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(|1208332|; |1790412|) \\ &= 1790412.0000 \text{ MPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 1790412 / (0.36 \cdot (434 - 0)^2) \\ &= 158.7477 \text{ MPa} \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138 \\ &= 276.40 \text{ MPa} \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 1117437 / (0.36 \cdot (260 - 0)^2) \\ &= 275.4908 \text{ MPa} \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 260 + 3 + 13 \\ &= 276.2706 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(276; 49.5) \\ &= 276.2706 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |18 - 14.3| \\ &= 3.720 \text{ MPa} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.33 \cdot 434 / 1736 \cdot \min(0.8 \cdot 138; 0.533 \cdot 246) \\ &= 36.660 \text{ MPa} \end{aligned}$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.33 \cdot 111) \cdot (4 \cdot 1695580 / 5132) \cdot |18 - 14.3| + 16 \\ &= 49.5305 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 109.000 mm
Min. Channel len. adjacent to the tubesheet is: 788.840mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$\begin{aligned} &= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t \\ &= 1762^2 / (4 \cdot 109 (1762 + 109) \cdot \cos(0)) \cdot 14.3 \\ &= 54.351 \text{ MPa} \end{aligned}$$



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Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot kc/tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do/h^3 \cdot (Cm\Theta + h \cdot \beta_c/2 \cdot Cm\Delta) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 38486400/109^2 \cdot [(0.034101211 \cdot 1 \cdot 0.004 - 0.000 \cdot 1) \cdot 14.3 - 6((1 - 0.32)/66955) \cdot 1736/434^3 \cdot (1 + 0.004 \cdot 434/2 \cdot 1) \cdot (1208332 + 1736^2/32 \cdot (18 - 14.3))]$$

$$= -155.864 \text{ MPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot Sc)$$

$$= 210.214 \text{ MPa (Should be } \leq 193.68)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$= Ec \cdot \sqrt{(1.5 \cdot Sc / \sigma_c)}$$

$$= 179315 \cdot \sqrt{(1.5 \cdot 129/210)}$$

$$= 172120.0 \text{ MPa}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 1813574 / (0.36 \cdot (434 - 0)^2)$$

$$= 160.8 \text{ MPa (Should be } \leq 276.4)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + Cats + Catc = 267 + 3 + 13 = 283 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(HreqB; HreqS) = \max(283; 49.5) = 283 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (19 - 3.05) 3.05$$

$$= 153.228 \text{ mm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 3.05 (19 - 3.05) 119$$

$$= 18232.701 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 119$$

$$= 18232.7012 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 119/119$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$



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Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 119$$

$$= 12989.8496 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) 1 \cdot 1} - 0.75 \cdot 19 \right)$$

$$= 3.445 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm

Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 18233 is $\geq$ tube strength 18233 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	160.8 $\leq$	276.4 MPa	Ok
Tubesheet shear stress	8.5 $\leq$	110.6 MPa	Ok
Stress in Channel at Tubesheet	210.2 $\leq$	389.4 MPa	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	282.655	450.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	276.613 343.000	101	276	32 111	Ps+Pt D1	Ok
D2uc	433.444 ...	259	276	40 111	Ps+Pt D2	Ok
D3uc	271.927 ...	155	276	8 111	Ps+Pt D3	Ok
D1c	283.547 343.000	103	276	33 111	Ps+Pt-c D1	Ok
D2c	444.933 ...	270	276	41 111	Ps+Pt-c D2	Ok
D3c	282.655 ...	161	276	9 111	Ps+Pt-c D3	Ok
Max:	444.9330 343.000 mm	0.977	0.374	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures



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[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	Shell	Channel	Shell Stress	Reqd. Thk.	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	...	...	247	389	Ok
D2uc	...	...	...	...	...	335	389	Ok
D3uc	...	...	...	...	...	193	194	Ok
D1c	...	...	...	...	...	261	389	Ok
D2c	...	...	...	...	...	362	389	Ok
D3c	...	...	...	...	...	210	389	Ok
Max	...	...	...	...	...	0.997		

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	444.933	450.000	Ok
Tubesheet thickness flanged extension :	343.000	413.000	Ok
Tube thickness :	2.165	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.000	Ok
Tube-tubesheet groove weld leg :	1.829	3.000	Ok

Given Shell Thickness: 132.0000 mm

Given Channel Thickness: 112.0000 mm

Min Channel length of thk, (112.000) adj. to tubesheet: 798.259 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	38.097	1.000	19.094	1.000
Tubesheet Shear Stress	48.152	1.000	48.152	1.000
Tubesheet-Channel Junction	21.332	1.000	19.094	1.042
Tube Pressure Stress	43.670	1.000	25.339	1.000
Tubesheet Extension Stress	No Calc	No Calc	18.707	...
Minimum MAWP	21.332		18.707	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	41.694	1.000	20.440	1.000
Tubesheet Shear Stress	53.468	1.000	53.467	1.000
Tubesheet-Channel Junction	30.032	1.000	20.440	1.000
Tube Pressure Stress	50.641	1.000	36.520	1.000
Tubesheet Extension Stress	No Calc	No Calc	18.707	...
Minimum MAPnc	30.032		18.707	



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ASME TS Calc: DESIGN 1

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(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:***Channel Side MDMT calculation:***

Governing thickness on the shell side per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(450/4, min(450, 112))
= 112.500 mm

Thickness Ratio = 0.96, Temperature Reduction per Fig. UCS 66.1 = 2 C

Min Metal Temp. w/o impact per UCS-66, Curve B	34 C
Min Metal Temp. at Required thickness (UCS 66.1)	32 C

Shell Side MDMT calculation:

Governing thickness for the shell side:
= tubesheet thickness/4
= 450/4
= 112.500 mm

Thickness Ratio = 0.96, Temperature Reduction per Fig. UCS 66.1 = 2 C

Min Metal Temp. w/o impact per UCS-66, Curve B	34 C
Min Metal Temp. at Required thickness (UCS 66.1)	32 C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(14.3/30, 18/18.7)
= 0.962

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ASME TS Calc: DESIGN 2

Case: 2 4:12p Apr 10,2026

Input Echo, Tubesheet Number 2, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	18.00 MPa
Shell Maximum Operating Pressure	Psox,max	15.93 MPa
Shell Minimum Operating Pressure	Psox,min	0.00 MPa
Shell Thickness	ts	132.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1756.000 mm
Shell Temperature for Internal Pressure	Ts	479.96 C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	110.72 MPa
Shell Allowable Stress at Ambient		148.00 MPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	14.28 MPa
Channel Maximum Operating Pressure	Ptox,max	12.55 MPa
Channel Minimum Operating Pressure	Ptox,min	0.00 MPa
Channel Thickness	tc	112.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1756.000 mm
Channel Design Temperature	TEMPC	342.97 C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	129.12 MPa
Channel Allowable Stress at Ambient		138.00 MPa

Tube Data:

Number of Tube Holes	Nt	2156
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	8500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8050.00 mm
Design Temperature of the Tubes		399.97 C
Tube Material		SA-213 TP321
Tube Material UNS Number		S32100
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		119.00 MPa
Tube Allowable Stress At Ambient		138.00 MPa
Tube Yield Stress At design Temperature	Syt	132.00 MPa
Tube Pitch (Center to Center Spacing)	P	28.5000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	858.460 mm
Largest Center-to-Center Tube Distance	Ul	76.2000 mm



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ASME TS Calc: DESIGN 2

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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	1695580.0 mm2
Perimeter of Tube Layout	Cp	5131.8 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	369.97 C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	138.20 MPa
Tubesheet Allowable Stress at Ambient	Tt	148.00 MPa
Thickness of Tubesheet	h	450.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2610.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	1847.00 mm
Step 1 Depth on the Shellside	hs1	9.00 mm
Step 2 Diameter on the Shellside	ds2	1751.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2610.00 mm

Note: Tubesheet diameter is now: 2610.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	192024.0 mm2
---------------------------	----	--------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2610.000 mm
Flange Inside Diameter	B	1760.000 mm
Flange Face Outside Diameter	Fod	1850.000 mm
Flange Face Inside Diameter	Fid	1760.000 mm
Gasket Outside Diameter	Go	1826.000 mm
Gasket Inside Diameter	Gi	1780.000 mm
Small end Hub thk.	g0	132.0000 mm
Large end Hub thk.	g1	206.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	69.00 MPa
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option	Program Selects	

Bolting Information:

Diameter of Bolt Circle	C	2410.000 mm
Nominal Bolt Diameter	dB	95.2500 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-540 B21
Bolt Allowable Stress At Temperature	Sb	228.00 MPa
Bolt Allowable Stress At Ambient	Sa	228.00 MPa
Weld between Flange and Shell/Channel		0.0000 mm



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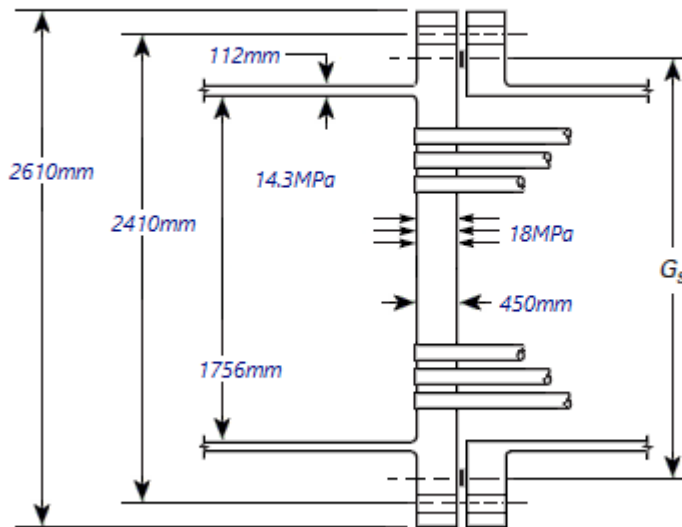
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ASME TS Calc: DESIGN 2

Case: 2 4:12p Apr 10, 2026

Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 413.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group D

Elastic Mod. at Design Temperature	480.0 C	0.17671E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	343.0 C	0.17931E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.20271E+06 MPa

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesht. Design Temp.	370.0 C	0.17107E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.19513E+06 MPa

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	370.0 C	0.18626E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (14.3 \cdot (19/2 - 0)) / (119 \cdot 1 + 0.4 \cdot 14.3)$
 $= 1.0907 + 0.0000 = 1.0907 \text{ mm}$



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ASME TS Calc: DESIGN 2

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Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 8908 / \sqrt{(9.52 \cdot 3.05)}$$

$$= 1653.342$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (19/3.05)$$

$$= 0.0880$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.088 \cdot 168489 (3.05/19)$$

$$= 3795.723 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3795.7/132.0$$

$$= 28.756$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 131.999 \text{ MPa}$$

For non-Carbon and Low Alloy Steel Materials re-compute [Fic]:

Factor A [A]:

$$= F/E_{mod}, \text{ see 2286-6 Step 1}$$

$$= 3796/168489$$

$$= 0.02253$$

From the External Pressure Chart, HA-2, determine the Stress [B]:

$$= 83.760 \text{ MPa}$$

The Tangent Modulus from 3-D.5.2, [Et]:

$$= 2B/A$$

$$= 2 \cdot 83.8/0.023$$

$$= 7436.1 \text{ MPa}$$

Predicted Buckling Stress [Fic]:

$$= (F \cdot E_t) / (F_S \cdot E_{mod})$$

$$= (3796 \cdot 7436) / (1 \cdot 168489)$$

$$= 167.519 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [168] \geq F_y [132], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y/FS$$

$$= 131.999/1.667$$

$$= 79.184 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$

$$= 2 \cdot 79.2 (3.05/19)$$

$$= 25.339 \text{ MPa}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.1653 mm
Allowable Internal Pressure at Corroded thickness	43.67 MPa
Required Internal Design Pressure	14.28 MPa
Allowable External Pressure at Corroded thickness	25.34 MPa



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ASME TS Calc: DESIGN 2

Case: 2 4:12p Apr 10,2026

Required External Design Pressure 18.00 MPa
Required Thickness due to Shell Side pressure 2.1653 mm

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	276.613	343.000	101	276	32	111	Ps+Pt D1	Ok
D2uc	433.444	...	259	276	40	111	Ps+Pt D2	Ok
D3uc	271.927	...	155	276	8	111	Ps+Pt D3	Ok
D1c	283.547	343.000	103	276	33	111	Ps+Pt-c D1	Ok
D2c	444.933	...	270	276	41	111	Ps+Pt-c D2	Ok
D3c	282.655	...	161	276	9	111	Ps+Pt-c D3	Ok
Max:	444.9330	343.000 mm		0.977		0.374	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk.		Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
	Shell	Channel					
D1uc	...	...	...	...	247	389	Ok
D2uc	...	...	...	...	335	389	Ok
D3uc	...	...	...	...	193	194	Ok
D1c	...	...	...	...	261	389	Ok
D2c	...	...	...	...	362	389	Ok
D3c	...	...	...	...	210	389	Ok
Max	...	...		...		0.997	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	444.933	450.000	Ok
Tubesheet thickness flanged extension :	343.000	413.000	Ok
Tube thickness :	2.165	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.000	Ok
Tube-tubesheet groove weld leg :	1.829	3.000	Ok

Given Shell Thickness: 132.0000 mm



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ASME TS Calc: DESIGN 2

Case: 2 4:12p Apr 10,2026

Given Channel Thickness: 112.0000 mm

Min Channel length of thk, (112.000) adj. to tubesheet: 798.259 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	38.097	1.000	19.094	1.000
Tubesheet Shear Stress	48.152	1.000	48.152	1.000
Tubesheet-Channel Junction	21.332	1.000	19.094	1.042
Tube Pressure Stress	43.670	1.000	25.339	1.000
Tubesheet Extension Stress	No Calc	No Calc	18.707	...
Minimum MAWP	21.332		18.707	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	41.694	1.000	20.440	1.000
Tubesheet Shear Stress	53.468	1.000	53.467	1.000
Tubesheet-Channel Junction	30.032	1.000	20.440	1.000
Tube Pressure Stress	50.641	1.000	36.520	1.000
Tubesheet Extension Stress	No Calc	No Calc	18.707	...
Minimum MAPnc	30.032		18.707	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

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ASME TS Calc: DESIGN 3-STE

Case: 3 4:12p Apr 10,2026

Input Echo, Tubesheet Number 3, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	18.00 MPa
Shell Maximum Operating Pressure	Psox,max	15.93 MPa
Shell Minimum Operating Pressure	Psox,min	0.00 MPa
Shell Thickness	ts	132.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1756.000 mm
Shell Temperature for Internal Pressure	Ts	479.96 C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	110.72 MPa
Shell Allowable Stress at Ambient		148.00 MPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	14.28 MPa
Channel Maximum Operating Pressure	Ptox,max	12.55 MPa
Channel Minimum Operating Pressure	Ptox,min	0.00 MPa
Channel Thickness	tc	112.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1756.000 mm
Channel Design Temperature	TEMPC	342.97 C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	129.12 MPa
Channel Allowable Stress at Ambient		138.00 MPa

Tube Data:

Number of Tube Holes	Nt	2156
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	8500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8050.00 mm
Design Temperature of the Tubes		399.97 C
Tube Material		SA-213 TP321
Tube Material UNS Number		S32100
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		119.00 MPa
Tube Allowable Stress At Ambient		138.00 MPa
Tube Yield Stress At design Temperature	Syt	132.00 MPa
Tube Pitch (Center to Center Spacing)	P	28.5000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	858.460 mm
Largest Center-to-Center Tube Distance	U1	76.2000 mm



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ASME TS Calc: DESIGN 3-STE

Case: 3 4:12p Apr 10,2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	1695580.0 mm2
Perimeter of Tube Layout	Cp	5131.8 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	369.97 C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	138.20 MPa
Tubesheet Allowable Stress at Ambient	Tt	148.00 MPa
Thickness of Tubesheet	h	450.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2610.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	1847.00 mm
Step 1 Depth on the Shellside	hs1	9.00 mm
Step 2 Diameter on the Shellside	ds2	1751.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2610.00 mm

Note: Tubesheet diameter is now: 2610.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	192024.0 mm2
---------------------------	----	--------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2610.000 mm
Flange Inside Diameter	B	1760.000 mm
Flange Face Outside Diameter	Fod	1850.000 mm
Flange Face Inside Diameter	Fid	1760.000 mm
Gasket Outside Diameter	Go	1826.000 mm
Gasket Inside Diameter	Gi	1780.000 mm
Small end Hub thk.	g0	132.0000 mm
Large end Hub thk.	g1	206.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	69.00 MPa
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option	Program Selects	

Bolting Information:

Diameter of Bolt Circle	C	2410.000 mm
Nominal Bolt Diameter	dB	95.2500 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-540 B21
Bolt Allowable Stress At Temperature	Sb	228.00 MPa
Bolt Allowable Stress At Ambient	Sa	228.00 MPa
Weld between Flange and Shell/Channel		0.0000 mm



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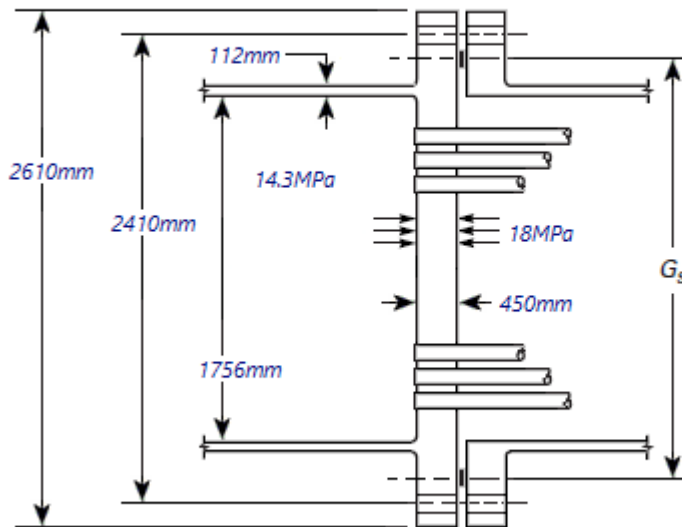
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ASME TS Calc: DESIGN 3-STE

Case: 3 4:12p Apr 10, 2026

Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 413.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group D

Elastic Mod. at Design Temperature	480.0 C	0.17671E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	343.0 C	0.17931E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.20271E+06 MPa

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesht. Design Temp.	370.0 C	0.17107E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.19513E+06 MPa

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	370.0 C	0.18626E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (14.3 \cdot (19/2 - 0)) / (119 \cdot 1 + 0.4 \cdot 14.3)$
 $= 1.0907 + 0.0000 = 1.0907 \text{ mm}$



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ASME TS Calc: DESIGN 3-STE

Case: 3 4:12p Apr 10, 2026

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 8908 / \sqrt{(9.52 \cdot 3.05)}$$

$$= 1653.342$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (19/3.05)$$

$$= 0.0880$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.088 \cdot 168489 (3.05/19)$$

$$= 3795.723 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3795.7/132.0$$

$$= 28.756$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 131.999 \text{ MPa}$$

For non-Carbon and Low Alloy Steel Materials re-compute [Fic]:

Factor A [A]:

$$= F/E_{mod}, \text{ see 2286-6 Step 1}$$

$$= 3796/168489$$

$$= 0.02253$$

From the External Pressure Chart, HA-2, determine the Stress [B]:

$$= 83.760 \text{ MPa}$$

The Tangent Modulus from 3-D.5.2, [Et]:

$$= 2B/A$$

$$= 2 \cdot 83.8/0.023$$

$$= 7436.1 \text{ MPa}$$

Predicted Buckling Stress [Fic]:

$$= (F \cdot Et) / (FS \cdot E_{mod})$$

$$= (3796 \cdot 7436) / (1 \cdot 168489)$$

$$= 167.519 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [168] \geq F_y [132], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y/FS$$

$$= 131.999/1.667$$

$$= 79.184 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$

$$= 2 \cdot 79.2 (3.05/19)$$

$$= 25.339 \text{ MPa}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.1653 mm
Allowable Internal Pressure at Corroded thickness	43.67 MPa
Required Internal Design Pressure	14.28 MPa
Allowable External Pressure at Corroded thickness	25.34 MPa



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 ASME TS Calc: DESIGN 3-STE Case: 3 4:12p Apr 10,2026

Required External Design Pressure 18.00 MPa
 Required Thickness due to Shell Side pressure 2.1653 mm

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	276.613	343.000	101	276	32	111	Ps+Pt D1	Ok
D2uc	433.444	...	259	276	40	111	Ps+Pt D2	Ok
D3uc	271.927	...	155	276	8	111	Ps+Pt D3	Ok
D1c	283.547	343.000	103	276	33	111	Ps+Pt-c D1	Ok
D2c	444.933	...	270	276	41	111	Ps+Pt-c D2	Ok
D3c	282.655	...	161	276	9	111	Ps+Pt-c D3	Ok
Max:	444.9330	343.000 mm		0.977		0.374	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk.		Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
	Shell	Channel					
D1uc	...	...	...	...	247	389	Ok
D2uc	...	...	...	...	335	389	Ok
D3uc	...	...	...	...	193	194	Ok
D1c	...	...	...	...	261	389	Ok
D2c	...	...	...	...	362	389	Ok
D3c	...	...	...	...	210	389	Ok
Max	...	...		...		0.997	

Note:
 Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	444.933	450.000	Ok
Tubesheet thickness flanged extension :	343.000	413.000	Ok
Tube thickness :	2.165	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.000	Ok
Tube-tubesheet groove weld leg :	1.829	3.000	Ok

Given Shell Thickness: 132.0000 mm



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ASME TS Calc: DESIGN 3-STE

Case: 3 4:12p Apr 10,2026

Given Channel Thickness: 112.0000 mm

Min Channel length of thk, (112.000) adj. to tubesheet: 798.259 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	38.097	1.000	19.094	1.000
Tubesheet Shear Stress	48.152	1.000	48.152	1.000
Tubesheet-Channel Junction	21.332	1.000	19.094	1.042
Tube Pressure Stress	43.670	1.000	25.339	1.000
Tubesheet Extension Stress	No Calc	No Calc	18.707	...
Minimum MAWP	21.332		18.707	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	41.694	1.000	20.440	1.000
Tubesheet Shear Stress	53.468	1.000	53.467	1.000
Tubesheet-Channel Junction	30.032	1.000	20.440	1.000
Tube Pressure Stress	50.641	1.000	36.520	1.000
Tubesheet Extension Stress	No Calc	No Calc	18.707	...
Minimum MAPnc	30.032		18.707	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

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ASME TS Calc: DESIGN 4

Case: 4 4:12p Apr 10,2026

Input Echo, Tubesheet Number 4, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	18.00 MPa
Shell Maximum Operating Pressure	Psox,max	15.93 MPa
Shell Minimum Operating Pressure	Psox,min	0.00 MPa
Shell Thickness	ts	132.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1756.000 mm
Shell Temperature for Internal Pressure	Ts	369.97 C
Shell Material		SA-387 22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K21590
Shell Allowable Stress at Temperature	Ss	138.20 MPa
Shell Allowable Stress at Ambient		148.00 MPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	14.28 MPa
Channel Maximum Operating Pressure	Ptox,max	12.55 MPa
Channel Minimum Operating Pressure	Ptox,min	0.00 MPa
Channel Thickness	tc	112.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1756.000 mm
Channel Design Temperature	TEMPC	342.97 C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	129.12 MPa
Channel Allowable Stress at Ambient		138.00 MPa

Tube Data:

Number of Tube Holes	Nt	2156
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	8500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8050.00 mm
Design Temperature of the Tubes		399.97 C
Tube Material		SA-213 TP321
Tube Material UNS Number		S32100
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		119.00 MPa
Tube Allowable Stress At Ambient		138.00 MPa
Tube Yield Stress At design Temperature	Syt	132.00 MPa
Tube Pitch (Center to Center Spacing)	P	28.5000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	858.460 mm
Largest Center-to-Center Tube Distance	Ul	76.2000 mm



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ASME TS Calc: DESIGN 4

Case: 4 4:12p Apr 10,2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	1695580.0 mm2
Perimeter of Tube Layout	Cp	5131.8 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	369.97 C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	138.20 MPa
Tubesheet Allowable Stress at Ambient	Tt	148.00 MPa
Thickness of Tubesheet	h	450.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2610.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	1847.00 mm
Step 1 Depth on the Shellside	hs1	9.00 mm
Step 2 Diameter on the Shellside	ds2	1751.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2610.00 mm

Note: Tubesheet diameter is now: 2610.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	192024.0 mm2
---------------------------	----	--------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2610.000 mm
Flange Inside Diameter	B	1760.000 mm
Flange Face Outside Diameter	Fod	1850.000 mm
Flange Face Inside Diameter	Fid	1760.000 mm
Gasket Outside Diameter	Go	1826.000 mm
Gasket Inside Diameter	Gi	1780.000 mm
Small end Hub thk.	g0	132.0000 mm
Large end Hub thk.	g1	206.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	69.00 MPa
Flange Facing Sketch	Code	Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2410.000 mm
Nominal Bolt Diameter	dB	95.2500 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-540 B21
Bolt Allowable Stress At Temperature	Sb	228.00 MPa
Bolt Allowable Stress At Ambient	Sa	228.00 MPa
Weld between Flange and Shell/Channel		0.0000 mm



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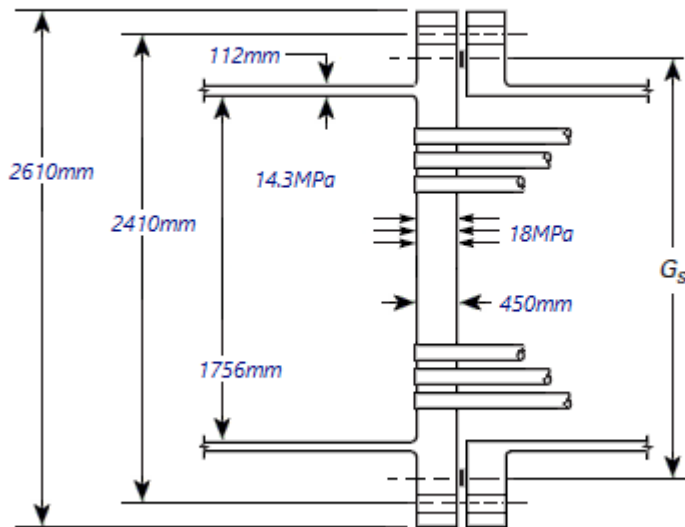
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ASME TS Calc: DESIGN 4

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Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 413.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group D

Elastic Mod. at Design Temperature	370.0 C	0.18626E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	343.0 C	0.17931E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.20271E+06 MPa

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesheet Design Temp.	370.0 C	0.17107E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.19513E+06 MPa

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	370.0 C	0.18626E+06 MPa
Elastic Mod. at Ambient Temperature	21.1 C	0.21099E+06 MPa

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (14.3 \cdot (19/2 - 0)) / (119 \cdot 1 + 0.4 \cdot 14.3)$
 $= 1.0907 + 0.0000 = 1.0907 \text{ mm}$



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Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 8908 / \sqrt{(9.52 \cdot 3.05)}$$

$$= 1653.342$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (19/3.05)$$

$$= 0.0880$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.088 \cdot 168489 (3.05/19)$$

$$= 3795.723 \text{ MPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 3795.7/132.0$$

$$= 28.756$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 131.999 \text{ MPa}$$

For non-Carbon and Low Alloy Steel Materials re-compute [Fic]:

Factor A [A]:

$$= F/E_{mod}, \text{ see 2286-6 Step 1}$$

$$= 3796/168489$$

$$= 0.02253$$

From the External Pressure Chart, HA-2, determine the Stress [B]:

$$= 83.760 \text{ MPa}$$

The Tangent Modulus from 3-D.5.2, [Et]:

$$= 2B/A$$

$$= 2 \cdot 83.8/0.023$$

$$= 7436.1 \text{ MPa}$$

Predicted Buckling Stress [Fic]:

$$= (F \cdot E_t) / (F_S \cdot E_{mod})$$

$$= (3796 \cdot 7436) / (1 \cdot 168489)$$

$$= 167.519 \text{ MPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [168] \geq F_y [132], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y/FS$$

$$= 131.999/1.667$$

$$= 79.184 \text{ MPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$

$$= 2 \cdot 79.2 (3.05/19)$$

$$= 25.339 \text{ MPa}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.1653 mm
Allowable Internal Pressure at Corroded thickness	43.67 MPa
Required Internal Design Pressure	14.28 MPa
Allowable External Pressure at Corroded thickness	25.34 MPa



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Required External Design Pressure 18.00 MPa
 Required Thickness due to Shell Side pressure 2.1653 mm

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	276.613	343.000	101	276	32	111	Ps+Pt D1	Ok
D2uc	433.444	...	259	276	40	111	Ps+Pt D2	Ok
D3uc	271.927	...	155	276	8	111	Ps+Pt D3	Ok
D1c	283.547	343.000	103	276	33	111	Ps+Pt-c D1	Ok
D2c	444.933	...	270	276	41	111	Ps+Pt-c D2	Ok
D3c	282.655	...	161	276	9	111	Ps+Pt-c D3	Ok
Max:	444.9330	343.000 mm		0.977		0.374	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk.		Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
	Shell	Channel					
D1uc	...	...	...	...	247	389	Ok
D2uc	...	...	...	...	335	389	Ok
D3uc	...	...	...	...	193	194	Ok
D1c	...	...	...	...	261	389	Ok
D2c	...	...	...	...	362	389	Ok
D3c	...	...	...	...	210	389	Ok
Max	...	...		...		0.997	

Note:
 Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	444.933	450.000	Ok
Tubesheet thickness flanged extension :	343.000	413.000	Ok
Tube thickness :	2.165	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.000	Ok
Tube-tubesheet groove weld leg :	1.829	3.000	Ok

Given Shell Thickness: 132.0000 mm



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ASME TS Calc: DESIGN 4

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Given Channel Thickness: 112.0000 mm

Min Channel length of thk, (112.000) adj. to tubesheet: 798.259 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	38.097	1.000	19.094	1.000
Tubesheet Shear Stress	48.152	1.000	48.152	1.000
Tubesheet-Channel Junction	21.332	1.000	19.094	1.042
Tube Pressure Stress	43.670	1.000	25.339	1.000
Tubesheet Extension Stress	No Calc	No Calc	18.707	...
Minimum MAWP	21.332		18.707	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	41.694	1.000	20.440	1.000
Tubesheet Shear Stress	53.468	1.000	53.467	1.000
Tubesheet-Channel Junction	30.032	1.000	20.440	1.000
Tube Pressure Stress	50.641	1.000	36.520	1.000
Tubesheet Extension Stress	No Calc	No Calc	18.707	...
Minimum MAPnc	30.032		18.707	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

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Problems/Failures Summary:

Step: 39 4:12pm Apr 10, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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